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Compensating Gas Sensor Drift via Domain Adaptation for Robust Gas Discrimination

Bachelor Thesis

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Submitted in	Munich, Germany 20 October 2026

Abstract

Electronic noses (e-noses) are susceptible to sensor drift, which alters the distribution of sensor responses collected over time and degrades classification performance. Domain adaptation aims to reduce this degradation by using existing labeled data together with subsequently collected unlabeled measurements.

This thesis builds on the domain-adversarial neural network (DANN) and the conditional domain-adversarial network (CDAN), which use adversarial learning to reduce distribution discrepancies between domains. However, adversarial alignment does not necessarily ensure that features of the same class are aligned across domains. Therefore, we introduce latent class-center alignment (LC) to promote within-class consistency between the source and target domains. To investigate the limitations of model-derived target information, we analyzed prediction errors and evaluated multi-seed probability averaging (MSPA). This ensemble approach averages softmax probabilities at inference from models trained with different random seeds.

On the short-term drift dataset, LC improved the mean classification accuracy of both DANN and CDAN across different target-data access settings. Seven-member MSPA changed short-term mean accuracy by -0.02 to $+0.89$ percentage points across the four methods under Protocol 1. On the long-term gas sensor drift dataset from the University of California, Irvine (UCI) Machine Learning Repository, the original LC yielded only modest gains. Analysis revealed persistent target prediction errors and differences across random seeds. MSPA improved the mean accuracy of all four methods under the long-term setting, although the benefit of LC remained model-dependent. Under the medium-term setting, MSPA improved the DANN variants on average but slightly reduced the accuracy of the CDAN variants.

The observed effects of class-center alignment and prediction averaging varied with the drift setting and adaptation model.

Keywords: Electronic nose; Drift compensation; Domain adaptation; Adversarial learning; Latent class-center alignment; Ensemble learning

Contents

1 Introduction **1**

1.1 Research Context . . . 1
1.2 Motivation . . . 1
1.3 Objectives and Contributions . . . 2
1.4 Research Questions . . . 3
1.5 Outline . . . 3

2 Related Work **4**

2.1 Electronic Noses and Drift Compensation . . . 4
2.2 From Distribution Matching to Class-Aware Adaptation . . . 4
2.3 Domain Adaptation for Electronic Noses and Positioning of This Work . . . 4
2.3.1 Ensemble Learning and Prediction Aggregation . . . 5

3 Methodology **6**

3.1 Problem Formulation . . . 6
3.2 Domain-Adversarial Neural Network (DANN) . . . 7
3.2.1 Network Architecture and Source Supervision . . . 7
3.2.2 Domain Discrimination and Adversarial Learning . . . 7
3.2.3 Gradient Reversal and Training . . . 8
3.3 Conditional Domain-Adversarial Network (CDAN) . . . 9
3.4 Latent Class-Center Alignment (LC) . . . 10
3.4.1 Class-Center Estimation . . . 10
3.4.2 Alignment Loss . . . 11
3.4.3 Integration with DANN and CDAN . . . 12
3.5 Role of Target Pseudo-Labels . . . 12
3.6 Multi-Seed Probability Averaging . . . 13

4 Datasets and Experimental Settings **14**

4.1 Short-Term Drift Dataset . . . 14
4.1.1 Hardware and Data Acquisition . . . 14
4.1.2 Dataset and Features . . . 15
4.1.3 Setting A: Short-Term Adaptation . . . 18
4.2 UCI Gas Sensor Drift Dataset . . . 18
4.2.1 Hardware and Data Acquisition . . . 18
4.2.2 Dataset and Features . . . 19
4.2.3 Settings B and C: Long- and Medium-Term Adaptation . . . 21
4.3 Data Preprocessing . . . 22

5 Short-Term Drift Compensation **24**
5.1 Evaluation Setups . . . 24
5.1.1 Implementation Details . . . 24
5.1.2 Evaluation Metrics and Procedures . . . 26
5.2 Classification Performance and Feature Representations . . . 26
5.2.1 Overall Classification Accuracy . . . 26
5.2.2 Class-Balanced Metrics and Class-Specific Performance . . . 28
5.2.3 Feature Visualization . . . 30
5.2.4 Quantitative Analysis of Class Structure . . . 32
5.3 Effect of Target Data Availability . . . 34
5.3.1 Classification Performance Across Protocols . . . 34
5.3.2 Discussion of Data Availability and Adaptation Performance . . . 35
5.4 Parameter Sensitivity Analysis . . . 36
5.4.1 Effect of the LC Loss Weight . . . 36
5.4.2 Effect of the Domain-Adversarial Loss Weight . . . 37
5.5 Multi-Seed Probability Averaging . . . 38
5.5.1 Evaluation Setups . . . 38
5.5.2 Overlap of Prediction Errors . . . 38
5.5.3 Classification Results . . . 39
5.5.4 Sensitivity Analysis . . . 40
5.5.5 Analysis of Short-Term MSPA Performance . . . 42
5.6 Chapter Summary . . . 43

6 Long-Term Drift Compensation **44**
6.1 Evaluation Setups . . . 44
6.1.1 Implementation Details . . . 44
6.1.2 Evaluation Metrics and Procedures . . . 46
6.2 Classification Performance . . . 46
6.2.1 Classification Results . . . 46
6.2.2 Effect of LC on Classification Performance . . . 47
6.2.3 Parameter Sensitivity Analysis . . . 48
6.3 Analysis of Pseudo-Label Reliability . . . 49
6.3.1 Pseudo-Label Error During Training . . . 49
6.3.2 Pseudo-Label Confidence During Training . . . 50
6.3.3 Effect of Entropy Conditioning . . . 51
6.3.4 Class-Wise Pseudo-Label Errors . . . 52
6.4 Multi-Seed Probability Averaging . . . 53
6.4.1 Evaluation Setups . . . 53
6.4.2 Overlap of Prediction Errors . . . 54
6.4.3 Classification Results . . . 55
6.4.4 Sensitivity Analysis . . . 57
6.4.5 Analysis of Long-Term MSPA Performance . . . 59
6.5 Chapter Summary . . . 60

7 Medium-Term Drift Compensation **61**
7.1 Single-Model Evaluation 61
7.1.1 Classification Results 61
7.1.2 Parameter Sensitivity Analysis 62
7.2 Multi-Seed Probability Averaging 63
7.2.1 Evaluation Setups 63
7.2.2 Overlap of Prediction Errors 63
7.2.3 Classification Results 64
7.2.4 Sensitivity Analysis 65
7.2.5 Analysis of Medium-Term MSPA Performance 67
7.3 Chapter Summary 68

8 Discussion and Conclusions **69**
8.1 Discussion Across the Three Drift Settings 69
8.2 Limitations and Future Work 70
8.3 Final Conclusions 71

Bibliography **72**

1 Introduction

1.1 Research Context

Electronic noses (e-noses) are artificial olfactory systems that combine gas sensor arrays with pattern recognition algorithms. The sensor array converts interactions between gases and sensing materials into electrical signals. Following data acquisition, preprocessing, and feature extraction, a predictive model uses the array's response patterns to identify gas species or estimate their concentrations. E-noses have been applied in areas such as healthcare research [1], [2] and the identification and quantification of harmful gases [3].

However, maintaining consistent sensor responses is difficult over long-term operation and across different devices. Sensor aging can alter responses to the same gas over time, resulting in time-varying drift. Fluctuations in temperature and humidity can further affect measurements [4], [5]. Manufacturing differences can produce distinct response characteristics among e-nose devices, even when they use the same types of sensors. Measurements obtained from different devices may follow different data distributions [5], [6]. Both temporal drift and instrumental variation can cause distribution shifts between training data and measurements collected during deployment, reducing the predictive performance of previously trained models.

Domain adaptation (DA) provides a machine learning approach to mitigating performance degradation caused by sensor drift. In gas sensor drift scenarios, labeled data collected at an earlier stage can serve as the source domain. Data collected later, with a different distribution, form the target domain [7]. Unsupervised domain adaptation uses labeled source data and unlabeled target data to improve prediction performance in the target domain by adapting feature representations or classification models. The aim is to preserve class-discriminative information while reducing the impact of distribution differences between domains. This allows adaptation to new measurements without using target-domain labels during training [8].

Domain-adversarial learning has been used for domain adaptation to learn discriminative features while reducing distribution differences between domains [9]. This approach has also been applied to electronic noses to improve gas classification across domains [10]. These studies provide a basis for algorithmic drift compensation and motivate further investigation into the relationship between domain alignment and class discrimination.

1.2 Motivation

Domain-adversarial learning reduces feature distribution differences between the source and target domains, enabling classification knowledge learned from the source domain to be transferred to the target domain. The domain-adversarial neural network (DANN) primarily performs domain alignment at the level of overall feature distributions, with the objective of learning domain-invariant features that con-

fuse the domain discriminator [9]. However, it does not directly constrain the correspondence between features of the same gas class across domains. The conditional domain-adversarial network (CDAN) combines feature representations with class predictions, allowing domain discrimination to exploit class-related information and better account for the structure of distributions with multiple classes [11]. Another line of research explicitly aligns corresponding class-specific feature centers between the source and target domains to promote cross-domain class consistency [12]. Both approaches exploit class information, but impose constraints on feature distributions in different ways. Therefore, this thesis investigates whether explicit class-center constraints can complement existing domain-adversarial learning in e-nose drift tasks. In particular, it examines whether they can provide additional benefits when conditional alignment is already employed.

Class alignment also depends on the reliability of target-domain class information. In unsupervised domain adaptation, ground-truth target labels are unavailable during training, so target-domain class correspondence and class centers generally need to be estimated from model predictions. When these predictions are incorrect, erroneous pseudo-labels can bias the estimation of target class centers and consequently affect cross-domain class alignment. This dependence motivates the analysis of target prediction reliability. Models trained with different random seeds may produce different predictions, raising the question of whether aggregating their outputs at inference can improve classification performance.

1.3 Objectives and Contributions

This thesis evaluates methods for reducing drift-related gas classification errors using existing labeled data and subsequently collected unlabeled measurements. It also examines the conditions under which class-structure constraints and multi-seed probability averaging (MSPA) improve classification. The evaluation focuses on classification under distribution shifts across batches. It considers per-batch classification performance, performance balance across classes, variability across random seeds, and prediction confidence. The experiments cover short-term drift tasks based on data collected in this study, as well as medium- and long-term drift tasks constructed from the gas sensor drift dataset available through the University of California, Irvine (UCI) Machine Learning Repository.

This thesis combines DANN and CDAN with latent class-center alignment (LC) to evaluate the complementary effects of explicit class-center constraints under different domain-adversarial alignment mechanisms. The combination of DANN with LC is denoted as DANNLC, while the combination of CDAN with LC is denoted as CDANLC. Comparing these four models allows the effects of LC to be evaluated under both overall and conditional domain alignment.

Following the class-centroid alignment principle of Xie et al. [12], this thesis evaluates an LC implementation using mini-batch centers and detached soft target probabilities.

The work comprises three main components:

1. **Implementation and controlled evaluation of class-center alignment.** Four models (DANN, DANNLC, CDAN, and CDANLC) were compared under a common experimental protocol. The comparisons assessed the effects of LC on classification performance across different drift tasks and target-data access settings. Twenty independent training runs using different random seeds were conducted to evaluate the consistency of the observed performance differences.

2. **Parameter sensitivity analysis.** Controlled experiments examined the effects of the LC and domain-adversarial loss weights on classification performance to assess their influence under the evaluated training configurations.

3. **Prediction reliability and multi-seed probability averaging.** Target prediction errors, confidence, and class-wise differences were analyzed to assess the reliability of model-derived target information. MSPA was evaluated for all four models. At inference, the predicted class probabilities were averaged across models trained with different random seeds. Matched ensemble sizes were used to compare aggregation gains and the contribution of LC under consistent evaluation conditions.

1.4 Research Questions

The evaluation is organized around three global research questions (RQs):

RQ1: To what extent does LC improve DANN and CDAN classification performance across short-, long-, and medium-term drift settings?

RQ2: How does model performance vary with target-data availability, loss weights, and target prediction reliability?

RQ3: To what extent does MSPA improve classification across drift settings, and what limits its benefit?

Chapters 5–7 address these RQs through chapter-specific key evaluation questions (KEQs). Chapter 8 combines the findings to answer the global RQs.

1.5 Outline

The remainder of this thesis is organized as follows.

Chapter 2 reviews electronic-nose drift compensation and domain adaptation. Chapter 3 presents DANN, CDAN, LC, and MSPA, while Chapter 4 describes the datasets, protocols, and preprocessing. Chapters 5–7 evaluate short-, long-, and medium-term drift, respectively, examining classification performance and parameter sensitivity. Chapter 5 also compares target-data access protocols, and Chapter 6 analyzes pseudo-label reliability. All three experimental chapters evaluate MSPA. Chapter 8 synthesizes the findings across the three drift settings and discusses limitations, future work, and conclusions.

2 Related Work

2.1 Electronic Noses and Drift Compensation

Electronic noses combine cross-sensitive sensor arrays with pattern recognition to discriminate gases. Early studies demonstrated that combined sensor responses could distinguish odors, establishing the basis for integrating sensing and classification [13], [14]. Subsequent work developed portable recognition systems and quantitative gas analysis [3], [15]. As applications expanded, attention shifted from recognition under fixed conditions to stability over time: sensor aging, contamination, and environmental changes can alter gas responses and undermine previously trained models.

Drift compensation initially relied on response correction or model updating, including direct standardization between measurement spaces and ensembles of classifiers trained at different times [4], [6]. Domain adaptation instead treats measurements acquired under different conditions as source and target domains. Domain adaptation extreme learning machine (DAELM) adapts the model using a small amount of labeled target data [16]. Domain regularized component analysis (DRCA) instead reduces domain differences through a shared subspace without requiring target class labels [7]. This thesis adopts the latter information setting: labeled source data and unlabeled target measurements.

2.2 From Distribution Matching to Class-Aware Adaptation

Changes in acquisition devices, illumination, and backgrounds also cause distribution shifts in computer vision. Early adaptation methods established connections between domains through statistical matching or shared representations, as exemplified by transfer component analysis [8]. DANN subsequently combined feature learning with domain-adversarial training, reducing domain distinguishability while retaining source classification information [9].

However, similar overall distributions do not guarantee correct class correspondence. CDAN incorporates classifier predictions into domain discrimination so that adaptation can account for class-related structure [11]. The Moving Semantic Transfer Network (MSTN) of Xie et al. explicitly aligns source and target class centroids using target pseudo-labels and exponential moving-average centroids [12]. These methods refine alignment through conditional discrimination and centroid matching, respectively, but both depend on inferred target class information. Prediction errors can therefore affect the direction of adaptation.

2.3 Domain Adaptation for Electronic Noses and Positioning of This Work

Electronic-nose adaptation has likewise progressed from overall distribution matching to preserving discriminative structure. Following DRCA, methods such as local discriminant subspace projection (LDSP) incorporated class information and local structure into shared-subspace learning [7], [17]. Wasserstein

distance learned feature representations (WDLFR) introduced Wasserstein-based adversarial representation learning for electronic-nose drift compensation, connecting this application to nonlinear domain adaptation [18].

Feature entropy domain adaption (FEDA) and the entropy minimization model based on adversarial networks (EMAD) are particularly relevant to this thesis. Both combine adversarial adaptation with additional classification-related constraints. FEDA uses a separate domain discriminator and introduces feature-norm and target conditional-entropy losses [10]. EMAD reuses the classifier for discrimination, connecting classifier predictions to domain-discrepancy estimation while retaining norm and entropy constraints [19]. Both address discriminability, but these constraints do not explicitly match corresponding class centroids across domains. Conditional adversarial learning has also been applied to electronic-nose drift suppression [20].

This thesis builds on DANN and CDAN and adapts the class-centroid matching principle of MSTN through latent class-center alignment (LC) [9], [11], [12]. Relative to FEDA and EMAD, the focus is on directly constraining the mean representations of corresponding classes. Unlike MSTN's moving-average centroids, LC estimates centers from the current mini-batch using true source labels and soft target-probability weights. Matched comparisons with and without LC across drift tasks of different time spans evaluate the benefits and limitations of this explicit alignment mechanism.

2.3.1 Ensemble Learning and Prediction Aggregation

Ensemble learning combines predictions from multiple classifiers to improve classification performance and reduce dependence on individual models. Common prediction aggregation methods include majority voting and probability averaging. Lam and Suen analyzed the classification performance of majority voting [21], while Kittler et al. investigated several classifier combination rules, including the sum rule, which provides a theoretical basis for probability averaging [22]. Ensemble methods can exploit differences among model predictions, although aggregation does not necessarily improve performance when multiple models make correlated errors.

Ensemble learning has also been applied to electronic-nose drift compensation. Vergara et al. combined support vector machine (SVM) classifiers trained at different time points through weighted aggregation to reduce performance degradation caused by long-term sensor drift [4]. More recently, Jiang et al. combined multi-level feature classifiers with domain adaptation, using maximum mean discrepancy (MMD) to weight their predictions [23]. These approaches construct ensembles by exploiting differences in training periods or feature representations.

An alternative strategy is to aggregate predictions from independently trained models with the same network architecture but different random seeds. Lakshminarayanan et al. demonstrated the effectiveness of this approach in deep learning through Deep Ensembles [24]. Building on this established principle, this thesis investigates multi-seed probability averaging (MSPA), which averages the softmax probabilities of independently trained DANN, DANNLC, CDAN, and CDANLC models at inference. The evaluation examines its classification performance and limitations under different sensor drift conditions.

3 Methodology

This chapter describes the adaptation framework used for gas sensor drift compensation. DANN provides the common network and adversarial training mechanism; CDAN introduces class information into the domain-discrimination branch; latent class-center alignment (LC) adds an explicit constraint on corresponding class centers. These components lead to the four models evaluated in this thesis: DANN, CDAN, DANNLC, and CDANLC. MSPA combines predictions from separately trained members of each model type at inference.

3.1 Problem Formulation

Let n_s and n_t denote the numbers of source and target observations, respectively. Superscripts s and t indicate the source and target domains, and i and j index their observations. Let m denote the input dimension and K the number of gas classes. A sensor-feature vector is $\mathbf{x} \in \mathbb{R}^m$, and its gas-class label is $y \in \{1, \ldots, K\}$. The labeled source dataset and the unlabeled target dataset are

$$\mathcal{D}_s = \{(\mathbf{x}_i^s, y_i^s)\}_{i=1}^{n_s}, \qquad \mathcal{D}_t = \{\mathbf{x}_j^t\}_{j=1}^{n_t}. \tag{3.1}$$

The two domains share the same input representation and class dictionary, but their data distributions can differ. A particular target batch need not contain observations from every class. Source labels are available for training, whereas target gas labels are reserved for evaluation. Domain membership is known from the dataset split and can be used without revealing a target sample's gas identity.

Let F_θ denote the feature extractor with parameters θ, and let $\mathbf{h} \in \mathbb{R}^d$ be its latent feature vector, where d is the latent dimension. Let C_ϕ denote the gas classifier with parameters ϕ; it outputs K logits, i.e., unnormalized class scores. The softmax function converts these logits into the predicted class-probability vector $\mathbf{p}(\mathbf{x}) \in \mathbb{R}^K$:

$$\mathbf{h} = F_\theta(\mathbf{x}) \in \mathbb{R}^d, \qquad \mathbf{p}(\mathbf{x}) = \mathrm{softmax}(C_\phi(\mathbf{h}))\,. \tag{3.2}$$

These predicted probabilities are model estimates of the class posterior probabilities and are not guaranteed to equal the true posterior probabilities. Both domains pass through the same extractor and classifier. The adaptation objective is to retain the class information learned from labeled source observations while making that information useful for target observations. This formulation follows the treatment of drift compensation as unsupervised domain adaptation in FEDA and EMAD [10], [19]. The following sections specify the losses used in this thesis; dataset-dependent network dimensions and training settings are given in the experimental chapters.

3.2 Domain-Adversarial Neural Network (DANN)

3.2.1 Network Architecture and Source Supervision

DANN augments the prediction network with a domain discriminator D_ψ with parameters ψ [9]. The classifier distinguishes gas categories, while the discriminator distinguishes source from target observations in the learned feature space. The extractor is shared by these two tasks. Its representation should support gas discrimination while providing less information about the domain of origin. Figure 3.1 illustrates the resulting training paths. In the figure, R_α denotes the gradient reversal layer (GRL) with gradient-reversal scale α, and $\mathcal{L}_{\text{dom}}$ denotes the domain-discrimination loss. Their definitions follow in Sections 3.2.3 and 3.2.2, respectively. Sample indices are omitted in the schematic.

At each training step, a mini-batch, i.e., a subset of observations used for one gradient update, is sampled from each domain. Let B_s and B_t denote the index sets of the source and target mini-batches, with sizes b_s and b_t, respectively. Let $\mathbb{I}(\cdot)$ denote the indicator function and $p_k(\mathbf{x})$ the predicted probability of class k for input $\mathbf{x}$. The supervised classification loss is

$$\mathcal{L}_{\text{cls}} = -\frac{1}{b_s} \sum_{i \in B_s} \sum_{k=1}^{K} \mathbb{I}(y_i^s = k) \log p_k(\mathbf{x}_i^s). \tag{3.3}$$

This loss trains the classifier and extractor using the known source gas labels. Target observations do not contribute a ground-truth classification term.

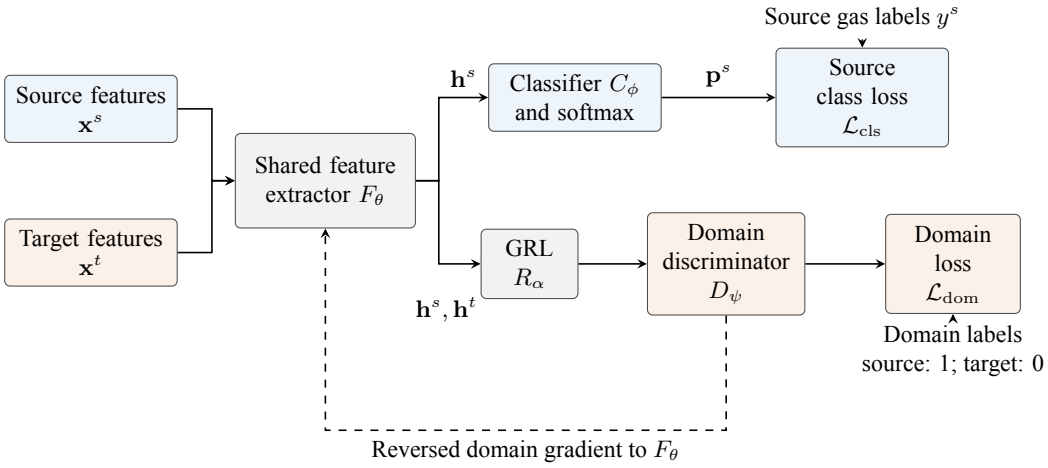


Figure 3.1 DANN training framework.

In this thesis, gas classification and domain discrimination use separate output heads. The DANN baseline combines the source classification loss with the domain-adversarial loss defined below.

3.2.2 Domain Discrimination and Adversarial Learning

Let $D_\psi(\mathbf{h})$ denote the predicted probability that a feature vector belongs to the source domain. Source observations are assigned domain label 1 and target observations label 0. Here, $\mathbf{h}_i^s = F_\theta(\mathbf{x}_i^s)$ and $\mathbf{h}_j^t =$

$F_\theta(\mathbf{x}_j^t)$ denote the latent feature vectors of source and target observations, respectively. The domain loss is the equally weighted average of the binary cross-entropy in the two domains:

$$\begin{aligned}\mathcal{L}_{\text{dom}} = -\frac{1}{2}\Bigg[&\frac{1}{b_s}\sum_{i\in B_s}\log D_\psi(\mathbf{h}_i^s) \\ &+\frac{1}{b_t}\sum_{j\in B_t}\log\left(1-D_\psi(\mathbf{h}_j^t)\right)\Bigg].\end{aligned} \tag{3.4}$$

In the implementation, the discriminator outputs logits and binary cross-entropy is evaluated directly from those logits. Equation 3.4 gives the equivalent probability-based expression. Averaging each domain separately prevents its sample count from determining its weight in this objective.

The discriminator minimizes $\mathcal{L}_{\text{dom}}$ to distinguish source from target features. In contrast, the feature extractor is encouraged to make domain discrimination more difficult while preserving source classification performance. Let $\lambda_d \geq 0$ denote the domain-loss weight. These opposing optimization objectives can be conceptually expressed as

$$\min_{\theta,\phi}\max_{\psi}\left[\mathcal{L}_{\text{cls}}(\theta,\phi) - \lambda_d\mathcal{L}_{\text{dom}}(\theta,\psi)\right]. \tag{3.5}$$

Maximization with respect to ψ minimizes the domain cross-entropy, whereas minimization with respect to θ encourages an increase in this loss while reducing the source classification loss. The classifier parameters ϕ are optimized through the source classification objective. This formulation illustrates the opposing optimization directions; the actual gradient scaling also depends on the gradient reversal layer, as detailed in Section 3.2.3. Source supervision is necessary because domain confusion alone does not preserve distinctions between gas classes. For example, mapping all observations to the same representation would remove domain information but also prevent useful classification.

3.2.3 Gradient Reversal and Training

The opposing optimization directions are implemented using a gradient reversal layer (GRL), introduced by Ganin and Lempitsky [25]. The GRL leaves its input unchanged during the forward pass. During backpropagation, the GRL maps an incoming domain-gradient vector $\mathbf{v}$ to $-\alpha\mathbf{v}$, where α is the gradient-reversal scale:

$$R_\alpha(\mathbf{h}) = \mathbf{h} \quad \text{(forward)}, \qquad \mathbf{v} \longmapsto -\alpha\mathbf{v} \quad \text{(backward)}. \tag{3.6}$$

Thus, the discriminator sees the current feature representation, while the extractor receives a reversed training signal. The backward expression is a custom differentiation rule for the GRL.

The training code forms the positive loss sum $\mathcal{L}_{\text{cls}} + \lambda_d\mathcal{L}_{\text{dom}}$ and differentiates it through the GRL. Ignoring optimizer-specific momentum and regularization, let $\mathbf{g}_\theta$, $\mathbf{g}_\phi$, and $\mathbf{g}_\psi$ denote the gradient signals for the extractor, classifier, and discriminator, respectively. The operator ∇ denotes differentiation with respect to the indicated parameter block. These signals are

$$\begin{aligned}\mathbf{g}_\theta &= \nabla_\theta\mathcal{L}_{\text{cls}} - \lambda_d\alpha\nabla_\theta\mathcal{L}_{\text{dom}}, \\ \mathbf{g}_\phi &= \nabla_\phi\mathcal{L}_{\text{cls}}, \qquad \mathbf{g}_\psi = \lambda_d\nabla_\psi\mathcal{L}_{\text{dom}}.\end{aligned} \tag{3.7}$$

Here, the domain derivative on the right-hand side is evaluated before gradient reversal. Each parameter block is updated in the negative direction of its corresponding $\mathbf{g}$. The domain-loss weight λ_d scales both the discriminator gradient and the domain-adversarial gradient sent to the extractor, whereas the gradient-reversal scale α acts only on the latter. Consequently, the effective domain-adversarial gradient coefficient for the extractor is $\lambda_d\alpha$. When $\alpha = 1$, these gradients correspond to the minimax formulation in Equation 3.5. For other values of α, Equation 3.5 describes the opposing optimization directions, while Equation 3.7 specifies the actual gradient scaling. The numerical values and scheduling of λ_d and α are part of the experimental configuration.

Each training step draws labeled source observations and unlabeled target observations, computes their shared features, and evaluates the two losses. Backpropagation then updates the extractor, classifier, and discriminator according to their respective roles. After adaptation, a target observation is classified as $\hat{y} = \arg\max_k p_k(\mathbf{x})$, where $\hat{y}$ is the predicted class and $\arg\max$ returns the class index with the largest probability. The domain branch is used only during training.

3.3 Conditional Domain-Adversarial Network (CDAN)

CDAN retains the source classification task and adversarial training mechanism, but changes the information supplied to the domain discriminator [11]. In a multiclass problem, the feature distribution contains contributions from several gas categories. A discriminator receiving only $\mathbf{h}$ does not explicitly observe the classifier's estimate of this class structure. CDAN addresses this limitation by conditioning domain discrimination on both the feature vector and its predicted class probabilities.

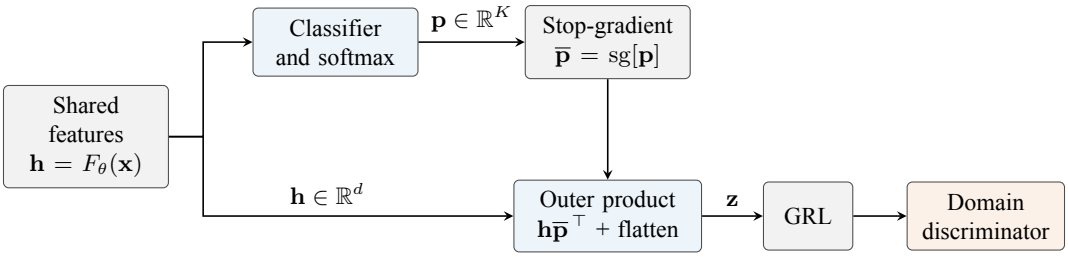
Here, sg denotes stop-gradient: its forward value is unchanged, but no gradient is propagated through it. The operator $\mathrm{flatten}$ stacks the matrix entries into a vector, and $\top$ denotes transposition. Let $\overline{\mathbf{p}}$ denote the detached probability vector and $\mathbf{z}$ the conditional representation. The implementation used here constructs the exact feature–probability outer product:

$$\mathbf{z} = \mathrm{flatten}\Big(\mathbf{h}\,\overline{\mathbf{p}}^{\top}\Big) \in \mathbb{R}^{dK}, \qquad \overline{\mathbf{p}} = \mathrm{sg}[\mathbf{p}(\mathbf{x})]\,. \tag{3.8}$$

Before flattening, the outer product has entries $h_r\overline{p}_k$, where $r \in \{1, \ldots, d\}$ indexes latent coordinates and $k \in \{1, \ldots, K\}$ indexes gas classes. Consequently, a latent feature contributes differently to the conditional representation depending on the predicted gas class. This multiplicative interaction makes class-related information available to domain discrimination. Figure 3.2 illustrates the implemented multilinear conditioning, in which feature–probability interactions replace the feature-only input of DANN [11].

The domain loss in Equation 3.4 is evaluated with $D_\psi(\mathbf{z})$ in place of $D_\psi(\mathbf{h})$. The classifier, source cross-entropy, and GRL retain their roles. Stopping the probability gradient means that the domain loss can update the extractor through $\mathbf{h}$, but cannot directly update the classifier through $\mathbf{p}$. Probabilities are recomputed at every training step. This version uses multilinear conditioning without entropy-based sample weighting or a randomized approximation of the outer product.

CDAN therefore makes domain alignment sensitive to predicted class information. It does not directly minimize the distance between corresponding class centers, and its conditioning remains dependent on the quality of target predictions. LC adds an explicit class-center constraint to either DANN or CDAN.



DANN: $\mathbf{h}$ enters the GRL. CDAN: $\mathrm{flatten}(\mathbf{h}\overline{\mathbf{p}}^\top)$ enters the GRL.

The conditional branch is applied to both domains; source classification is retained.

Figure 3.2 Conditional input to the CDAN domain discriminator.

3.4 Latent Class-Center Alignment (LC)

3.4.1 Class-Center Estimation

LC encourages observations of the same gas class to have similar mean representations across domains. This follows the principle of class-specific centroid alignment explored by Xie et al. [12]. Their method maintains moving centroids. Here, centers are recalculated from the current mini-batches at every step, using true source labels and soft target probabilities.

For source class k, let $n_{s,k} = \sum_{i \in B_s} \mathbb{I}(y_i^s = k)$ denote its sample count in the source mini-batch. Define the set of source classes present in that mini-batch as $\mathcal{K}_s = \{k : n_{s,k} > 0\}$. The source class center $\boldsymbol{\mu}_{s,k}$ is

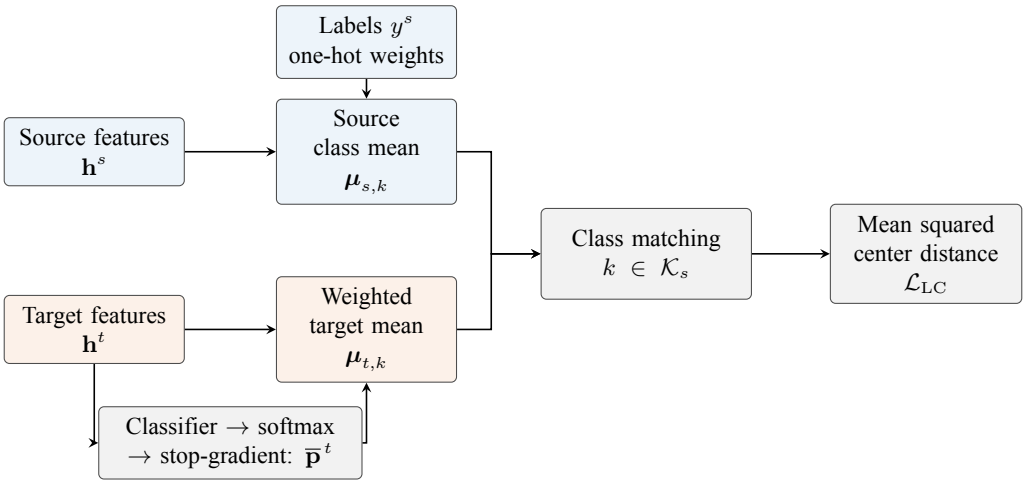
$$\boldsymbol{\mu}_{s,k} = \frac{\sum_{i \in B_s} \mathbb{I}(y_i^s = k)\mathbf{h}_i^s}{n_{s,k}}, \qquad k \in \mathcal{K}_s. \tag{3.9}$$

Source observations of other classes do not enter this average. Restricting alignment to $\mathcal{K}_s$ avoids constructing a class center from an empty source group.

For the target domain, define detached probabilities $\overline{p}_{jk} = \mathrm{sg}[p_k(\mathbf{x}_j^t)]$ and probability mass $m_{t,k} = \sum_{j \in B_t} \overline{p}_{jk}$. Let $\varepsilon > 0$ be a numerical safeguard that stabilizes division when the predicted mass is very small. In the implementation, ε was set to 10^{-8}. The target class center $\boldsymbol{\mu}_{t,k}$ is

$$\boldsymbol{\mu}_{t,k} = \frac{\sum_{j \in B_t} \overline{p}_{jk}\mathbf{h}_j^t}{\max(m_{t,k}, \varepsilon)}. \tag{3.10}$$

Each target observation contributes to a class center in proportion to its predicted probability for that class. No hard pseudo-label or confidence threshold is required in this formulation. Figure 3.3 denotes the resulting center-alignment loss by $\mathcal{L}_{\mathrm{LC}}$, defined in Equation 3.11. The predicted mass is a sum of soft assignments, rather than a count of target observations whose true class is known.



LC updates both feature sets; target probability weights are detached.

Figure 3.3 LC loss construction.

3.4.2 Alignment Loss

The LC loss is the mean squared discrepancy between corresponding centers, where $|\mathcal{K}_s|$ is the number of source classes present in the mini-batch and $\|\cdot\|_2$ denotes the Euclidean norm:

$$\mathcal{L}_{\mathrm{LC}} = \frac{1}{|\mathcal{K}_s| d} \sum_{k \in \mathcal{K}_s} \left\| \boldsymbol{\mu}_{s,k} - \boldsymbol{\mu}_{t,k} \right\|_2^2 . \tag{3.11}$$

The factor $1/d$ is included because the implementation averages over both the retained classes and the latent coordinates. Each retained class contributes equally after its center has been estimated. Figure 3.3 shows the computation of the two centers and their contribution to the loss.

For fixed probability weights, the loss in Equation 3.11 penalizes the mean squared discrepancy between corresponding source and target centers. A decrease in this aggregate loss does not necessarily imply a decrease in every class-specific discrepancy. Both source and target features receive gradients, so the source center is not a fixed anchor. For example, the direct gradient with respect to a target feature is

$$\frac{\partial \mathcal{L}_{\mathrm{LC}}}{\partial \mathbf{h}_j^t} = \frac{2}{|\mathcal{K}_s| d} \sum_{k \in \mathcal{K}_s} \frac{\overline{p}_{jk}}{\max(m_{t,k}, \varepsilon)} \left(\boldsymbol{\mu}_{t,k} - \boldsymbol{\mu}_{s,k} \right) . \tag{3.12}$$

Larger normalized weights give a class-center discrepancy more influence on an observation's update. Detaching the weights directs LC through the features rather than through the classifier's probability output. Predictions can still change at later steps as the representation and the source-supervised classifier are updated.

Matching class centers does not guarantee covariance alignment or class separation. Incorrect target predictions may bias center estimates and introduce inappropriate alignment signals. A class absent from a target batch may still receive nonzero predicted probability mass, which cannot be detected by the source-presence mask. These limitations motivate the analysis of target prediction reliability in Section 3.5 and the corresponding experiments in Chapter 6.

3.4.3 Integration with DANN and CDAN

The total training loss $\mathcal{L}_{\text{train}}$ combines LC with the other objectives using a nonnegative weight λ_{LC}:

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{cls}} + \lambda_d \mathcal{L}_{\text{dom}} + \lambda_{\text{LC}} \mathcal{L}_{\text{LC}}. \tag{3.13}$$

The domain branch contains the GRL, so Equation 3.13 denotes the loss differentiated by the implementation. With detached probabilities in both the conditional domain input and target-center weights, the extractor receives

$$\mathbf{g}_\theta = \nabla_\theta \mathcal{L}_{\text{cls}} - \lambda_d \alpha \nabla_\theta \mathcal{L}_{\text{dom}} + \lambda_{\text{LC}} \nabla_\theta \mathcal{L}_{\text{LC}}. \tag{3.14}$$

The classifier is directly supervised by the source classification term, and the discriminator by the domain term. LC adds no prediction head and does not change the inference procedure.

The combined model is DANNLC for feature-only domain discrimination and CDANLC for conditional domain discrimination. Setting $\lambda_{\text{LC}} = 0$ recovers the corresponding baseline. Comparing each pair tests whether explicitly aligning class centers adds useful information to its adversarial objective.

3.5 Role of Target Pseudo-Labels

In unsupervised domain adaptation, target gas labels are unavailable during training, so class-related target information must be inferred from model predictions. Hard pseudo-labels assign samples to their most probable classes, whereas soft pseudo-labels retain the predicted class probabilities. The CDAN and LC implementations in this thesis use soft probabilities, but their roles differ from direct supervision with ground-truth labels.

DANN: no explicit target pseudo-label term. DANN uses source gas labels for classification and domain labels for adversarial alignment [9]. As shown in Equations 3.3 and 3.4, neither objective directly uses predicted target gas classes or their probabilities. Although DANN can produce incorrect target predictions, these predictions are not explicitly reused in its training objective.

CDAN: predictions condition domain discrimination. CDAN incorporates target class probabilities into the conditional representation used for domain discrimination [11]. As defined in Equation 3.8, incorrect predictions can therefore alter the class-related information supplied to the discriminator. With the probability branch detached, domain-adversarial gradients still depend on the predicted probabilities through the conditional feature representation. Unlike direct pseudo-label supervision, this influence is mediated by the conditional representation and domain-adversarial objective.

LC: predictions determine target class-center estimates. LC uses soft target probabilities as class-membership weights in Equation 3.10. Incorrect predictions can bias the estimated target centers and consequently affect the alignment directions. As shown in Equation 3.12, the contribution of each class-center discrepancy to a target feature gradient depends on its normalized probability weight. Therefore, the effect of prediction errors depends not only on incorrect class assignments but also on prediction probabilities and feature distributions. Although the probability weights are detached, LC still updates the feature representations, which may in turn affect subsequent predictions and center estimates.

3.6 Multi-Seed Probability Averaging

Different random initializations and mini-batch sequences can lead models with the same architecture to make different target predictions. Combining their probability vectors may reduce dependence on an individual run. This follows the ensemble-learning principle that sufficiently accurate classifiers can benefit from complementary errors [40]. MSPA applies this principle at inference, after each member has been trained with its original objective.

Let M denote the number of separately trained models of the same type, indexed by $r \in \{1, \ldots, M\}$, and let s_r denote the random seed of member r. For a target input $\mathbf{x}$ and gas class $k \in \{1, \ldots, K\}$, let $p_k^{(r)}(\mathbf{x})$ denote the probability assigned by member r. Here, K is the number of gas classes defined in Section 3.1. The ensemble probability $p_{\text{ens},k}(\mathbf{x})$ and predicted class $\hat{y}_{\text{ens}}(\mathbf{x})$ are defined by

$$p_{\text{ens},k}(\mathbf{x}) = \frac{1}{M} \sum_{r=1}^{M} p_k^{(r)}(\mathbf{x}), \qquad \hat{y}_{\text{ens}}(\mathbf{x}) = \arg \max_{k \in \{1,\ldots,K\}} p_{\text{ens},k}(\mathbf{x}). \tag{3.15}$$

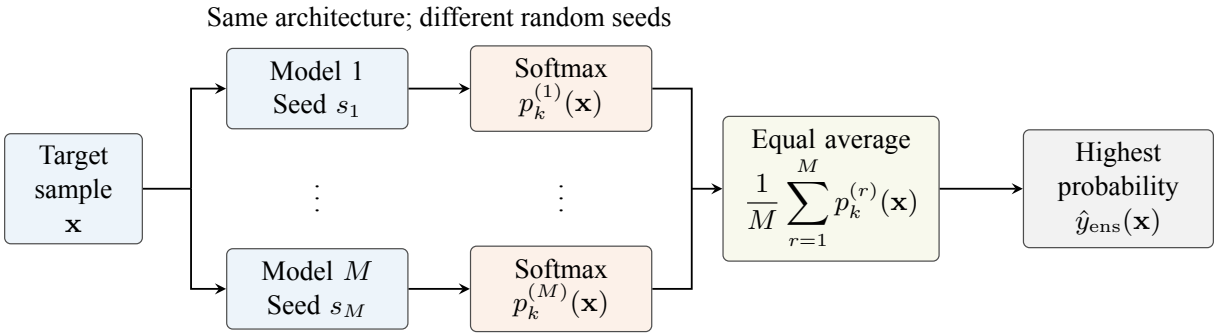


Figure 3.4 Multi-seed probability averaging (MSPA) with M models.

Figure 3.4 illustrates probability averaging across M models. MSPA combines full softmax vectors rather than voting on hard labels.

The averaging rule does not depend on drift duration or require a particular number of gas classes. Its benefit depends on the member predictions and must therefore be evaluated for each task. Chapters 5–7 evaluate MSPA separately for DANN, DANNLC, CDAN, and CDANLC. The short-term evaluation tests whether averaging improves the single-model predictions.

4 Datasets and Experimental Settings

This chapter introduces the datasets, feature representations, and experimental protocols used to study gas sensor drift over different time scales. Two datasets were used: a self-collected short-term (ST) drift dataset and the publicly available Gas Sensor Array Drift at Different Concentrations dataset from the University of California, Irvine (UCI) Machine Learning Repository. Three experimental settings (Settings A, B, and C) are defined to investigate gas classification under short-, medium-, and long-term drift.

Table 4.1 summarizes the datasets and drift time scales of the three experimental settings. Setting A comprises three protocols: offline pooled (Protocol 1), independent (Protocol 2), and cumulative (Protocol 3), as described in Section 4.1.3. Settings B and C are described in Section 4.2.3.

Table 4.1 Overview of the experimental settings.

Experimental setting	Dataset	Drift time scale
Setting A (Protocols 1–3)	ST	Short-term
Setting B	UCI	Long-term
Setting C	UCI	Medium-term

This chapter also describes the data preprocessing methods applied to both datasets before model training.

4.1 Short-Term Drift Dataset

4.1.1 Hardware and Data Acquisition

The ST measurements were acquired with a custom electronic nose developed by the Mobile Robotics and Olfaction (MRO) Lab at the Centre for Applied Autonomous Sensor Systems (AASS). Figure 4.1 shows the sensor board and the Arduino Micro-based acquisition and control board, connected by a ribbon cable. The hardware uses a 5 V supply and five metal-oxide sensor packages: MiCS-4514, MiCS-2714, MiCS-5914, MiCS-5524, and MiCS-2614. The MiCS-4514 contains two independent sensing elements [26], giving six measurement channels in total. In the firmware, channels 1–2 correspond to MiCS-4514, and channels 3–6 correspond to MiCS-2714, MiCS-5914, MiCS-5524, and MiCS-2614, respectively.

The circuit uses three AD5242 dual digital potentiometers for adjustable load resistances and provides pulse-width-modulated heater control. The Arduino firmware sequentially reads the six analog inputs A0–A5 and sends their digitized values, a millisecond timestamp, and the load and heater control settings to a computer over a Universal Serial Bus (USB) serial connection. The supplied firmware inserts a 250 ms delay after each acquisition cycle, corresponding to a nominal rate of approximately 4 Hz; reading

and transmission add to this interval. These readings are analog-to-digital converter (ADC) counts, rather than calibrated gas concentrations. The six recorded response channels provide the signals from which the 54 ST features are extracted.

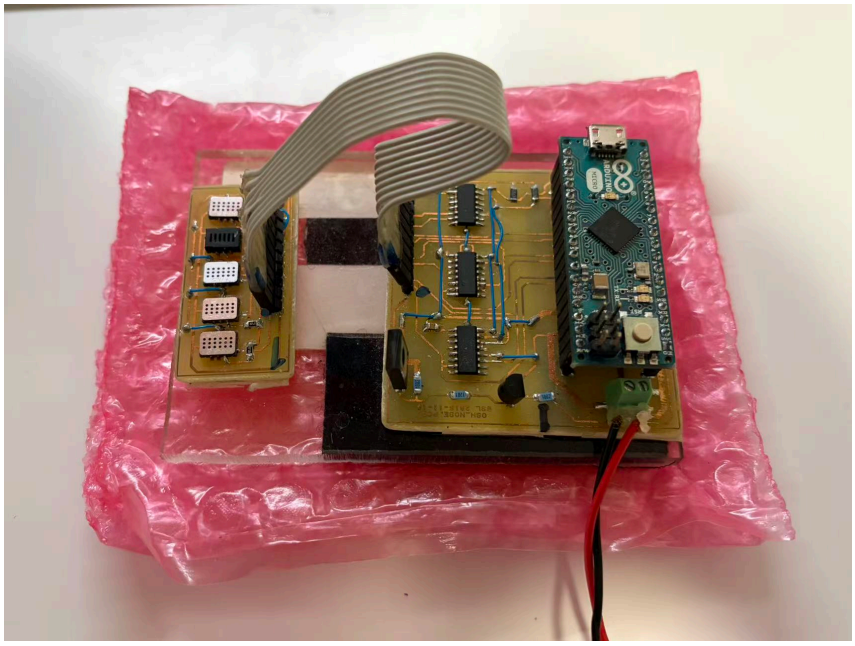


Figure 4.1 Electronic-nose hardware used for ST data acquisition. The sensor board (left) is connected to the acquisition and control board with an Arduino Micro (right).

The ST dataset was collected in a real-world environment rather than under controlled laboratory conditions. In this thesis, drift is distinguished in a narrow and a broad sense. In the narrow sense, it refers to changes in sensor response over time under otherwise unchanged measurement conditions, for example due to sensor aging. In the broad sense, it also encompasses response variations associated with environmental and acquisition conditions. Ye et al. [27] distinguish between natural and secondary drift. This study adopts the broad interpretation: effects associated with temperature, humidity, gas concentration, and acquisition procedures are included within the scope of drift compensation.

Data acquisition consisted of three phases: air baseline recording, gas exposure, and recovery in air. For most gas samples, these phases lasted 60 s, 60 s, and 60 s, respectively. Some samples had shorter gas exposure periods of 5 s or 10 s. Response variations associated with these differences in exposure duration are also treated as part of drift under the broad definition adopted here.

4.1.2 Dataset and Features

The ST dataset was used to evaluate gas classification performance across acquisition batches. The dataset contains 303 samples from three classes: air, alcohol, and acetone. Each sample corresponds to a separate acquisition experiment and is represented by a 54-dimensional feature vector extracted from six sensor channels. The source batch B_1 consists of 30 samples obtained from 30 separate acquisition experiments conducted during the first collection period, from 27 to 29 April 2026. The remaining 273 samples were divided chronologically into five target batches (B_2–B_6). Table 4.2 summarizes the sample distribution.

Table 4.2 Acquisition dates and sample counts of the ST dataset.

Batch	Date	Air	Alcohol	Acetone	Total
B_1	27 Apr – 29 Apr	6	12	12	30
B_2	14 May – 15 May	8	10	10	28
B_3	19 May – 25 May	18	33	33	84
B_4	27 May – 4 Jun	31	25	25	81
B_5	12 Jun – 15 Jun	14	16	16	46
B_6	20 Jun – 22 Jun	10	12	12	34
Total	—	**87**	**108**	**108**	**303**

Note: All dates are in 2026.

For each sample and sensor channel, the baseline median was subtracted from the readings, and the result was divided by the absolute baseline median. This produced a dimensionless relative response. Nine statistical features were then extracted from each of the six channels to describe the gas response, recovery process, and overall signal variation. Table 4.3 defines these features. Class labels, acquisition dates, and sample identifiers were excluded from the model inputs. In Table 4.3, n denotes the number of observations in the corresponding response or recovery window, and $\lceil \cdot \rceil$ denotes rounding up to the nearest integer.

Table 4.3 Features extracted from each ST sensor channel.

No.	Feature	Definition
1	Response mean	Mean of the observations within the response window.
2	Maximum absolute response	Maximum absolute value within the response window.
3	Response range	Difference between the maximum and minimum values within the response window.
4	Response tail mean	Mean of the final $\lceil 0.25n \rceil$ observations within the response window.
5	Mean absolute response	Mean of the absolute values within the response window.
6	Recovery mean	Mean of the observations within the recovery window.
7	Recovery tail mean	Mean of the final $\lceil 0.25n \rceil$ observations within the recovery window.
8	Recovery slope	Slope of the least-squares fit of the recovery response against elapsed time (s).
9	Overall standard deviation	Population standard deviation from the start of the record to the end of the recovery window.

The tail means use the final quarter of each corresponding window, with the number of observations rounded up. Features 1–7 and 9 are dimensionless. The recovery slope is expressed in s^{-1} because the relative response was fitted against elapsed time in seconds.

The extracted features underwent the common preprocessing procedure described in Section 4.3, consisting of an inverse hyperbolic sine (asinh) transformation followed by z-score standardization using source-domain statistics.

Figure 4.2 presents two-dimensional principal component analysis (PCA) visualizations of the six ST dataset batches before domain adaptation. PCA was fitted separately to each batch. In Figure 4.3, all batches were projected onto a common PCA basis fitted to the pooled samples. Both figures include all 303 samples after preprocessing using source statistics from Batch 1. PC1 and PC2 denote the first and second principal component scores, respectively; both are dimensionless after standardization. The shared PCA space supports comparisons across batches, whereas the separate projections describe within-batch structure.

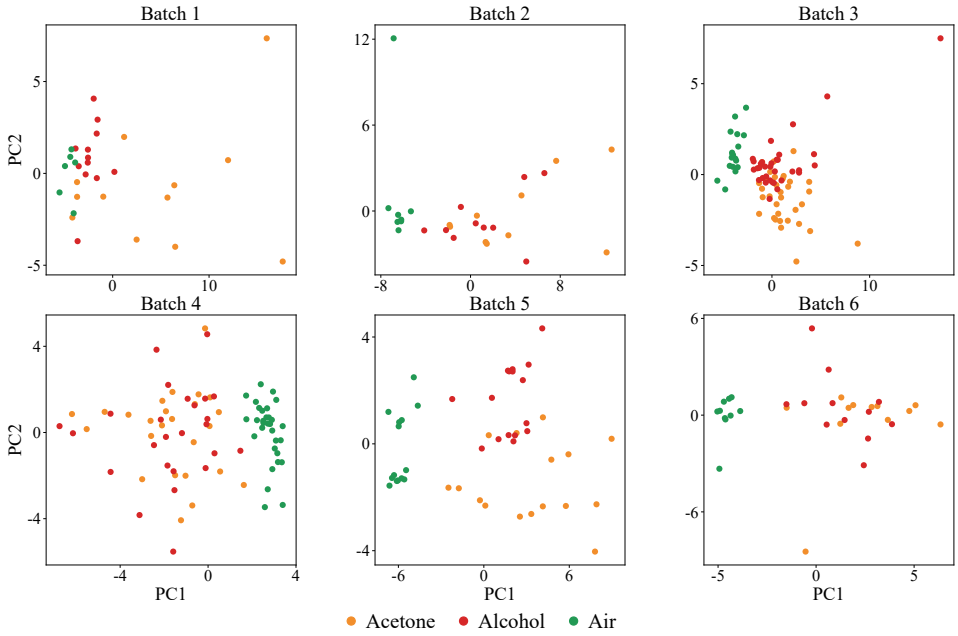


Figure 4.2 Distribution of principal components of the ST dataset (individual batches).

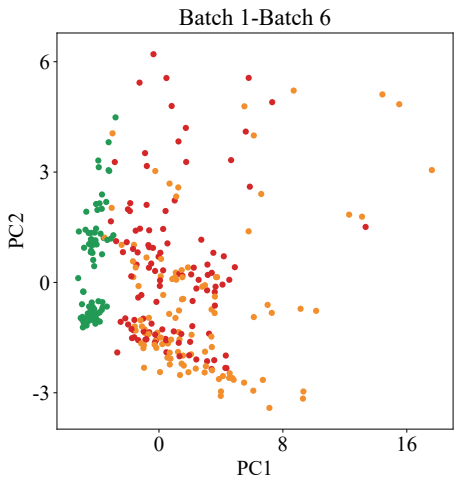


Figure 4.3 Distribution of principal components of the ST dataset (all batches). Colors follow Figure 4.2.

4.1.3 Setting A: Short-Term Adaptation

Within Setting A, three adaptation protocols (Protocols 1–3) are defined to investigate how the availability of unlabeled target data affects drift compensation. All three protocols use labeled data from Batch 1 (B_1) as the source domain and evaluate classification performance on Batches 2–6 (B_2–B_6). They share the same network architectures and training hyperparameters but differ in the target data available during adaptation. In Table 4.4, B_k denotes Batch k, where $k = 2, \ldots, 6$.

Table 4.4 Target-data access protocols for the ST dataset.

Protocol	Labeled source	Unlabeled target
Protocol 1: Offline pooled	B_1	$B_2, \ldots, B_6$
Protocol 2: Independent	B_1	Current batch B_k
Protocol 3: Cumulative	B_1	$B_2, \ldots, B_k$

Protocol 1 trains one model using all target data and represents an offline scenario in which all target measurements have already been collected. Protocol 2 trains a separate model for each target batch using only its unlabeled data. Protocol 3 accumulates target data chronologically, allowing historical measurements to support adaptation to the current batch. In Protocols 2 and 3, each model was initialized from scratch and trained independently for its corresponding task.

In Protocols 1 and 3, all available target batches are pooled and treated as a single target domain for alignment with the source domain B_1. Training mini-batches were drawn from this combined pool, without computing separate alignment losses for individual target batches. For example, when evaluating B_4, Protocol 3 pools B_2, B_3, and B_4 for adaptation but evaluates the resulting model only on B_4.

All three protocols use unlabeled inputs from the current evaluation batch during adaptation and therefore follow a transductive domain adaptation setting.

4.2 UCI Gas Sensor Drift Dataset

The Gas Sensor Array Drift at Different Concentrations dataset was obtained from the UCI Machine Learning Repository [28]. The dataset was collected at the ChemoSignals Laboratory, University of California San Diego, between 2008 and 2011, over a period of approximately 36 months.

4.2.1 Hardware and Data Acquisition

The original measurement platform used 16 Figaro metal-oxide gas sensors: four each of TGS2600, TGS2602, TGS2610, and TGS2620. Their heaters were operated at a constant 5 V. The array was housed in a 60 mL airtight chamber connected to a computer-controlled gas-delivery system with three digital mass-flow controllers and calibrated gas cylinders. The total flow was maintained at 200 mL/min, and the responses were recorded at 100 Hz. The assignment of sensors to acquisition channels remained unchanged throughout data collection [29].

Each measurement included 50 s of dry-air baseline recording, 100 s of exposure to an analyte diluted in the carrier gas, and 200 s of recovery in clean dry air [29]. These acquisition conditions describe the

original UCI measurement system; the present study used its released feature vectors rather than operating this hardware.

4.2.2 Dataset and Features

The dataset contains 13,910 samples obtained by exposing an array of 16 chemical sensors to six gases at different concentrations: acetone (Acn), acetaldehyde (Aca), ethanol (Eta), ethylene (Ety), ammonia (Amm), and toluene (Tol) [4]. Eight features were extracted from each sensor, giving a 128-dimensional feature vector for each sample. The samples were divided chronologically into ten batches (B_1–B_{10}). Table 4.5 summarizes the collection periods and sample counts.

Table 4.5 Collection periods and sample counts of the UCI dataset.

Batch	Months	Acn	Aca	Eta	Ety	Amm	Tol	Total
B_1	1, 2	70	30	90	98	83	74	445
B_2	3, 4, 8–10	532	109	164	334	100	5	1,244
B_3	11–13	275	240	365	490	216	0	1,586
B_4	14, 15	12	30	64	43	12	0	161
B_5	16	63	46	28	40	20	0	197
B_6	17–20	606	29	514	574	110	467	2,300
B_7	21	630	744	649	662	360	568	3,613
B_8	22, 23	143	33	30	30	40	18	294
B_9	24, 30	78	75	61	55	100	101	470
B_{10}	36	600	600	600	600	600	600	3,600
Total	—	**3,009**	**1,936**	**2,565**	**2,926**	**1,641**	**1,833**	**13,910**

Note: Months are numbered relative to the start of data collection; month 1 is the first collection month.

According to Vergara et al. [4] and Rodríguez-Luján et al. [30], the 128-dimensional representation consists of eight features from each of the 16 sensors, including two steady-state features and six transient features. The steady-state features describe the change in sensor resistance relative to the baseline and its normalized value. The transient features are obtained by applying exponential moving average (EMA) filtering to the resistance differences between successive samples. The maximum values during the rising response and the minimum values during the recovery phase are then extracted. Three smoothing parameters are used for each phase, giving six transient features. Table 4.6 lists the eight features and their definitions.

Following the original feature definitions, $R[k]$ denotes sensor resistance at a discrete sample index k, and $R_{\max}$ and $R_{\min}$ denote the maximum and minimum resistance over the relevant recording interval. In this convention, $R_{\min}$ is the baseline reference [4]. The smoothing parameter is $\alpha \in \{0.001, 0.01, 0.1\}$; here, α refers to EMA smoothing rather than the GRL scale in Chapter 3. The filtered resistance-difference signal $y_\alpha[k]$ is defined by

$$y_\alpha[k] = (1-\alpha)y_\alpha[k-1] + \alpha\big(R[k] - R[k-1]\big), \qquad y_\alpha[0] = 0. \tag{4.1}$$

The index k in this recurrence starts at 1. Resistance, resistance change, and the filtered differences are expressed in ohms (Ω); normalized resistance change and α are dimensionless.

Table 4.6 Features extracted from each UCI sensor channel.

No.	Feature	Definition
1	Resistance change	$\Delta R = R_{\max} - R_{\min}$
2	Normalized resistance change	$\Delta R_{\text{norm}} = (R_{\max} - R_{\min})/R_{\min}$
3	Rising transient, $\alpha = 0.001$	Maximum of $y_{0.001}[k]$ in the rising phase
4	Rising transient, $\alpha = 0.01$	Maximum of $y_{0.01}[k]$ in the rising phase
5	Rising transient, $\alpha = 0.1$	Maximum of $y_{0.1}[k]$ in the rising phase
6	Decaying transient, $\alpha = 0.001$	Minimum of $y_{0.001}[k]$ in the recovery phase
7	Decaying transient, $\alpha = 0.01$	Minimum of $y_{0.01}[k]$ in the recovery phase
8	Decaying transient, $\alpha = 0.1$	Minimum of $y_{0.1}[k]$ in the recovery phase

The eight features from each sensor are arranged in the order shown in Table 4.6 and concatenated across the 16 sensors to form a 128-dimensional feature vector. The feature values supplied with the UCI dataset were used directly; they were not re-extracted from the raw sensor signals. The common preprocessing procedure described in Section 4.3 was applied before model training.

Figure 4.4 presents two-dimensional PCA visualizations of the ten UCI dataset batches before domain adaptation. PCA was fitted separately to each batch. In Figure 4.5, all batches were projected onto a common PCA basis fitted to the pooled samples. Both figures include all 13,910 samples. The visualization data were preprocessed as described in Section 4.3, using source statistics from Batch 1. As in the ST figures, PC1 and PC2 are dimensionless principal component scores.

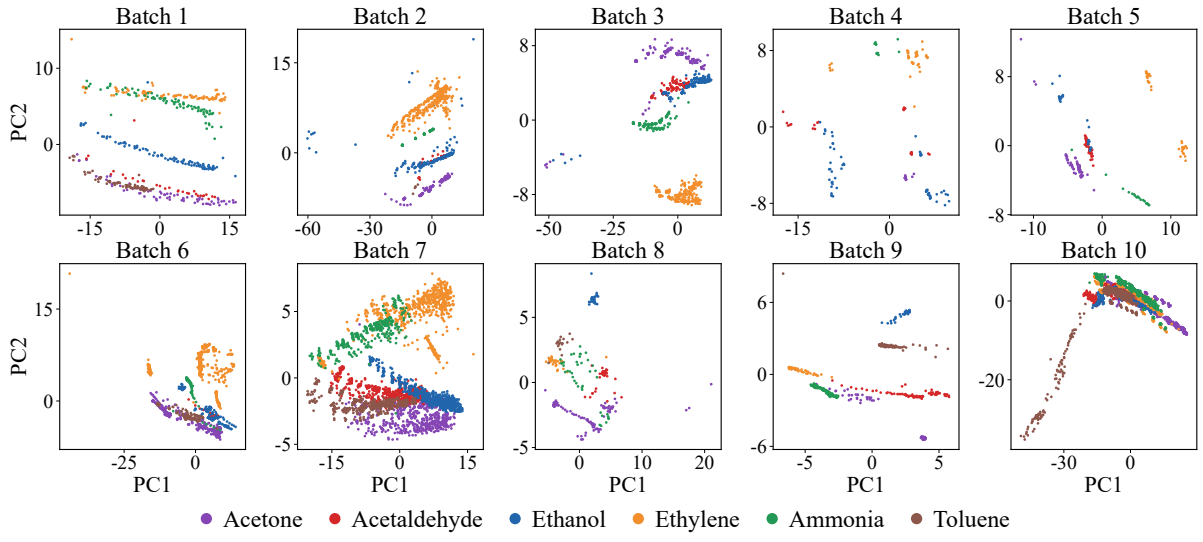


Figure 4.4 Distribution of principal components of the UCI dataset (individual batches).

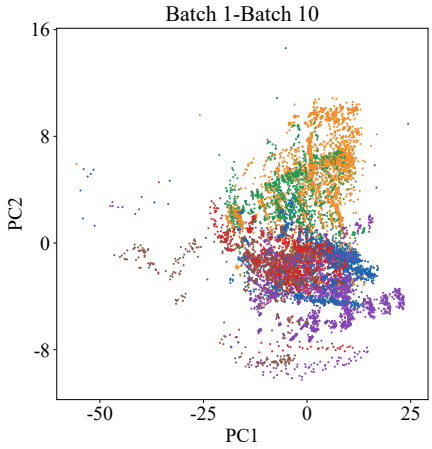


Figure 4.5 Distribution of principal components of the UCI dataset (all batches). Colors follow Figure 4.4.

4.2.3 Settings B and C: Long- and Medium-Term Adaptation

Two domain adaptation settings are defined using the ten batches of the UCI dataset. Setting B uses a fixed source domain, while Setting C uses adjacent batches as the source and target domains. Table 4.7 summarizes the source–target assignments for both settings.

Table 4.7 Source–target assignments for the UCI dataset.

Setting	Labeled source	Unlabeled target
Setting B: fixed-source	B_1	B_k, $k = 2, \ldots, 10$
Setting C: adjacent batches	B_k	B_{k+1}, $k = 1, \ldots, 9$

In Setting B, B_1 remains the source domain, while B_2–B_{10} serve as separate target domains, forming nine adaptation tasks. In Setting C, each batch serves as the source domain for the following batch, forming nine tasks from $B_1 \rightarrow B_2$ to $B_9 \rightarrow B_{10}$. Setting C assumes that ground-truth labels are available for each source batch. The classifier retains six output classes even when some source batches do not contain samples from all gas classes.

In both settings, each model was initialized from scratch and trained independently for its corresponding task, without reusing parameters from previous tasks. Unlabeled target samples were used during adaptation, while target labels were reserved for evaluation. Therefore, both follow a transductive unsupervised domain adaptation setting.

Batch sizes and class distributions also vary (Table 4.5). Therefore, performance differences between Settings B and C reflect not only the temporal separation between source and target batches but also changes in the source reference.

4.3 Data Preprocessing

The two datasets contain features with different numerical ranges and scales. Therefore, all models used the same two-stage preprocessing procedure: the asinh transformation followed by feature-wise z-score standardization using source-domain statistics. All 54 features of the ST dataset and all 128 features of the UCI dataset were retained.

Inverse hyperbolic sine transformation. Let $x_{d,ij}$ denote the original numerical value of feature j for sample i in domain d, where $d \in \{s, t\}$, with s and t denoting the source and target domains, respectively. Here, d is a local domain index, distinct from the latent dimension in Chapter 3. The transformation is applied to numerical feature values in the units specified above. The asinh-transformed feature $u_{d,ij}$ is defined as

$$u_{d,ij} = \operatorname{asinh}(x_{d,ij}) = \ln\left(x_{d,ij} + \sqrt{x_{d,ij}^2 + 1}\right). \tag{4.2}$$

This transformation is approximately linear near zero and compresses values with large absolute magnitudes. Unlike the ordinary logarithm, asinh can handle zero and negative values without requiring an additional positive offset. It is therefore suitable for sensor features with different numerical ranges and both positive and negative values.

Source-based z-score standardization. To further reduce scale differences between features, feature-wise z-score standardization was applied after the asinh transformation. The mean and standard deviation were calculated using only source-domain samples. Let n_s denote the number of source samples and m the input dimension, as in Chapter 3, where $m = 54$ for ST and $m = 128$ for UCI. For feature j, with $j = 1, \ldots, m$, the source-domain mean $\mu_{s,j}$ and standard deviation $\sigma_{s,j}$ are defined as

$$\begin{aligned} \mu_{s,j} &= \frac{1}{n_s}\sum_{i=1}^{n_s} u_{s,ij}, \\ \sigma_{s,j} &= \sqrt{\frac{1}{n_s}\sum_{i=1}^{n_s}(u_{s,ij} - \mu_{s,j})^2}. \end{aligned} \tag{4.3}$$

The population standard deviation was used, with n_s as the denominator in the variance calculation. For any constant source feature, its standard deviation was replaced with 1 to avoid division by zero. The standardized feature $\tilde{x}_{d,ij}$ was then calculated for both domains using the same source statistics:

$$\widetilde{x}_{d,ij} = \frac{u_{d,ij} - \mu_{s,j}}{\sigma_{s,j}}, \qquad d \in \{s, t\}, \quad j = 1, \ldots, m. \tag{4.4}$$

This procedure uses the source statistics for both domains rather than separately centering and scaling the target domain. It therefore does not force the target features to have zero mean and unit variance.

Similar two-stage preprocessing procedures have also been used in other measurement domains. For example, Hunter et al. [31] applied an asinh transformation with a cofactor of one followed by z-score standardization in their imaging mass cytometry study to reduce batch-related differences. The present study used the same transformation sequence.

In Setting A, the 30 samples in ST Batch 1 (B_1) were used to calculate the standardization statistics, which were then applied to B_1–B_6. In Setting B, UCI Batch 1 (B_1) provided the statistics for all long-term adaptation tasks. In Setting C, the statistics were calculated separately from the source batch B_k for each task and applied to both B_k and its corresponding target batch B_{k+1}, where $k = 1, \ldots, 9$. Target samples were never used to fit the standardization parameters.

The same preprocessing procedure was applied to DANN, DANNLC, CDAN, and CDANLC to ensure that all models used identical input features within each task.

5 Short-Term Drift Compensation

This chapter investigates the effect of introducing LC into DANN and CDAN for short-term gas sensor drift compensation. The experiments were conducted on the ST dataset under Setting A (Subsection 4.1.3), which comprises three experimental protocols. The central question is whether this additional class-level alignment constraint can further improve gas classification performance across short-term acquisition batches.

This chapter addresses RQ1 through the LC comparison, RQ2 through target-data availability and loss-weight analyses, and RQ3 through the short-term MSPA evaluation. The following KEQs organize these experiments:

KEQ 5.1: Does LC improve classification accuracy under short-term drift? (Section 5.2)

KEQ 5.2: How sensitive is model performance to the availability and organization of unlabeled target data? (Section 5.3)

KEQ 5.3: How does varying the LC and domain-adversarial loss weights around their selected values affect classification performance? (Section 5.4)

KEQ 5.4: To what extent does MSPA improve classification under short-term drift? (Section 5.5)

5.1 Evaluation Setups

5.1.1 Implementation Details

This chapter compares DANN, DANNLC, CDAN, and CDANLC under a common experimental configuration. All four models used the same feature extractor and classifier architecture. DANNLC and CDANLC incorporate the latent class-center alignment (LC) loss defined in Chapter 3 into their respective baselines without introducing additional trainable modules.

The feature extractor used batch normalization (BN) and a rectified linear unit (ReLU) activation. Table 5.1 summarizes the network architectures of the four models. The domain discriminator in DANN received a 200-dimensional feature vector, while the discriminator in CDAN received a 600-dimensional conditional representation constructed from the feature vector and predicted class probabilities.

Table 5.2 summarizes the training parameters used for all four models. During training, the source domain provided labeled samples, while the target domain provided unlabeled samples.

Let t denote the optimization step, α_t the gradient reversal layer (GRL) coefficient at step t, and p_t the normalized progress of the coefficient schedule, capped at 1. In these experiments, α_t increased during the first 150 optimization steps and remained constant thereafter. It is defined as

$$\alpha_t = \frac{2}{1+\exp(-3p_t)} - 1, \qquad p_t = \frac{\min(t-1, 149)}{149}, \qquad t = 1, \ldots, 300. \tag{5.1}$$

All models were evaluated using the final checkpoint after 300 optimization steps.

Table 5.1 Network architectures for the ST experiments.

Network component	Architecture
Feature extractor	54 – 200 – BN – ReLU
Classifier	200 – 3
DANN / DANNLC domain discriminator	200 – 32 – ReLU – 32 – ReLU – 1
CDAN / CDANLC domain discriminator	600 – 32 – ReLU – 32 – ReLU – 1

Numbers indicate layer dimensions.

Table 5.2 Training settings for the ST experiments.

Training setting	Value
Optimizer	Adaptive moment estimation (Adam)
Learning rate	0.005 (constant)
Weight decay	10^{-4}
Domain-adversarial loss weight λ_d	0.9
LC loss weight λ_{LC}	1.2 for DANNLC and CDANLC; 0 otherwise
Number of optimization steps	300
Mini-batch size (source / target)	20 / 20
Gradient reversal coefficient	Sigmoidal increase over steps 1–150; constant thereafter
Upper bound on the spectral norm of discriminator weight matrices	0.75
Model used for evaluation	Final model after 300 optimization steps

5.1.2 Evaluation Metrics and Procedures

Classification accuracy was used to evaluate gas recognition performance. Under the three protocols of Setting A, accuracy was calculated separately for the five target batches (B_2–B_6). The results were then summarized using equal-weighted mean accuracy and sample-weighted mean accuracy. For a given random seed and target-batch index $k \in \{2, \ldots, 6\}$, let a_k denote the classification accuracy on batch B_k and n_k the number of samples in that batch. Let A_{equal} and A_{weighted} denote the equal-weighted and sample-weighted mean accuracies, respectively. They are defined as

$$\begin{aligned} A_{\text{equal}} &= \frac{1}{5} \sum_{k=2}^{6} a_k, \\ A_{\text{weighted}} &= \frac{\sum_{k=2}^{6} n_k a_k}{\sum_{k=2}^{6} n_k}. \end{aligned} \tag{5.2}$$

The equal-weighted mean was used as the primary evaluation metric. It assigns the same weight to each target batch, preventing larger batches from dominating the overall results. The sample-weighted mean is reported as a supplementary metric to reflect overall classification performance across all target samples.

For each model and protocol, 20 independent training runs were performed using random seeds 0–19. Batch-wise accuracy and the corresponding mean accuracy across batches were first calculated separately for each seed. Means and sample standard deviations were calculated across the 20 seeds. Tables 5.3–5.5 report accuracy as mean ± sample standard deviation. The model metrics and paired gains in Table 5.7 use the same format. Both gain means and standard deviations are expressed in percentage points.

To evaluate the additional contribution of LC, DANNLC was compared with DANN, and CDANLC with CDAN, using the same random seeds and protocols. Paired differences in equal-weighted and sample-weighted mean accuracy were calculated, and accuracy gains are expressed in percentage points (pp). The paired gains are reported as the mean and sample standard deviation of the 20 differences between runs with matching seeds. Variation across seeds reflects training randomness under fixed data and training configurations.

5.2 Classification Performance and Feature Representations

5.2.1 Overall Classification Accuracy

Tables 5.3–5.5 present the classification results under the three protocols of Setting A. Bold values identify the highest equal-weighted mean accuracy, the primary metric, within each table; boldface does not indicate statistical significance.

Table 5.6 summarizes the paired LC gains under both averaging methods. In Tables 5.3–5.5, sample-weighted mean accuracy was lower than equal-weighted mean accuracy in all 12 model–protocol combinations. This reflects the greater influence of larger and relatively challenging target batches, particularly B_4, in the sample-weighted calculation.

The paired results in Table 5.6 demonstrate positive mean accuracy gains from LC under all three protocols and both averaging methods. The gains ranged from 1.23 to 3.79 percentage points for equal-weighted mean accuracy and from 1.30 to 4.25 percentage points for sample-weighted mean accuracy.

Table 5.3 Classification accuracy (%) on the ST dataset under Protocol 1.

Target batch	DANN	DANNLC	CDAN	CDANLC
B_2	93.04 ± 2.95	95.36 ± 2.04	96.07 ± 1.10	95.54 ± 1.59
B_3	89.23 ± 2.06	92.02 ± 0.78	89.76 ± 1.12	91.07 ± 1.66
B_4	77.35 ± 2.13	83.95 ± 2.63	81.11 ± 2.60	85.19 ± 2.60
B_5	85.98 ± 7.00	92.61 ± 1.48	93.48 ± 0.71	93.48 ± 0.00
B_6	90.59 ± 1.54	91.18 ± 0.00	89.26 ± 1.44	90.59 ± 1.21
Equal-weighted mean	87.23 ± 2.24	91.02 ± 0.82	89.94 ± 0.67	$\mathbf{91.17 \pm 0.87}$
Sample-weighted mean	85.71 ± 1.93	89.96 ± 0.93	88.41 ± 0.92	90.13 ± 1.18

Table 5.4 Classification accuracy (%) on the ST dataset under Protocol 2.

Target batch	DANN	DANNLC	CDAN	CDANLC
B_2	96.61 ± 1.82	94.46 ± 2.16	87.68 ± 2.16	93.57 ± 2.74
B_3	89.40 ± 1.54	92.62 ± 1.53	89.76 ± 1.41	90.71 ± 1.26
B_4	78.77 ± 2.22	82.53 ± 1.76	85.74 ± 2.02	86.67 ± 3.81
B_5	91.85 ± 1.56	93.26 ± 0.67	92.83 ± 1.02	93.26 ± 0.67
B_6	90.59 ± 1.21	91.18 ± 0.00	88.82 ± 1.54	89.26 ± 1.44
Equal-weighted mean	89.44 ± 0.89	$\mathbf{90.81 \pm 0.75}$	88.97 ± 0.88	90.70 ± 1.11
Sample-weighted mean	87.55 ± 0.98	89.74 ± 0.82	88.75 ± 0.97	90.05 ± 1.28

Table 5.5 Classification accuracy (%) on the ST dataset under Protocol 3.

Target batch	DANN	DANNLC	CDAN	CDANLC
B_2	96.61 ± 1.82	94.46 ± 2.16	87.68 ± 2.16	93.57 ± 2.74
B_3	87.74 ± 2.81	91.67 ± 1.09	89.17 ± 1.85	89.76 ± 1.98
B_4	77.78 ± 2.08	84.51 ± 2.17	83.09 ± 3.23	86.17 ± 1.36
B_5	87.61 ± 4.68	91.52 ± 2.43	93.59 ± 0.49	93.48 ± 0.00
B_6	90.59 ± 1.54	91.18 ± 0.00	89.26 ± 1.44	90.59 ± 1.21
Equal-weighted mean	88.06 ± 1.26	90.67 ± 0.89	88.56 ± 0.89	$\mathbf{90.71 \pm 0.82}$
Sample-weighted mean	86.03 ± 1.36	89.74 ± 0.93	87.97 ± 1.24	89.82 ± 0.89

Table 5.6 Paired LC gains across the three protocols.

Protocol	Weighting	DANNLC gain (pp)	CDANLC gain (pp)
1	Equal	$+3.79 \pm 2.19$	$+1.23 \pm 0.91$
1	Sample	$+4.25 \pm 1.91$	$+1.72 \pm 1.28$
2	Equal	$+1.37 \pm 1.07$	$+1.73 \pm 1.14$
2	Sample	$+2.20 \pm 1.16$	$+1.30 \pm 1.16$
3	Equal	$+2.60 \pm 1.40$	$+2.16 \pm 0.95$
3	Sample	$+3.72 \pm 1.34$	$+1.85 \pm 1.23$

Gains are computed as DANNLC minus DANN and CDANLC minus CDAN, respectively.

The largest gains were achieved by DANNLC under Protocol 1, with improvements of 3.79 and 4.25 percentage points, respectively.

5.2.2 Class-Balanced Metrics and Class-Specific Performance

For each gas class, precision measures the proportion of predictions assigned to that class that are correct, whereas recall measures the proportion of samples in that class that are correctly classified. The F1-score is the harmonic mean of class precision and recall. Precision, recall, and F1-score are expressed as percentages; gains are expressed in percentage points.

To further evaluate the effect of LC on the recognition of different gas classes, precision, recall, and F1-score were used as supplementary evaluation metrics. Since LC aims to align the feature centers of corresponding classes between the source and target domains, macro-averaging was adopted to assign equal importance to the three gas classes: air, alcohol, and acetone. The results were also averaged equally across the five target batches (B_2–B_6), preventing larger batches from dominating the evaluation.

Table 5.7 presents the results for the three protocols of Setting A. In each protocol, DANNLC and CDANLC achieved higher precision, recall, and F1-scores than their respective baselines. This indicates that the benefits of LC extend beyond overall classification accuracy to improvements in class-balanced classification performance.

Table 5.7 Macro-averaged classification metrics (%) under the three protocols.

Protocol	Metric	DANN	DANNLC	Gain (pp)	CDAN	CDANLC	Gain (pp)
1	Precision	88.59±2.08	92.13±0.73	+3.54±1.99	90.91±0.61	91.99±0.85	+1.08±0.83
1	Recall	87.18±2.13	90.89±0.85	+3.71±2.06	89.81±0.67	91.05±0.86	+1.24±0.91
1	F1-score	86.99±2.25	90.93±0.84	+3.94±2.18	89.64±0.74	91.01±0.89	+1.38±0.95
2	Precision	90.25±0.90	91.73±0.77	+1.48±1.03	89.97±0.79	91.20±1.18	+1.23±1.17
2	Recall	89.29±0.90	90.69±0.74	+1.40±1.11	89.02±0.82	90.64±1.16	+1.62±1.16
2	F1-score	89.27±0.92	90.74±0.76	+1.47±1.10	88.85±0.89	90.53±1.14	+1.68±1.17
3	Precision	89.24±1.29	91.83±0.88	+2.60±1.50	89.85±0.76	91.48±0.82	+1.63±0.89
3	Recall	88.00±1.16	90.60±0.90	+2.60±1.35	88.58±0.87	90.79±0.80	+2.21±0.94
3	F1-score	87.87±1.24	90.69±0.88	+2.82±1.39	88.40±0.94	90.61±0.83	+2.21±1.02

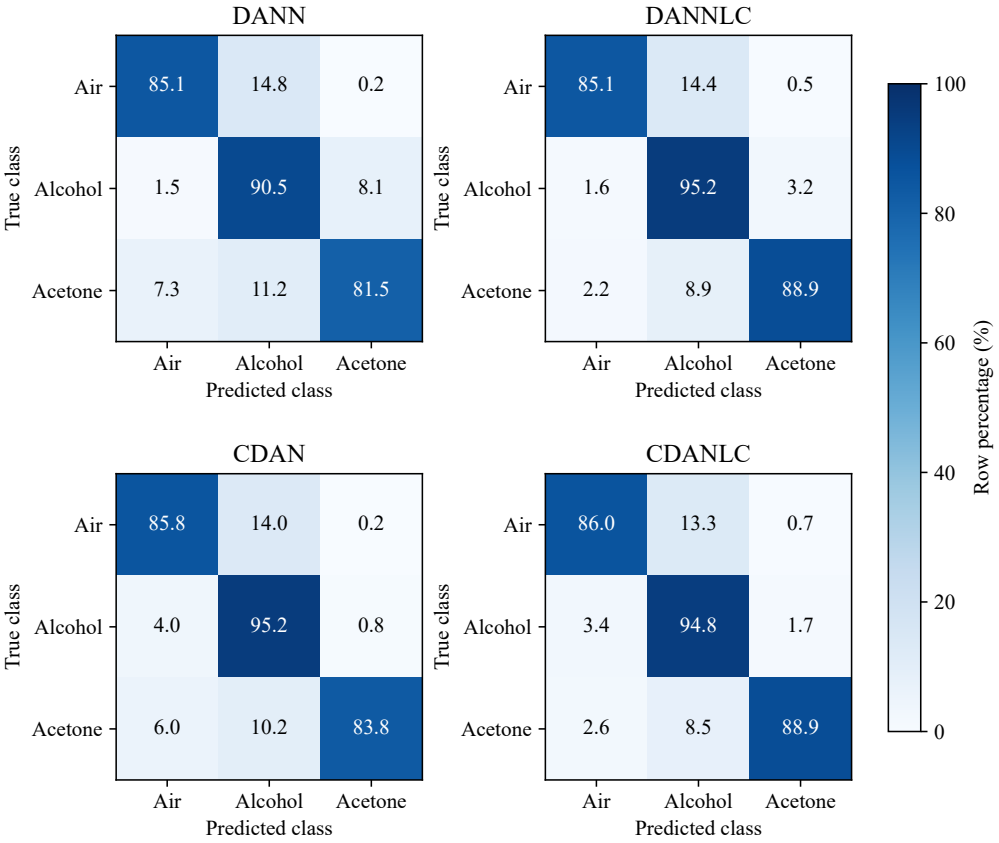


Figure 5.1 Confusion matrices of the four methods under Protocol 1.

Figure 5.1 presents the confusion matrices of the four models under Protocol 1. The matrices were constructed by aggregating predictions across five target batches (B_2–B_6) and 20 training runs. Each row was normalized by the total number of pooled predictions for samples belonging to the corresponding true class. The diagonal entries therefore represent the recall of each gas class, while the off-diagonal entries reveal misclassification patterns. Unlike the equal-weighted batch averages reported in Table 5.7, these matrices are based on pooled predictions, including repeated evaluations of the same target samples across training runs.

In Figure 5.1, both DANNLC and CDANLC reduced the proportion of acetone samples misclassified as air or alcohol. DANNLC also improved alcohol recall while leaving air recall unchanged. In contrast, CDANLC showed a slight decrease in alcohol recall. The effects of LC varied across gas classes.

5.2.3 Feature Visualization

To visualize the effect of LC on the learned feature representations, principal component analysis (PCA) was used to examine the feature distributions before and after domain adaptation. Figures 5.2–5.4 show the feature distributions of source batch B_1 and target batches B_2–B_6 under Protocol 1. Random seed 6 was used for these exploratory PCA visualizations. The adapted representations were obtained from models trained with this seed. For each representation, PCA was fitted to all 303 samples, and the source and target panels used the same fitted basis. PC1 and PC2 denote the first and second principal component scores. The percentages in parentheses indicate the proportions of variance explained by the corresponding components.

As shown in Figures 5.2–5.4, domain adaptation changed the distribution of features across gas classes. In the target features learned by DANN, alcohol and acetone samples overlapped in some regions. After LC was introduced, these two classes showed clearer separation in the two-dimensional projection, although some local overlap remained. Similarly, compared with CDAN, CDANLC showed a more distinct class distribution in the target domain.

These visualizations suggest that LC may improve the separability of different gas classes in the learned feature space. PCA was used for qualitative inspection of two-dimensional projections; the next subsection evaluates class structure quantitatively in the full feature space.

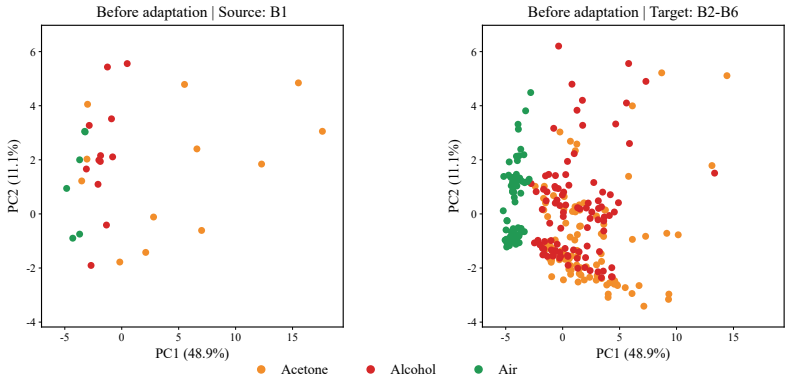


Figure 5.2 Distribution of principal components before adaptation.

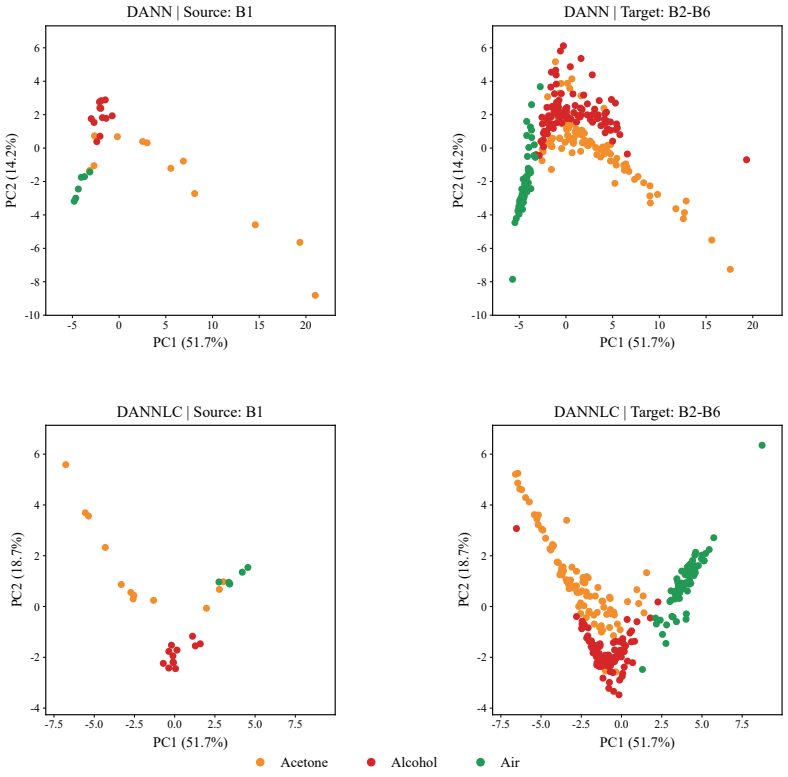


Figure 5.3 Distribution of principal components after DANN and DANNLC adaptation.

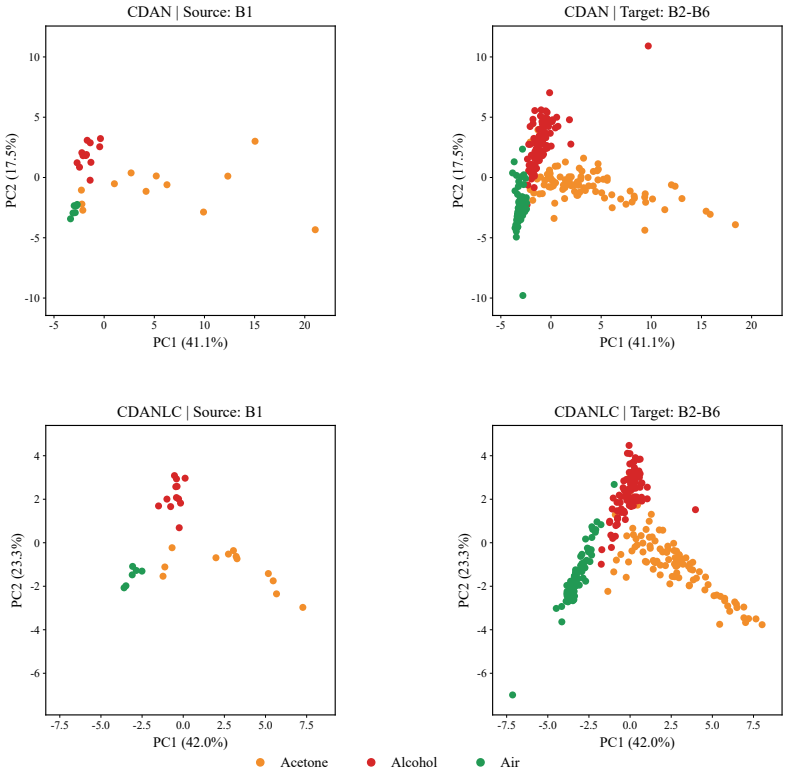


Figure 5.4 Distribution of principal components after CDAN and CDANLC adaptation.

5.2.4 Quantitative Analysis of Class Structure

To quantify the effect of LC on class structure in the feature space, three metrics were used: the silhouette coefficient (S), the Davies–Bouldin index (DB), and the Calinski–Harabasz index (CH). The groups were defined by the true gas labels rather than by a clustering algorithm. Let n denote the number of evaluated samples and K the number of gas classes. Sample indices are denoted by i and j, and class indices by c and r. Let $\mathbf{z}_i$ be the representation of sample i, c_i its class label, $\mathcal{I}_c$ the set of sample indices in class c, and $n_c = |\mathcal{I}_c|$ its sample count. The notation $\|\cdot\|_2$ denotes the Euclidean norm. Here, $n = 303$ and $K = 3$.

The silhouette coefficient compares each sample's proximity to its own class with its proximity to the nearest other class [32]. Let a_i denote the mean distance from sample i to the other samples in its class, and b_i the smallest mean distance from sample i to any other class. The sample coefficient s_i and its overall mean S are defined as

$$\begin{aligned} a_i &= \frac{1}{n_{c_i} - 1} \sum_{\substack{j \in \mathcal{I}_{c_i} \\ j \neq i}} \|\mathbf{z}_i - \mathbf{z}_j\|_2, \\ b_i &= \min_{r \neq c_i} \frac{1}{n_r} \sum_{j \in \mathcal{I}_r} \|\mathbf{z}_i - \mathbf{z}_j\|_2, \\ s_i &= \frac{b_i - a_i}{\max(a_i, b_i)}, \qquad S = \frac{1}{n} \sum_{i=1}^{n} s_i. \end{aligned} \tag{5.3}$$

A value of s_i close to 1 indicates that the sample is much closer, on average, to samples in its own class than to those in the nearest other class. A value near zero indicates similar mean distances to the two classes, and a negative value indicates greater mean proximity to another class. Thus, higher S indicates clearer class separation. All classes in this analysis contain more than one sample.

The Davies–Bouldin index evaluates class compactness relative to the separation of class centers [33]. Let $\boldsymbol{\mu}_c$ denote the mean representation of class c, δ_c the mean distance of samples in class c to $\boldsymbol{\mu}_c$, and d_{cr} the distance between the centers of classes c and r. Then

$$\begin{aligned} \boldsymbol{\mu}_c &= \frac{1}{n_c} \sum_{i \in \mathcal{I}_c} \mathbf{z}_i, \qquad \delta_c = \frac{1}{n_c} \sum_{i \in \mathcal{I}_c} \|\mathbf{z}_i - \boldsymbol{\mu}_c\|_2, \\ d_{cr} &= \|\boldsymbol{\mu}_c - \boldsymbol{\mu}_r\|_2, \\ \mathrm{DB} &= \frac{1}{K} \sum_{c=1}^{K} \max_{r \neq c} \frac{\delta_c + \delta_r}{d_{cr}}. \end{aligned} \tag{5.4}$$

For each class, the maximum selects the class with the largest ratio of combined dispersion to center distance. Averaging these ratios gives DB; lower values indicate more compact and better-separated classes.

The Calinski–Harabasz index compares between-class and within-class dispersion [34]. Let $\boldsymbol{\mu}$ denote the mean representation of all samples, B the between-class sum of squares, and W the within-class sum of squares. Using the class centers defined above, the index is

$$\begin{aligned} \boldsymbol{\mu} &= \frac{1}{n}\sum_{i=1}^{n}\mathbf{z}_i, \\ B &= \sum_{c=1}^{K} n_c\|\boldsymbol{\mu}_c-\boldsymbol{\mu}\|_2^2, \qquad W = \sum_{c=1}^{K}\sum_{i\in\mathcal{I}_c}\|\mathbf{z}_i-\boldsymbol{\mu}_c\|_2^2, \\ \mathrm{CH} &= \frac{B/(K-1)}{W/(n-K)}. \end{aligned} \tag{5.5}$$

The factors $K-1$ and $n-K$ normalize the two sums of squares by their respective degrees of freedom. Higher CH indicates greater separation of class centers relative to the dispersion within classes.

Under Protocol 1, all 303 samples from source batch B_1 and target batches B_2–B_6 were pooled for this analysis. The three metrics were calculated in the preprocessed 54-dimensional input space and the learned 200-dimensional feature space, without applying PCA. For this analysis, ground-truth target labels were used only for post-training class-structure evaluation. Results for the learned representations are reported as the mean and sample standard deviation over 20 random seeds.

Table 5.8 Class-structure metrics before and after adaptation. Arrows indicate whether higher (↑) or lower (↓) values are better.

Representation	**S ↑**	**DB ↓**	**CH ↑**
Preprocessed input features	0.143	2.369	60.12
DANN	0.213 ± 0.010	1.925 ± 0.095	81.47 ± 5.39
DANNLC	0.303 ± 0.015	1.336 ± 0.057	153.53 ± 11.25
CDAN	0.228 ± 0.013	1.513 ± 0.040	77.50 ± 4.89
CDANLC	0.323 ± 0.010	1.223 ± 0.033	139.09 ± 7.34

As shown in Table 5.8, the learned representations of all four models achieved higher S and CH values and lower DB values than the preprocessed input features. Compared with their respective baselines, DANNLC and CDANLC both achieved higher S and CH values and lower DB values, indicating improved class structure in the learned feature space after LC was introduced.

The three metrics did not rank the two LC models identically. CDANLC achieved the highest S and lowest DB, while DANNLC achieved the highest CH. The rankings reflect the different aspects of class structure measured by these metrics.

The full-space class-structure metrics were consistent with the PCA observations. They also complement the classification improvements reported in Sections 5.2.1 and 5.2.2.

5.3 Effect of Target Data Availability

5.3.1 Classification Performance Across Protocols

Electronic noses may be used under different data-access conditions. For example, adaptation can use observations pooled from several batches after data collection, observations from the current batch only, or observations from both the current and previously collected batches. To examine these conditions, this section compares three protocols on the self-collected ST dataset. The aim is to investigate whether access to more unlabeled target data improves classification performance and whether the benefits of LC depend on the availability of all target batches.

We compared the three target-data access protocols defined in Section 4.1.3: offline pooled, independent, and cumulative adaptation. Table 5.9 lists the number of unlabeled target samples available for each evaluation task.

Table 5.9 Number of unlabeled target samples available for adaptation under each protocol.

Evaluated batch	Protocol 1	Protocol 2	Protocol 3
B_2	273	28	28
B_3	273	84	112
B_4	273	81	193
B_5	273	46	239
B_6	273	34	273

Figure 5.5 compares the equal-weighted mean classification accuracy of the four models across the three protocols. DANN achieved its highest mean accuracy under Protocol 2, whereas CDAN performed best under Protocol 1. Thus, no single protocol provided the highest accuracy for all models. Both LC variants achieved their highest mean accuracy under Protocol 1. For each LC variant, the range across the three protocols was less than 0.5 percentage points and smaller than that of its corresponding baseline.

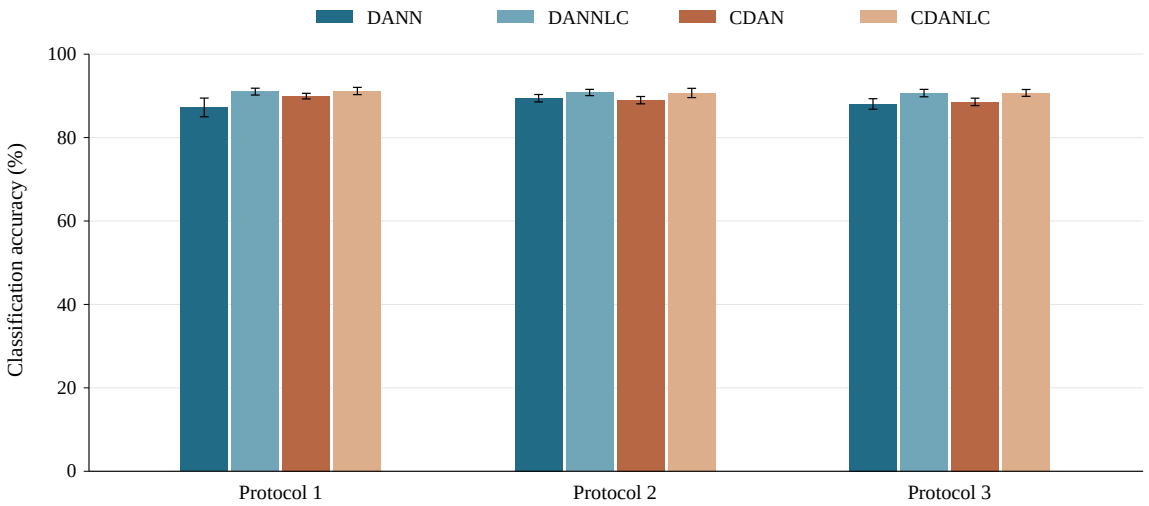


Figure 5.5 Classification accuracy of the four models across the three protocols. Bars show means over 20 seeds and error bars show one sample standard deviation. Accuracy is averaged equally across the five target batches within each seed.

As shown in Table 5.6, LC improved the mean accuracy of both baseline models under all three protocols. In particular, under Protocol 2, the two LC variants achieved mean accuracies close to those obtained with offline pooled adaptation, despite using no other target batches. These results indicate that the classification benefits of LC are not limited to conditions where all target data are available.

5.3.2 Discussion of Data Availability and Adaptation Performance

Target-data quantity and the way target domains are combined are separate factors in adaptation. Gholami et al. [35] investigated multi-target domain adaptation by considering shared and domain-specific information across target domains. Isobe et al. [36] further showed that directly pooling multiple target domains may perform worse than adapting to them separately, while collaborative consistency learning can provide additional benefits. These studies suggest that adaptation performance depends not only on the amount of available data but also on how information from different distributions is incorporated.

Protocols 1 and 3 combine the available target batches into a single target domain without applying separate alignment losses to individual batches. Enlarging the target data pool therefore increases the number of observations while also changing the distribution used for adaptation. In contrast, Protocol 2 uses fewer samples but focuses directly on the current evaluation batch. This suggests a possible trade-off between broader target-data coverage and more targeted batch-specific adaptation, which may help explain why no protocol consistently performed best across the tested models.

The fixed training budget is another factor to consider. All models used the same number of optimization steps and the same target mini-batch size per training run. Consequently, a larger target data pool does not automatically increase the total number of target samples processed during a run. Instead, it changes how sampling opportunities are distributed among the available observations. The results therefore reflect adaptation performance under a fixed training configuration, where both target-data quantity and composition vary.

5.4 Parameter Sensitivity Analysis

This section examines how the LC and domain-adversarial loss weights affect classification accuracy under Protocol 1.

Each condition used the same 300-update training procedure and random seeds 0–19. For each seed, accuracy was averaged equally across the five target batches, and the 20 seed-level values were summarized by their mean and sample standard deviation, with a denominator of $20 - 1 = 19$ in the sample variance. All other training settings were held fixed.

5.4.1 Effect of the LC Loss Weight

The domain-adversarial loss weight was fixed at $\lambda_d = 0.9$, while λ_{LC} was varied over 0, 0.1, 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, 2.1, 2.4, and 2.7. Setting $\lambda_{\mathrm{LC}} = 0$ recovers the corresponding baseline without LC. The extended range examines whether the improvements observed up to 1.2 continue at larger weights. Figure 5.6 presents the results.

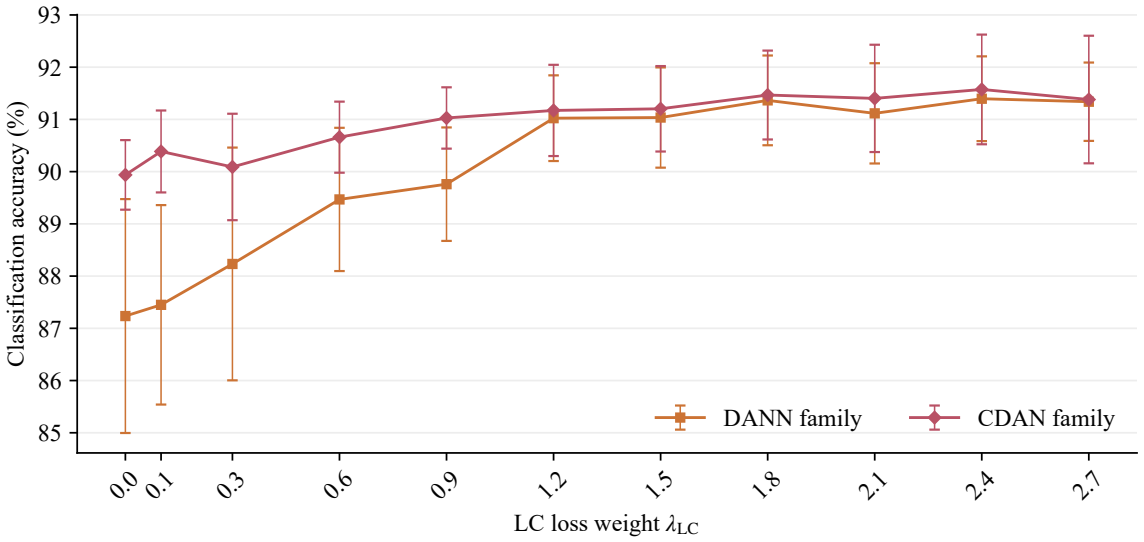


Figure 5.6 Classification accuracy under different LC loss weights at $\lambda_d = 0.9$. Points show the mean over 20 seeds and error bars show one sample standard deviation.

Both model families achieved higher mean accuracy with each tested positive LC weight than without LC. The DANN family improved from 87.23 % at zero LC weight to 91.02 % at 1.2, whereas the CDAN family improved from 89.94 % to 91.17 %. Increasing the weight further did not produce a monotonic improvement. Between 1.2 and 2.7, mean accuracy ranged from 91.02 % to 91.40 % for the DANN family and from 91.17 % to 91.57 % for the CDAN family, corresponding to ranges of 0.37 and 0.40 percentage points calculated from unrounded values.

As shown in Figure 5.6, mean classification accuracy tended to level off once λ_{LC} reached approximately 1.8, with only small upward and downward fluctuations as the weight increased further. Over the tested weights from 1.8 to 2.7, mean accuracy varied within 0.28 percentage points for the DANN family and 0.19 percentage points for the CDAN family. Thus, larger LC weights did not produce a consistent additional improvement in this range.

The main experiments retained $\lambda_{\mathrm{LC}} = 1.2$ and $\lambda_d = 0.9$ for consistency across protocols and analyses. The numerical loss weights alone also do not determine the relative gradient contributions of the two objectives.

5.4.2 Effect of the Domain-Adversarial Loss Weight

Domain-adversarial learning encourages transferable features by reducing the distinguishability between source and target domains, while LC further constrains the feature centers of corresponding classes. The following analysis tests whether LC reduces sensitivity to the domain-adversarial loss weight.

To investigate this question, the domain-adversarial loss weight λ_d was varied over 0, 0.1, 0.3, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0. The LC weight was fixed at $\lambda_{\mathrm{LC}} = 1.2$ for DANNLC and CDANLC, while the baseline models do not use LC. The zero-weight condition was included to examine the contribution of LC without an explicit domain-adversarial loss.

Figure 5.7 presents the results.

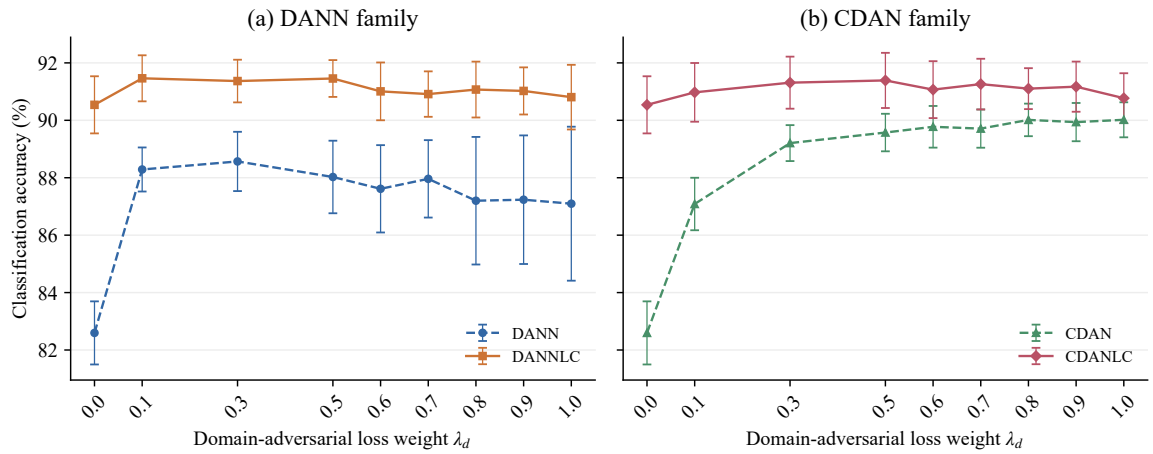


Figure 5.7 Mean classification accuracy under different domain-adversarial loss weights. Points show means over 20 seeds and error bars show one sample standard deviation. The LC variants use $\lambda_{\mathrm{LC}} = 1.2$.

Figure 5.7 shows that introducing LC changed how the models responded to the domain-adversarial loss weight. Across the tested nonzero domain-adversarial weights, the ranges of mean accuracy for DANNLC and CDANLC were 0.66 and 0.62 percentage points, respectively, compared with 1.47 and 2.93 percentage points for their corresponding baselines. These results indicate that, at $\lambda_{\mathrm{LC}} = 1.2$, LC improved classification performance and reduced the observed sensitivity of mean accuracy to the domain-adversarial loss weight under the tested configuration.

When $\lambda_d = 0$, the domain-adversarial loss no longer contributes to feature extractor optimization, and the differences between the DANN and CDAN domain discriminators no longer affect classification. Consequently, the two baseline models achieved identical accuracies, as did the two LC variants. LC improved classification accuracy even without the domain-adversarial loss. This suggests that class-center alignment can improve classification performance independently of explicit domain-adversarial training in the current experiments.

This observation is consistent with the distinct objectives of adversarial learning and class-level alignment. DANN and CDAN encourage domain alignment through domain discrimination, whereas LC introduces an additional constraint based on class correspondence [9], [11], [12]. LC may therefore help preserve class-discriminative information across different domain-adversarial strengths. Nevertheless, the LC variants achieved higher mean accuracies at nonzero domain-adversarial weights, indicating that adversarial learning still provided an additional benefit under the tested configuration.

5.5 Multi-Seed Probability Averaging

This section evaluates whether MSPA improves the short-term classification results reported above. It addresses KEQ 5.4 using the MSPA rule defined in Equation 3.15.

5.5.1 Evaluation Setups

We evaluated DANN, DANNLC, CDAN, and CDANLC separately under Protocol 1 of Setting A. Each evaluation used the final 300-update models trained with seeds 0–19 and the settings in Table 5.2. Target labels were used only to calculate the evaluation metrics after training.

The main comparison used seven members ($M = 7$) per model type. For group index $g \in \{0, \ldots, 19\}$ and seed offset δ, the member seeds were $(g+\delta) \bmod 20$, where $\bmod$ denotes the modulo operation and

$$\delta \in \{0, 7, 13, 3, 17, 5, 11\}.$$

These are the same 20 seven-member groups used in Chapters 6 and 7. Each seed appears in seven groups. No member was selected according to its target accuracy.

For each group, accuracy was calculated separately for B_2–B_6 and then averaged equally across the five batches. The mean and sample standard deviation were calculated across the 20 groups. Their standard deviation describes variation among overlapping compositions of the fixed seed pool.

5.5.2 Overlap of Prediction Errors

For each method under Protocol 1, target samples were classified into three categories using all 20 single-seed models: all models predicted correctly; some predicted correctly and others incorrectly; or all models predicted incorrectly. The proportions were calculated within each target batch and then averaged equally across the five batches. This analysis used 20 members at once, whereas the overlap analyses in Chapters 6 and 7 used seven-member groups.

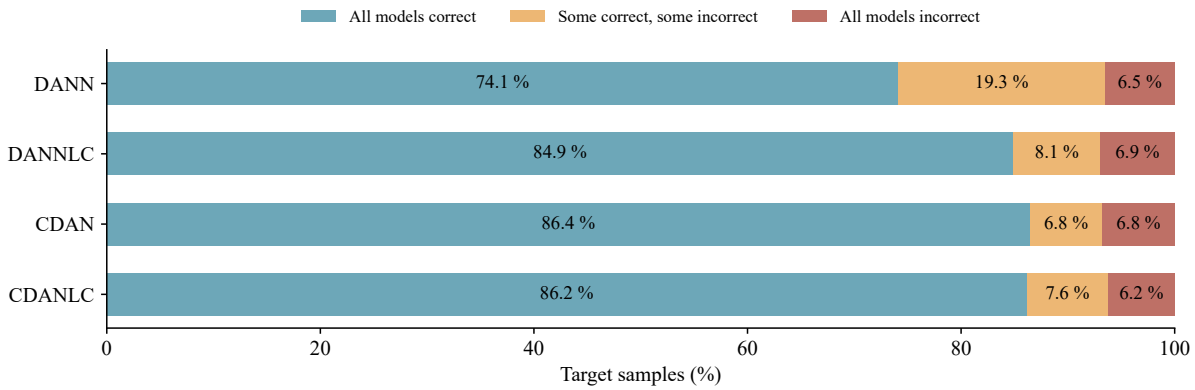


Figure 5.8 Agreement of 20 single-seed models on target predictions under Protocol 1 of Setting A.

Note: Each row uses one model type and seeds 0–19. Proportions were calculated before MSPA and averaged equally across B_2–B_6.

Figure 5.8 shows that most samples were classified correctly by all 20 models. Under Protocol 1, this proportion ranged from 74.13 % for DANN to 86.44 % for CDAN. Mixed correct and incorrect predictions occurred for 19.35 % of samples with DANN and 6.81 %–8.14 % with the other methods. The all-incorrect proportions ranged from 6.23 % to 6.95 %.

These results identify samples with correct predictions available from some seeds, but the effect of averaging also depends on the full probability vectors.

5.5.3 Classification Results

Table 5.10 reports the seven-member MSPA results under Protocol 1. Relative to the single-model means in Table 5.3, MSPA improved DANN, DANNLC, and CDANLC by 0.89, 0.84, and 0.40 percentage points, respectively; CDAN decreased by 0.02 points.

Table 5.10 Seven-member MSPA classification accuracy (%) under Protocol 1 of Setting A.

Method	B_2	B_3	B_4	B_5	B_6	Average	Gain (pp)
DANN	93.93	89.88	77.04	88.59	91.18	88.12 ± 0.73	$+0.89$
DANNLC	96.43	92.02	86.30	93.37	91.18	91.86 ± 0.28	$+0.84$
CDAN	96.43	90.24	80.62	93.48	88.82	89.92 ± 0.54	-0.02
CDANLC	96.43	90.71	86.05	93.48	91.18	91.57 ± 0.25	$+0.40$

Note: Batch entries are means across 20 overlapping seven-member groups. Average reports the mean $\pm$ sample standard deviation of each group's equal-weighted accuracy across B_2–B_6. Gain is relative to the corresponding unrounded single-model mean and is expressed in percentage points (pp).

5.5.4 Sensitivity Analysis

Number of ensemble members. We also evaluated $M \in \{1, 3, 5, 7, 10, 12, 15, 20\}$ using the nested seed groups specified in Section 6.4.4. At $M = 20$, all seeds form one unique ensemble, so no between-group standard deviation is reported. Figure 5.9 shows the resulting accuracies. Increasing the number of members did not consistently improve performance. For example, under Protocol 1, DANN achieved 88.12 % with seven members and 87.60 % with twenty, while DANNLC achieved 91.86 % and 91.83 %, respectively.

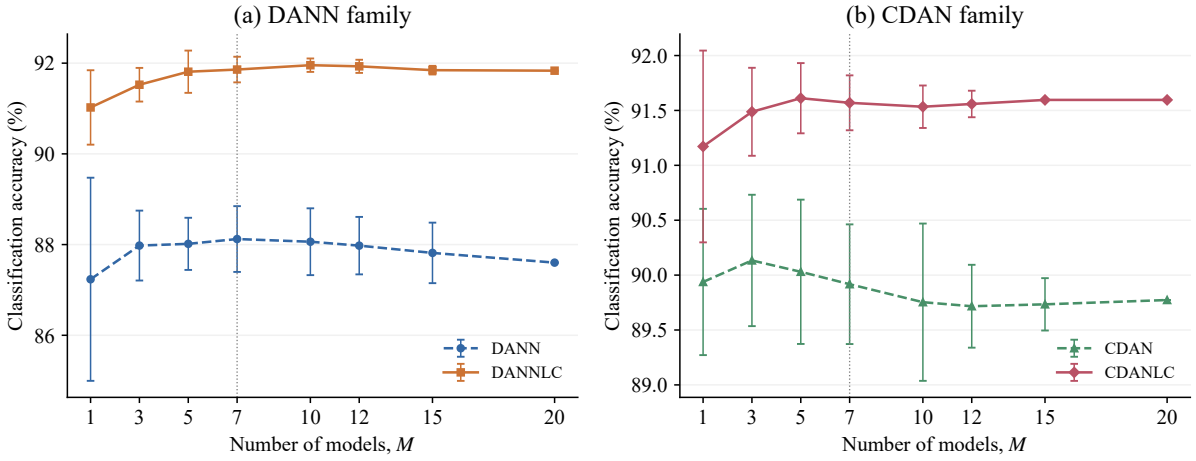


Figure 5.9 Effect of ensemble size on accuracy under Protocol 1 of Setting A.

Loss weights with seven-member MSPA. The second sensitivity analysis fixed $M = 7$ under Protocol 1 and varied one loss weight at a time. The LC sweep used weights 0, 0.1, 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, 2.1, 2.4, and 2.7, with $\lambda_d = 0.9$. The domain-loss sweep used weights 0, 0.1, 0.3, 0.5, 0.6, 0.7, 0.8, 0.9, and 1.0, with $\lambda_{\mathrm{LC}} = 1.2$ for the LC variants and zero for DANN and CDAN. These were the same grids as in Section 5.4. Each setting used members trained for 300 updates with seeds 0–19 and the same 20 overlapping seven-member groups as the main experiment. Within each group, probabilities were averaged before classification, and accuracy was then averaged equally across B_2–B_6. The reported means and sample standard deviations were calculated across groups. All other training and evaluation settings remained unchanged.

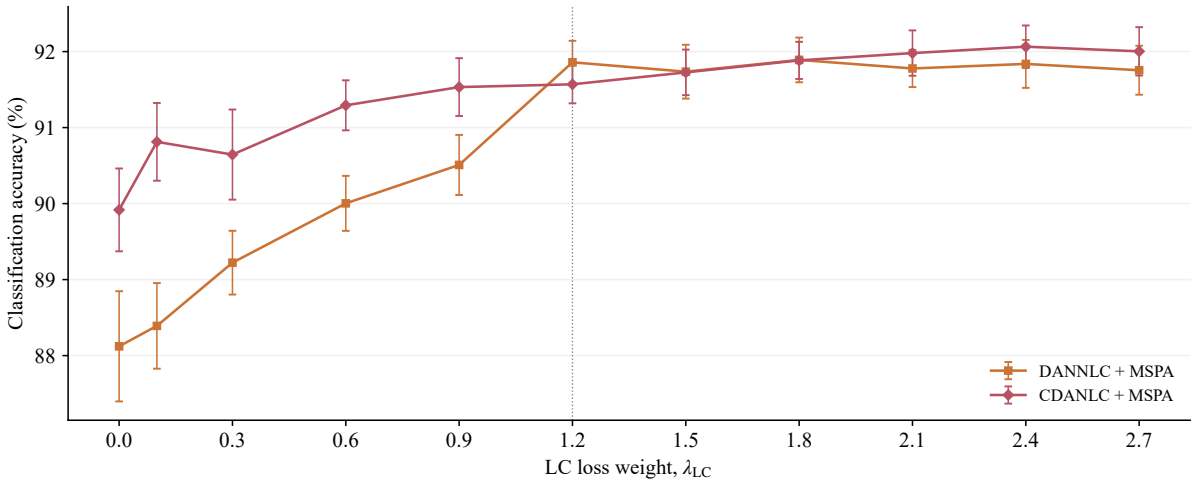


Figure 5.10 LC-weight sensitivity of seven-member MSPA under Protocol 1 of Setting A, with $\lambda_d = 0.9$. Error bars show one sample standard deviation across the 20 groups. The dotted line marks $\lambda_{\mathrm{LC}} = 1.2$.

Figure 5.10 shows that every tested positive LC weight produced a higher ensemble mean than the corresponding zero-LC baseline. DANNLC + MSPA increased from 88.12 % at zero LC weight to 91.86 % at 1.2, then remained between 91.74 % and 91.89 % over 1.5–2.7. CDANLC + MSPA increased from 89.92 % at zero LC weight to 91.57 % at 1.2 and reached its highest tested mean of 92.06 % at 2.4. Its mean decreased slightly to 92.00 % at 2.7. Thus, stronger LC beyond the retained weight brought little further improvement for the DANN ensemble and a smaller additional gain for the CDAN ensemble.

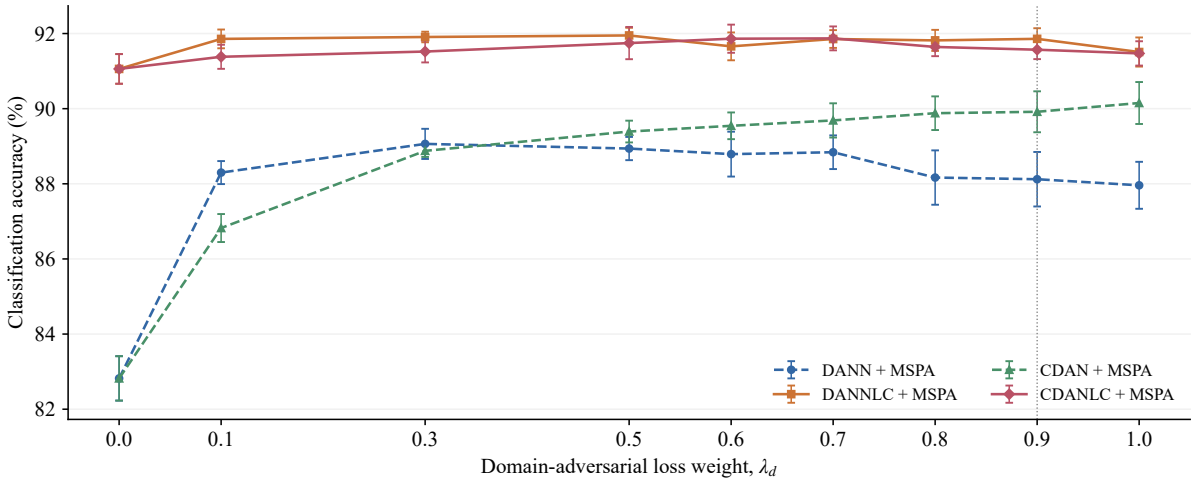


Figure 5.11 Domain-loss sensitivity of seven-member MSPA under Protocol 1 of Setting A. Solid curves show LC variants with $\lambda_{\mathrm{LC}} = 1.2$; dashed curves show DANN and CDAN without LC. Error bars show one sample standard deviation across the 20 groups.

Figure 5.11 shows different responses to the domain-loss weight. DANN + MSPA reached its highest tested mean of 89.06 % at $\lambda_d = 0.3$ and declined to 87.96 % at 1.0, whereas CDAN + MSPA increased from 82.82 % at zero domain weight to 90.15 % at 1.0. Both LC ensembles exceeded their corresponding baselines at every tested weight. Across the nonzero domain weights, DANNLC + MSPA ranged from 91.51 % to 91.95 %, and CDANLC + MSPA ranged from 91.38 % to 91.87 %. At $\lambda_d = 0$, target samples still affect shared BN and, for LC variants, the LC loss; this setting is therefore not a strictly source-only baseline.

5.5.5 Analysis of Short-Term MSPA Performance

MSPA provided small gains for DANN, DANNLC, and CDANLC, while CDAN remained almost unchanged. Increasing the ensemble size did not consistently yield further gains. The loss-weight experiments also showed that the ensemble results continued to depend on how the members were trained (Figures 5.10 and 5.11).

A possible explanation for the limited averaging gains is the high proportion of samples already classified correctly across all 20 seeds. In Figure 5.8, this proportion was 84.91 %, 86.44 %, and 86.16 % for DANNLC, CDAN, and CDANLC, respectively, while their mixed proportions were only 6.81 %–8.14 %. For samples already classified correctly by every member, probability averaging leaves the predicted class unchanged. The relatively small mixed subset may therefore limit the opportunities for correct predictions from some seeds to compensate for errors from others.

DANN had the largest mixed subset, but its accuracy did not improve steadily as more members were added (Figure 5.9). This result suggests that the availability of some correct predictions alone does not determine the benefit of a larger ensemble. MSPA averages the full probability vectors, so the outcome also depends on how much probability each member assigns to the competing classes. Adding members may preserve an incorrect prediction if they assign high probability to the same wrong class. Under these short-term conditions, more seeds therefore offered limited additional benefit beyond the seven-member comparison.

5.6 Chapter Summary

For KEQ 5.1, LC improved the mean classification accuracy of both DANN and CDAN under all three short-term protocols. Macro-averaged precision, recall, and F1-score also improved, while the PCA visualizations and full-space class-structure metrics supported better class discrimination. These results connect the classification gains with changes in the learned representations, supporting explicit class-center alignment as a complement to both overall and conditional domain alignment.

For KEQ 5.2, LC remained beneficial under all three target-data access protocols. The independent protocol achieved LC results close to those of offline pooled adaptation, so access to all target batches in advance was not required for the observed gains. No protocol gave the highest accuracy for every model. The results suggest that the composition of the target data used for alignment matters alongside the number of available observations.

For KEQ 5.3, increasing the LC weight improved accuracy over part of the tested range before the gains levelled off. With the LC weight fixed at the main setting, both LC models maintained higher mean accuracies and smaller variations across the tested nonzero domain-adversarial weights than their respective baselines. Stronger domain alignment did not uniformly improve performance.

For KEQ 5.4, MSPA provided small additional gains for DANN, DANNLC, and CDANLC, while CDAN remained almost unchanged. Most samples were already classified correctly across seeds, and larger ensembles did not consistently improve accuracy. The main short-term benefit came from the LC training objective, with probability averaging providing a smaller additional change.

The short-term experiments support LC across different target-data access conditions, while showing limited scope for further improvement through averaging. The next chapter evaluates LC and MSPA under long-term drift and examines whether unreliable target predictions help explain changes in their effectiveness.

6 Long-Term Drift Compensation

This chapter investigates LC under long-term drift using Setting B (Subsection 4.2.3) of the UCI gas sensor drift dataset. Compared with the short-term results in Chapter 5, the classification gains from LC were smaller and less consistent. To investigate this limitation, this chapter analyzes target prediction errors and confidence, then evaluates MSPA, defined in Section 3.6, as a way to improve final classification.

This chapter addresses RQ1 through the long-term LC comparison, RQ2 through loss-weight sensitivity and pseudo-label reliability, and RQ3 through prediction averaging. The following KEQs organize these experiments:

KEQ 6.1: Does LC improve classification accuracy under long-term drift? (Section 6.2)

KEQ 6.2: How reliable are target pseudo-labels during training, and does entropy conditioning improve classification performance? (Section 6.3)

KEQ 6.3: To what extent does MSPA reduce target prediction errors? (Section 6.4)

6.1 Evaluation Setups

6.1.1 Implementation Details

The initial experiments compared DANN, DANNLC, CDAN, and CDANLC under Setting B (Subsection 4.2.3). Batch B_1 is the labeled source domain, and each of B_2–B_{10} serves as a separate unlabeled target domain. Each model was trained from scratch for each source–target pair. All tasks used the preprocessing procedure described in Chapter 4, with transformation and standardization based on the source data.

All four models shared the same feature extractor and classifier architecture. The feature extractor mapped the 128 input features to a 200-dimensional representation using a linear layer, batch normalization (BN), and a rectified linear unit (ReLU) activation. The classifier had six outputs corresponding to the six gas classes. DANN used the 200-dimensional feature vector for domain discrimination, whereas CDAN used a 1200-dimensional conditional representation formed from the features and class probabilities. DANNLC and CDANLC add the LC loss to their respective baselines without introducing additional trainable modules. Table 6.1 summarizes the architectures used in these experiments.

Table 6.1 Network architectures for the UCI Setting B experiments.

Network component	Architecture
Feature extractor	128 – 200 – BN – ReLU
Classifier	200 – 6
DANN / DANNLC domain discriminator	200 – 64 – ReLU – 64 – ReLU – 1
CDAN / CDANLC domain discriminator	1200 – 64 – ReLU – 64 – ReLU – 1

Numbers indicate layer dimensions.

Table 6.2 summarizes the training settings. Training comprised 45 epochs with 10 optimization steps per epoch, giving 450 updates in total. Here, an epoch denotes a fixed block of 10 updates rather than a complete pass through the dataset. Each update used 16 source and 16 target samples. Source and target samples were concatenated for the shared feature-extraction step.

Table 6.2 Training settings for the UCI Setting B experiments.

Training setting	Value
Optimizer	Stochastic gradient descent (SGD); momentum 0.9
Initial learning rate	0.03
Learning-rate schedule	Cosine annealing; minimum 10^{-6}
Weight decay	0
Domain-adversarial loss weight λ_d	0.8
LC loss weight λ_{LC}	0.2 for DANNLC and CDANLC; 0 otherwise
Training epochs / steps per epoch	45 / 10 (450 updates in total)
Mini-batch size (source / target)	16 / 16
Gradient reversal coefficient	Sigmoidal increase over all 450 updates
Upper bound on the spectral norm of discriminator weight matrices	1.0
Model used for evaluation	Final model after 450 optimization steps

The learning rate was updated after each training epoch using cosine annealing. Let η_e denote the learning rate used during epoch e, where $e = 1, \ldots, 45$. With initial rate η_0 and minimum rate $\eta_{\min}$, the schedule is

$$\eta_e = \eta_{\min} + \frac{\eta_0 - \eta_{\min}}{2}\left[1 + \cos\left(\frac{\pi(e-1)}{45}\right)\right], \qquad \eta_0 = 0.03, \quad \eta_{\min} = 10^{-6}. \tag{6.1}$$

The GRL coefficient α_t increased throughout training. For optimization step t, let p_t denote normalized training progress. The schedule is defined as

$$\alpha_t = \frac{2}{1 + \exp(-10p_t)} - 1, \qquad p_t = \frac{t-1}{449}, \qquad t = 1, \ldots, 450. \tag{6.2}$$

After each update, the discriminator weight matrices were projected to enforce the spectral-norm bound in Table 6.2. All models were evaluated using the final checkpoint. Target labels were reserved for evaluation and were not used in model training.

6.1.2 Evaluation Metrics and Procedures

Classification accuracy was used to evaluate performance on each of the nine target batches B_2–B_{10}. For each random seed, let a_k denote classification accuracy (%) on target batch B_k, where $k = 2, \ldots, 10$. Overall performance was summarized by the equal-weighted mean accuracy A_{equal}:

$$A_{\text{equal}} = \frac{1}{9} \sum_{k=2}^{10} a_k. \tag{6.3}$$

Equal weighting prevents larger batches from dominating the overall result. Each model was trained independently for each source–target pair using 20 random seeds (0–19). Batch-wise accuracies and equal-weighted mean accuracies were averaged across seeds. To quantify the contribution of LC, DANNLC was compared with DANN, and CDANLC with CDAN, using matching seeds and tasks. Paired accuracy gains are expressed in percentage points.

Target pseudo-label error was analyzed over training to assess the reliability of the predicted class information. For this analysis, hard pseudo-labels were assigned to the class with the highest predicted probability, and their accuracy was measured against the ground-truth target labels.

MSPA was evaluated on the same target samples for all four models. Error overlap across random seeds was analyzed to assess the potential for complementary predictions. Ensemble performance was compared with single-model performance, with variability reported across the specified model groups.

6.2 Classification Performance

6.2.1 Classification Results

Table 6.3 summarizes the results under Setting B. The baselines included a support vector machine with a radial basis function kernel (RBF-SVM), a five-nearest-neighbor classifier (5NN), random forest, PCA, and linear discriminant analysis (LDA). The projection methods were DRCA [7], discriminative domain regularization component analysis (DDRCA) [37], domain-based subspace learning with maximizing label–feature dependency and minimizing feature redundancy (DMDMR) [38], and multi-constraint subspace projection (MCSP) [39]. PCA and LDA used an RBF-SVM classifier; DRCA, DDRCA, DMDMR, and MCSP used a common classification head with BN after projection. Here, MCSP denotes the projection component of the framework proposed by Liu et al., evaluated with the shared classification head. FEDA adds feature-norm and target-entropy losses to the DANN backbone [10].

Table 6.3 Classification accuracy (%) of all evaluated methods on the UCI dataset under Setting B.

Approach	B_2	B_3	B_4	B_5	B_6	B_7	B_8	B_9	B_{10}	Average
RBF-SVM	87.46	80.83	55.90	43.65	51.22	43.65	23.13	45.96	48.50	53.37
5NN	87.30	79.89	57.14	44.16	51.52	43.43	15.31	43.40	51.61	52.64
Random forest	69.98	72.76	58.57	43.27	49.09	40.28	23.28	41.94	34.36	48.17 ± 1.64
PCA	87.94	80.33	56.52	42.64	51.17	43.26	23.81	45.96	47.92	53.28
LDA	46.06	52.27	23.60	28.93	27.17	19.15	13.61	33.62	34.75	31.02
DRCA	57.71	54.96	50.78	42.51	47.80	37.01	31.05	35.36	35.19	43.60 ± 1.54
DDRCA	86.81	78.20	67.17	60.10	60.25	44.55	23.71	46.71	38.67	56.24 ± 1.25
DMDMR	87.89	70.93	56.93	74.92	86.45	66.83	53.86	44.62	58.74	66.80 ± 2.03
MCSP	87.54	70.51	57.27	75.03	84.85	65.86	52.52	45.98	58.63	66.46 ± 1.84
FEDA	89.09	85.83	64.47	75.61	92.48	73.16	63.08	54.32	61.04	73.23 ± 2.17
DANN	88.05	86.81	65.75	79.62	91.88	73.33	55.61	56.79	60.14	73.11 ± 1.52
DANNLC	88.14	84.64	63.60	86.45	91.69	74.34	57.89	57.36	**61.86**	74.00 ± 1.43
CDAN	89.56	**91.78**	**67.55**	**92.11**	94.40	**75.73**	70.44	58.84	58.99	77.71 ± 2.09
CDANLC	**89.83**	89.54	63.20	89.92	**95.01**	75.11	**78.27**	**66.13**	57.63	$\mathbf{78.29 \pm 1.49}$

Stochastic methods were evaluated using 20 random seeds (0–19). The Average column reports mean ± sample standard deviation across these seeds to indicate run-to-run variability, calculated from each seed's equal-weighted accuracy across B_2–B_{10}. Deterministic methods were evaluated once and are reported without a seed standard deviation. B denotes a target batch. Bold values identify the highest mean in each column and do not indicate statistical significance.

CDANLC achieved the highest average accuracy. Together with CDAN and DANNLC, it exceeded all additional baselines, while DANN performed slightly below FEDA. The stronger average performance of CDAN than DANN is consistent with a benefit from incorporating class information into domain alignment.

Accuracy varied across batches, and no model achieved the highest accuracy on every batch. Classification errors remained even for CDANLC, which had the highest overall mean. The next comparison isolates the incremental contribution of LC relative to each corresponding baseline.

6.2.2 Effect of LC on Classification Performance

The paired comparisons in Table 6.4 show small positive average LC gains in both model families, but considerable variation across random seeds. Compared with Chapter 5, the benefit was smaller and less consistent; differences in datasets and configurations mean that this contrast cannot be attributed solely to drift duration.

Table 6.4 Paired LC gains in equal-weighted mean accuracy under Setting B.

Comparison	**Gain (pp)**	**Seeds with positive gain**
DANNLC minus DANN	$+0.89 \pm 2.12$	15/20
CDANLC minus CDAN	$+0.58 \pm 2.85$	12/20

Gains were computed between runs with matching seeds and are reported as the mean ± sample standard deviation in percentage points.

In the CDAN family, improvements on B_8 and B_9 offset losses on five other batches. LC gains also varied across seeds.

A possible explanation concerns the reliability of the class information used for alignment. As discussed in Section 3.5, CDAN conditions domain discrimination on target predictions, while LC uses the same predicted probabilities to estimate target class centers. When these soft pseudo-labels are inaccurate, the additional LC constraint may reinforce incorrect class associations rather than correct them. Combining the two mechanisms therefore does not provide an independent check on the reliability of target predictions.

Section 6.3 examines target prediction errors during training to investigate this possible explanation.

6.2.3 Parameter Sensitivity Analysis

The two loss weights were varied one at a time, with all other settings unchanged. The LC sweep fixed $\lambda_d = 0.8$ and used $\lambda_{\mathrm{LC}} \in \{0, 0.1, 0.2, 0.3, 0.6, 0.9, 1.2\}$. At zero LC weight, each LC variant reduces to its corresponding baseline. The domain-loss sweep included all four models at $\lambda_d \in \{0, 0.1, 0.3, 0.6, 0.8, 1.0\}$; LC variants used $\lambda_{\mathrm{LC}} = 0.2$, whereas DANN and CDAN used zero LC weight. Every point used seeds 0–19 and the same equal-batch averaging as the main experiment.

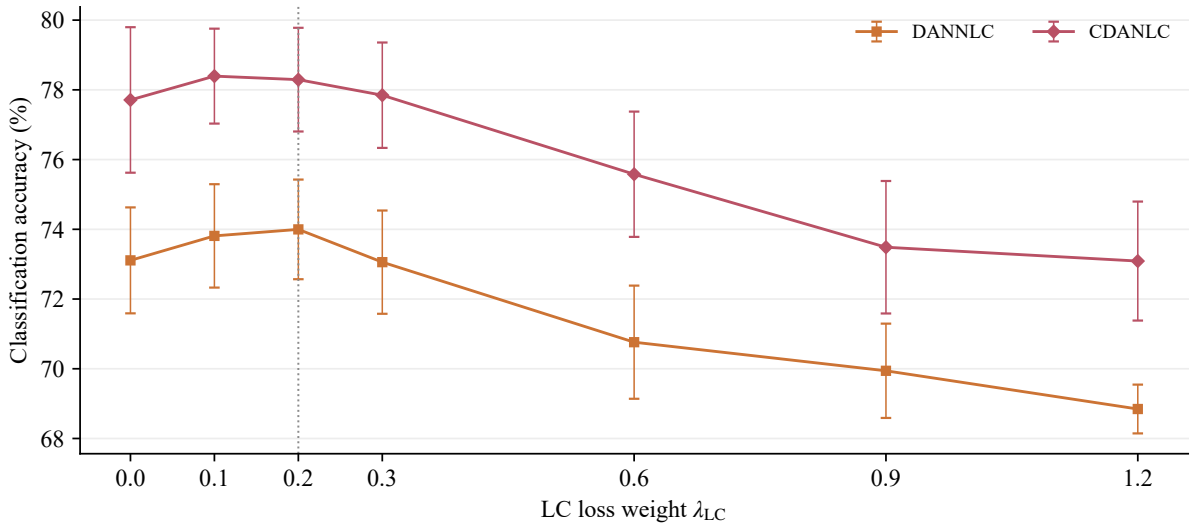


Figure 6.1 Single-model LC-weight sensitivity under Setting B, with $\lambda_d = 0.8$. Error bars show one sample standard deviation over seeds 0–19. The dotted line marks $\lambda_{\mathrm{LC}} = 0.2$.

Figure 6.1 shows that the retained LC weight was close to the best tested value for both model families. DANNLC reached 74.00 % at $\lambda_{\mathrm{LC}} = 0.2$, while CDANLC reached 78.39 % at 0.1 and 78.29 % at 0.2. Larger weights reduced both means, reaching 68.85 % and 73.09 %, respectively, at 1.2. Stronger class-center alignment therefore did not compensate for the limitations observed under Setting B.

The domain-loss response in Figure 6.2 differed between the two model families. The DANN variants attained their highest tested means near $\lambda_d = 0.3$, whereas the CDAN variants continued to improve up to 1.0. Thus, no single domain weight was optimal for all four models. At $\lambda_d = 0$, the domain-adversarial objective is absent, but target samples still affect shared BN and, for the LC variants, the LC loss. This setting is therefore not a strictly source-only baseline. The main settings were retained for the subsequent analyses.

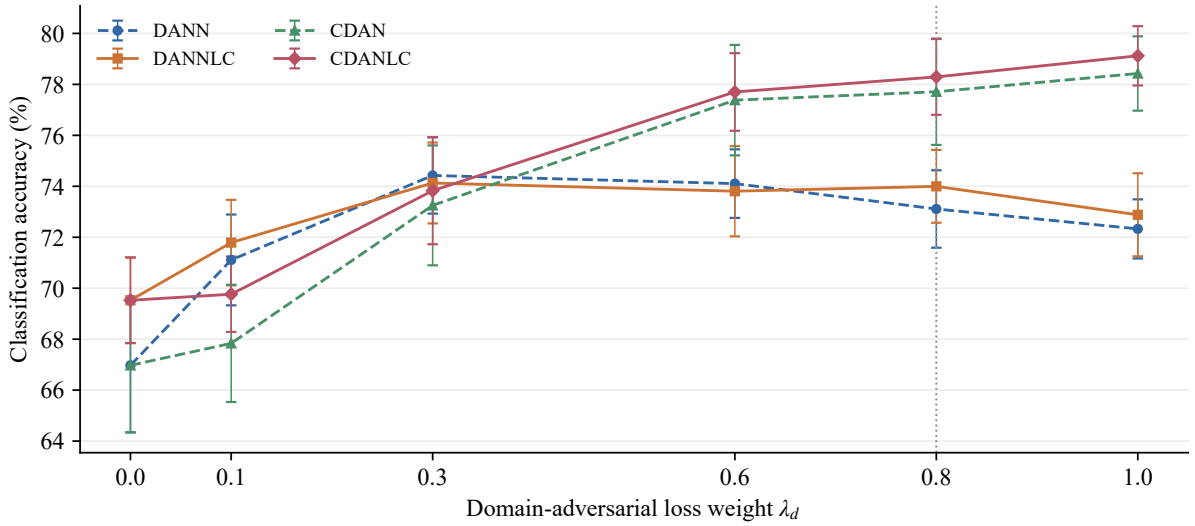


Figure 6.2 Single-model domain-loss sensitivity under Setting B. LC variants use $\lambda_{\mathrm{LC}} = 0.2$; dashed curves show models without LC. Error bars show one sample standard deviation over seeds 0–19. The dotted line marks $\lambda_d = 0.8$.

6.3 Analysis of Pseudo-Label Reliability

Section 6.2 shows that LC did not consistently improve classification performance. As discussed in Section 3.5, both CDAN and LC rely on target predictions to guide alignment. Inaccurate target predictions may help explain this inconsistent benefit.

This section examines how pseudo-label error and confidence change during training, whether entropy conditioning improves classification, and which gas classes are most often misclassified.

6.3.1 Pseudo-Label Error During Training

The hard pseudo-label is the class with the highest predicted probability. For example, a softmax output of $[0.10, 0.10, 0.50, 0.15, 0.10, 0.05]$ assigns the sample to the third gas class, since its probability of 0.50 is the largest. The pseudo-label error rate is the percentage of incorrectly classified target samples. This measures prediction quality, although CDAN and LC use soft probabilities during training. DANN serves as a reference without target pseudo-labels in its training loss.

For Figure 6.3, each complete target batch was evaluated after every epoch using the settings in Section 6.1. Error rates were averaged equally across B_2–B_{10} for each seed, then summarized across seeds 0–19. The final accuracies match the runs used in Table 6.3.

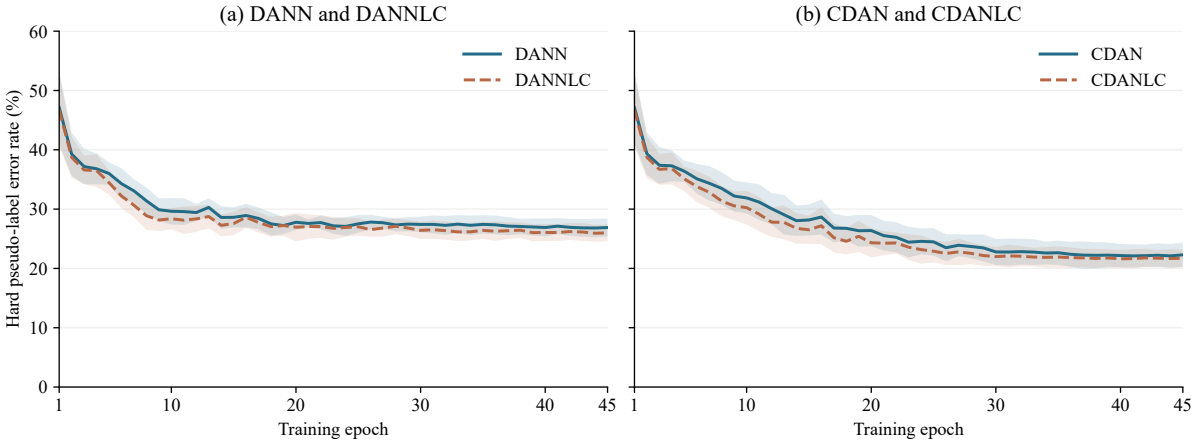


Figure 6.3 Hard pseudo-label error during training under UCI Setting B: (a) DANN and DANNLC; (b) CDAN and CDANLC. Lines show the mean over seeds 0–19; shaded bands show one sample standard deviation after equal averaging across the nine target batches.

All four models reduced their average error during training. DANN also reduced target prediction errors despite not using target pseudo-labels for training. This suggests that source-supervised classification and domain-adversarial alignment can improve target predictions without explicit pseudo-label-based supervision. The DANN variants improved mainly in the early epochs and changed relatively little afterwards, whereas the CDAN variants continued to improve until approximately epoch 30. LC generally produced lower mean error, but the difference remained modest. For CDAN, the reduction associated with LC was about 2.06 percentage points at epoch 20 and only 0.58 at the final epoch.

The final average error remained around 22 %–27 %, with improvement slowing toward the end of training.

The next subsection examines whether these incorrect predictions receive high confidence and may therefore provide misleading class information to CDAN and LC.

6.3.2 Pseudo-Label Confidence During Training

Unlike a hard pseudo-label, a soft pseudo-label retains the full softmax probability vector. For example, $[0.10, 0.10, 0.50, 0.15, 0.10, 0.05]$ and $[0.02, 0.02, 0.90, 0.02, 0.02, 0.02]$ both predict the third gas class, but the second prediction is more confident. If the true label is another class, it is a high-confidence error. Here, confidence is defined as the largest softmax probability.

At every epoch, target predictions were divided into correct and incorrect groups. Let $p_{i,c}$ denote the predicted probability of class $c \in \{1, \ldots, 6\}$ for target sample i, and let $q_i = \max_c p_{i,c}$ denote its confidence. Both probabilities and confidence are dimensionless. For target batch B_k, let $\mathcal{I}_{\text{correct},B_k}$ and $\mathcal{I}_{\text{incorrect},B_k}$ be the sets of correctly and incorrectly predicted samples, with $|\mathcal{I}|$ denoting the number of samples in a set. Their mean confidences, C_{correct,B_k} and $C_{\text{incorrect},B_k}$, are

$$C_{\text{correct},B_k} = \frac{1}{|\mathcal{I}_{\text{correct},B_k}|} \sum_{i \in \mathcal{I}_{\text{correct},B_k}} q_i, \tag{6.4}$$

$$C_{\text{incorrect},B_k} = \frac{1}{|\mathcal{I}_{\text{incorrect},B_k}|} \sum_{i \in \mathcal{I}_{\text{incorrect},B_k}} q_i. \tag{6.5}$$

Each mean uses only the samples belonging to that group. For the experimental curves, these batch-wise means were averaged equally across B_2–B_{10} for each seed. Figure 6.4 reports the mean and sample standard deviation over seeds 0–19. Group membership was updated each epoch.

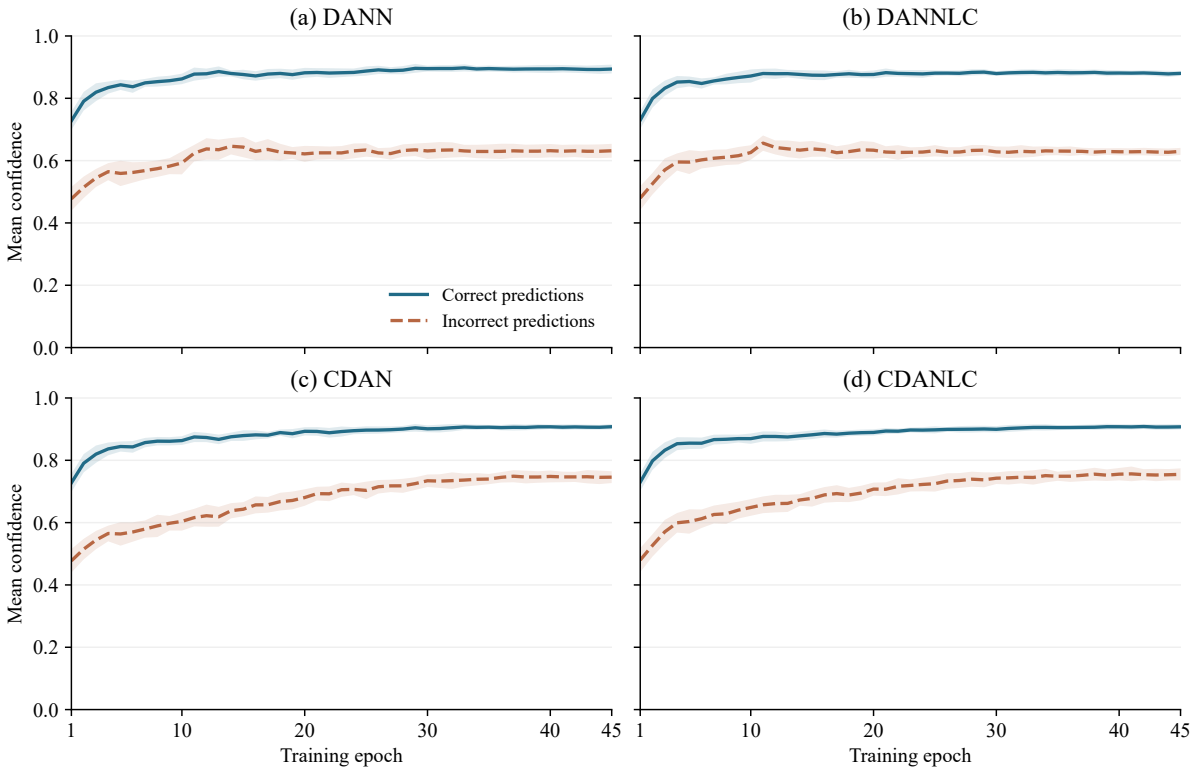


Figure 6.4 Mean confidence of correct and incorrect target predictions during training under Setting B. Lines and shaded bands show the mean and one sample standard deviation over seeds 0–19.

Correct predictions had higher average confidence in all four models. However, confidence also increased among incorrect predictions, reaching about 0.63 for the DANN variants and 0.75 for the CDAN variants. Compared with DANN and DANNLC, the increase in confidence among incorrect predictions was larger and continued for longer in CDAN and CDANLC. Together with Section 6.3.1, this shows that the CDAN variants reduced their overall error while becoming more confident among the predictions that remained incorrect. Because group membership changed each epoch, the curves describe averages over changing groups of incorrect predictions.

At the final epoch, approximately 33 % and 35 % of incorrect predictions in CDAN and CDANLC, respectively, had confidence of at least 0.90, using the same equal-batch averaging. Adding LC therefore did not remove high-confidence errors. High-confidence incorrect predictions may provide misleading class information to CDAN and LC. The next subsection tests whether explicitly emphasizing confident predictions improves adaptation performance.

6.3.3 Effect of Entropy Conditioning

Long et al. [11] introduced entropy conditioning (E) to reduce the potentially adverse influence of uncertain class predictions on conditional adversarial adaptation, giving greater weight to samples with more certain predictions in the domain-adversarial loss. We applied this mechanism to the UCI long-term drift experiments to examine whether it could reduce the effects of unreliable predicted class information. For

sample i, let $\mathbf{p}_i$ be its predicted probability vector and $p_{i,c}$ its probability for class c. Using the natural logarithm $\log$, the entropy $H(\mathbf{p}_i)$ and sample weight w_i are

$$H(\mathbf{p}_i) = -\sum_{c=1}^{6} p_{i,c} \log p_{i,c}, \qquad w_i = 1 + \exp[-H(\mathbf{p}_i)]. \tag{6.6}$$

Lower-entropy predictions receive greater weight in the domain-adversarial loss. Unlike the maximum probability used in Section 6.3.2, entropy summarizes the full probability vector. Source and target weights are normalized separately. Following the authors' implementation, normalization denominators are detached from gradient computation, and gradients through entropy are reversed using the existing gradient-reversal schedule. This also encourages more certain predictions; it does not identify whether they are correct.

We applied this mechanism to all four models, denoting the resulting variants by +E. DANN+E and DANNLC+E are extensions for this comparison. The architecture, training schedule, $\lambda_d = 0.8$, and $\lambda_{\mathrm{LC}} = 0.2$ for LC models remained unchanged; E modifies the domain loss while leaving LC unchanged. For all nine Setting B tasks, models were trained with seeds 0–19 and evaluated at epoch 45.

Table 6.5 Effect of entropy conditioning on classification accuracy under Setting B.

Model	Original (%)	+E (%)	Change (pp)	Improved seeds
DANN	73.11 ± 1.52	73.37 ± 2.35	$+0.26$	12/20
DANNLC	74.00 ± 1.43	75.32 ± 2.23	$+1.32$	14/20
CDAN	77.71 ± 2.09	70.82 ± 2.49	-6.89	0/20
CDANLC	78.29 ± 1.49	72.32 ± 2.98	-5.97	0/20

Original and +E report mean $\pm$ sample standard deviation over seeds 0–19, after equal averaging across the nine target batches. Change is the mean paired difference in percentage points (pp). The Improved seeds column counts positive paired differences.

Table 6.5 shows that the effect of E depended on the model family. The DANN variants improved on average, whereas both CDAN variants declined for every matched seed. Thus, the weighting mechanism was beneficial in some configurations but did not provide a consistent improvement under Setting B.

The confidence analysis helps interpret this limitation. For CDAN and CDANLC, mean confidence among incorrect predictions increased from 0.746 and 0.755, respectively, to approximately 0.875 with E, using the same equal-batch averaging. Greater certainty therefore accompanied lower accuracy. A confident error can receive a relatively high entropy-based weight, so this mechanism cannot ensure reliable class information for alignment.

6.3.4 Class-Wise Pseudo-Label Errors

We analyzed class-wise prediction errors using the final epoch-45 predictions from DANN, DANNLC, CDAN, and CDANLC without entropy conditioning. The analysis assessed differences among gases and the effect of LC on their recognition.

For each sample, the gas class with the highest predicted softmax probability is assigned as its hard pseudo-label. A prediction is counted as incorrect when this assigned class differs from the true gas class. Samples are grouped by their true gas class, and the error rate within each target batch is the number of incorrectly predicted samples in that group divided by the total number of samples in the group. For each

seed, these rates were averaged equally across batches containing the class. Figure 6.5 reports the mean and sample standard deviation over seeds 0–19.

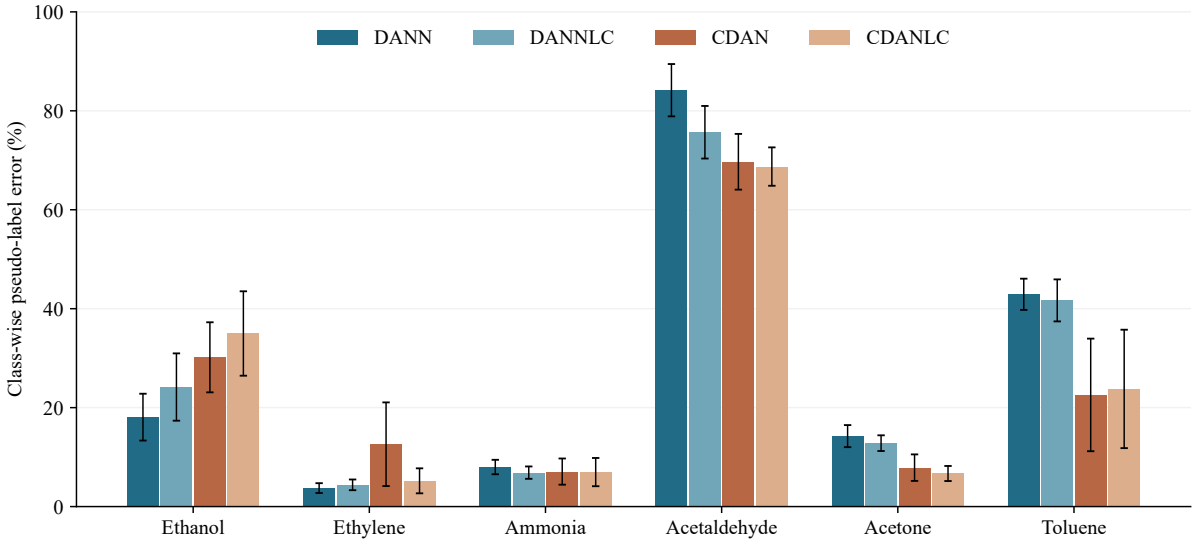


Figure 6.5 Class-wise pseudo-label error at epoch 45 under Setting B.
Note: Bars show the mean over 20 seeds (0–19), and error bars indicate one sample standard deviation. Toluene is absent from B_3–B_5. Its error rates were averaged over the six target batches containing it; rates for the other gases were averaged over all nine target batches.

Acetaldehyde had the highest error rate in all four models, ranging from approximately 69 % to 84 %, whereas ammonia remained below 8 %. Errors differed sharply by gas class. The CDAN variants reduced acetaldehyde and toluene errors relative to the DANN variants, but produced more ethanol errors. Their higher overall accuracy thus did not imply better predictions for every class.

LC also had different effects across gases. In DANN, it reduced the acetaldehyde error by 8.50 percentage points but increased the ethanol error by 6.09 points. In CDAN, its clearest improvement was for ethylene, while acetaldehyde remained difficult and ethanol error increased. LC therefore improved some classes while leaving substantial errors in others. This motivates investigating more reliable target predictions and evaluating improvements at both the overall and class levels, particularly for acetaldehyde.

6.4 Multi-Seed Probability Averaging

The preceding analysis showed that individual models still produced incorrect target predictions, including confident errors. This section evaluates MSPA, defined in Section 3.6, to examine whether averaging across random seeds reduces these errors.

6.4.1 Evaluation Setups

We evaluated all four methods under Setting B using the original final checkpoints from seeds 0–19. Each member was trained for 450 updates using the settings in Section 6.1, including $\lambda_d = 0.8$ and $\lambda_{\mathrm{LC}} = 0.2$ for LC models. A group contains only one model type; DANN, DANNLC, CDAN, and CDANLC were evaluated separately.

The main experiment used $M = 7$ and the 20 overlapping groups defined in Section 5.5.1. The same groups were used for all methods, and no member was selected according to its target accuracy. Seven members were adopted as the common reference size after exploratory comparisons under Setting B. Section 6.4.4 examines the effect of this choice; the short- and medium-term evaluations retained it for comparison.

For each group, accuracy was first averaged equally across the nine target batches. The mean and sample standard deviation across groups are reported. These groups share members, so their variation describes the choice of ensemble members within the fixed seed pool rather than uncertainty across independent ensembles. Seven-member inference requires seven model evaluations, and training one group from scratch requires seven member training runs. Existing checkpoints were reused here.

6.4.2 Overlap of Prediction Errors

In the terminology of Section 6.3, hard pseudo-labels from different seeds may be incorrect on different samples.

Figure 6.6 shows the overlap of individual-member prediction errors. For each method, we examined the seven-member groups defined in Section 6.4.1, using models trained with seeds 0–19. Within each group, a target sample is assigned to one of three categories: all seven models predict correctly; at least one predicts correctly and at least one incorrectly; or all seven predict incorrectly. The calculation used each model's highest-probability class and the true label for analysis only. The mixed category excludes disagreements where every model is wrong.

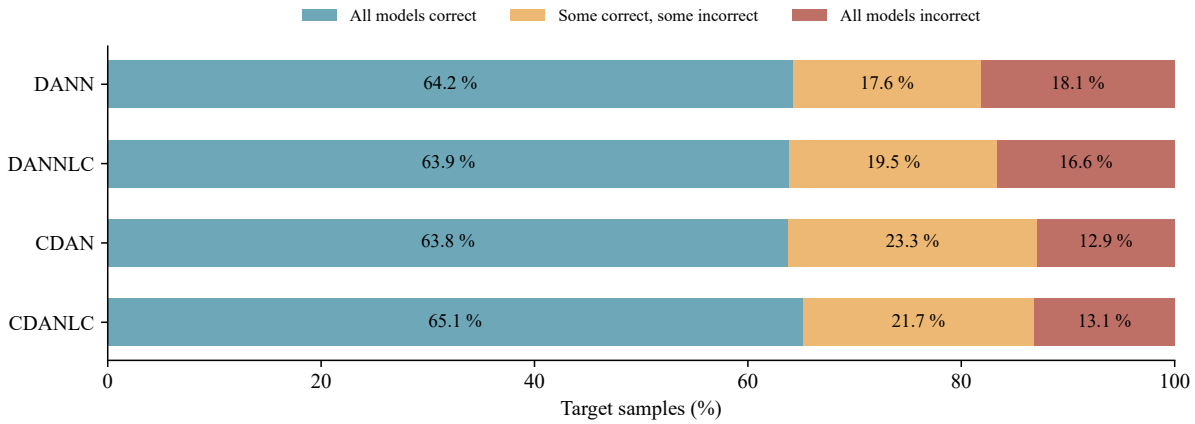


Figure 6.6 Overlap of prediction errors across random seeds.

Note: Each group contains seven final models. Sample proportions were calculated within each target batch, averaged equally across B_2–B_{10}, and then averaged over the 20 groups. The categories describe individual predictions before aggregation.

Mixed correct and incorrect predictions occurred for 17.6 %–23.3 % of target samples across the four methods, showing that their errors were not identical across seeds. This supports testing aggregation as a way to recover part of the available correct information. Nevertheless, 12.9 %–18.1 % of samples were misclassified by every member of a group.

6.4.3 Classification Results

Table 6.6 reports the seven-model results for each target batch. The single-model and ensemble means are compared in Figure 6.9. All four methods benefited on average from probability averaging, with the largest gain for CDAN.

Table 6.6 Seven-member MSPA classification accuracy (%) under Setting B.

Method	B_2	B_3	B_4	B_5	B_6	B_7	B_8	B_9	B_{10}	Average	Gain (pp)
DANN	88.95	88.23	65.96	77.34	92.52	74.38	54.54	59.67	60.84	73.60 ± 0.30	$+0.49$
DANNLC	88.85	88.23	63.88	89.24	91.91	75.43	55.58	59.89	62.24	75.03 ± 0.49	$+1.03$
CDAN	89.63	93.20	69.13	97.44	94.83	78.45	82.21	65.70	58.87	81.05 ± 0.64	$+3.34$
CDANLC	89.63	93.16	65.43	94.37	95.31	77.79	81.60	68.79	57.72	80.42 ± 0.61	$+2.13$

Note: B denotes a target batch. Seeds 0–19 define 20 seven-member groups as described in Section 6.4.1. Batch entries are group means. Average reports mean $\pm$ sample standard deviation of the 20 equal-weighted nine-batch group accuracies. Gain is relative to the corresponding single-model mean and is expressed in percentage points (pp).

Table 6.7 Change in batch-wise accuracy produced by seven-member MSPA under Setting B (percentage points).

Method	B_2	B_3	B_4	B_5	B_6	B_7	B_8	B_9	B_{10}	Average
DANN	$+0.90$	$+1.43$	$+0.22$	-2.28	$+0.64$	$+1.05$	-1.07	$+2.88$	$+0.70$	$+0.49$
DANNLC	$+0.71$	$+3.58$	$+0.28$	$+2.79$	$+0.22$	$+1.09$	-2.31	$+2.53$	$+0.39$	$+1.03$
CDAN	$+0.07$	$+1.42$	$+1.58$	$+5.33$	$+0.44$	$+2.72$	$+11.77$	$+6.86$	-0.12	$+3.34$
CDANLC	-0.19	$+3.62$	$+2.24$	$+4.44$	$+0.30$	$+2.68$	$+3.33$	$+2.66$	$+0.09$	$+2.13$

Note: Each entry is the mean seven-member MSPA accuracy in Table 6.6 minus the corresponding mean single-model accuracy in Table 6.3. Positive values indicate an improvement. Average is the equal-weighted mean of the nine batch-wise differences.

Table 6.7 shows that MSPA improved most model–batch combinations, but the gains were not uniform. The largest improvements occurred for CDAN on B_8 and B_9, whereas several DANN-family combinations showed small decreases. Probability averaging therefore reduced some seed-dependent errors without removing all task-specific weaknesses.

DANNLC exceeded DANN by 1.42 percentage points, whereas CDANLC fell below CDAN by 0.63 points. Thus, LC remained beneficial on average in the DANN family, but its small single-model advantage in the CDAN family did not persist after aggregation.

Figure 6.7 examines whether these improvements were distributed across gas classes. Error decreased for several model–gas combinations, including ethanol for all four methods and ethylene and toluene for the CDAN variants. Acetaldehyde remained the most difficult gas for every method, with errors ranging from 67.58 % to 85.86 % after MSPA.

To examine this persistent error, Figure 6.8 uses all 20 single-seed models and separates acetaldehyde samples according to prediction agreement. Only 2.09 %–4.49 % of acetaldehyde samples were correctly classified by all 20 models, whereas 44.09 %–58.76 % were misclassified by every model. The remaining samples received correct predictions from some members and incorrect predictions from others.

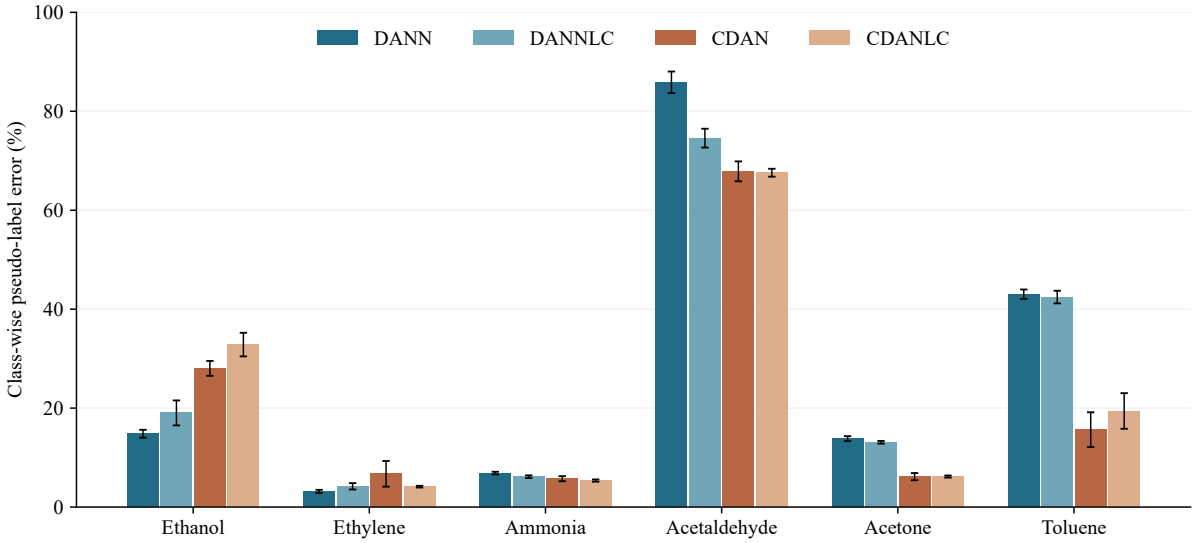


Figure 6.7 Class-wise pseudo-label error after seven-member MSPA under Setting B.

Note: Bars show the mean over 20 seven-member groups, and error bars indicate one sample standard deviation across groups. For each group, class error was first calculated within each target batch and then averaged equally across batches containing the class. Toluene was averaged over the six target batches containing it; the other gases were averaged over all nine target batches.

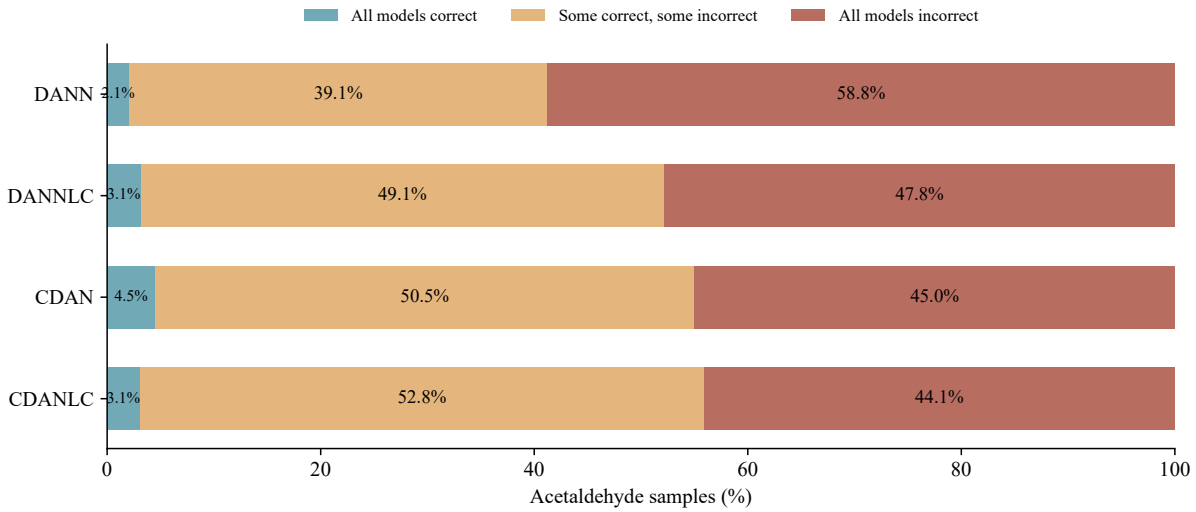


Figure 6.8 Agreement of 20 single-seed models on acetaldehyde predictions under Setting B.

Note: Categories were determined from each model's highest-probability class before aggregation. Percentages were calculated among acetaldehyde samples within each target batch and averaged equally across B_2–B_{10}. Target labels were used only for this evaluation.

The large all-incorrect proportion indicates that the high acetaldehyde error was primarily a shared weakness of the original models rather than an error confined to a few random seeds. This may help explain why class-wise error decreased for some gases while remaining high for acetaldehyde.

6.4.4 Sensitivity Analysis

Ensemble size. We tested $M \in \{1, 3, 5, 7, 10, 12, 15, 20\}$ using the existing models trained with seeds 0–19. For the first group, seeds were selected in the following fixed order:

$$0, 7, 13, 3, 17, 5, 11, 1, 9, 15,$$
$$2, 12, 4, 8, 16, 6, 10, 14, 18, 19.$$

An ensemble of size M uses the first M seeds: for example, $M = 1$ uses seed 0, $M = 3$ uses seeds $0, 7, 13$, and $M = 5$ uses seeds $0, 7, 13, 3, 17$. The other 19 groups add $1, \ldots, 19$, respectively, to every seed in this order, wrapping values of 20 or above back to 0–19. Each group again takes the first M seeds. At $M = 20$, every group contains all seeds, so the ensemble was evaluated once. Model parameters remained fixed throughout.

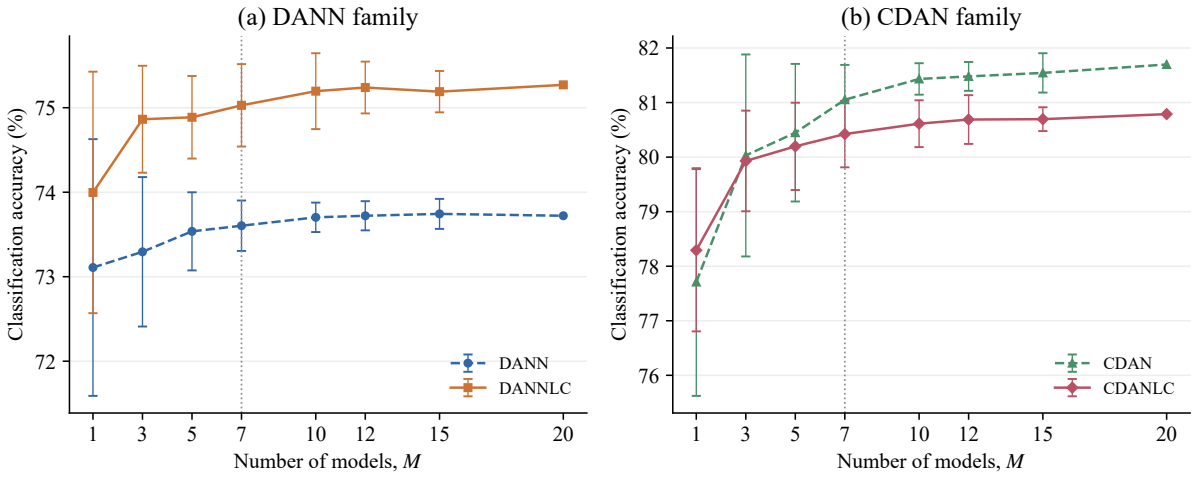


Figure 6.9 Effect of ensemble size on classification accuracy.

Note: Accuracy is averaged equally across B_2–B_{10}. Points show group means, and error bars show one sample standard deviation for $M < 20$. At $M = 1$, groups are individual seeds. At $M = 20$, only one unique ensemble exists, so no standard deviation is reported. The two panels use different vertical scales. The dotted line marks the seven-member reference setting.

Figure 6.9 shows that the main improvement occurred with relatively small ensembles, followed by diminishing returns. Increasing the number of members from seven to twenty improved accuracy by 0.12–0.65 percentage points across the four methods, while requiring almost three times as many model evaluations. Seven-member ensembles therefore retained much of the observed benefit at a lower inference cost in these experiments.

The variation across compositions generally decreased as more members were used. Larger groups also share more models, partly constraining variation across compositions. Across the tested ensemble sizes, DANNLC exceeded DANN, whereas CDAN exceeded CDANLC.

Loss weights with seven-member ensembles. To examine sensitivity to the member training objectives, we fixed $M = 7$ and varied one loss weight at a time. The LC sweep compared DANNLC and CDANLC; the domain-loss sweep included all four methods. The LC sweep used weights 0, 0.1, 0.2, 0.3, 0.6, 0.9, and 1.2, with $\lambda_d = 0.8$. The domain-loss sweep used weights 0, 0.1, 0.3, 0.6, 0.8, and 1.0, with $\lambda_{\mathrm{LC}} = 0.2$ for the LC variants and zero for DANN and CDAN. Each parameter setting used

separately trained members with seeds 0–19, 450 updates, and the same 20 seven-member groups as the main experiment. All other training and evaluation settings remained unchanged.

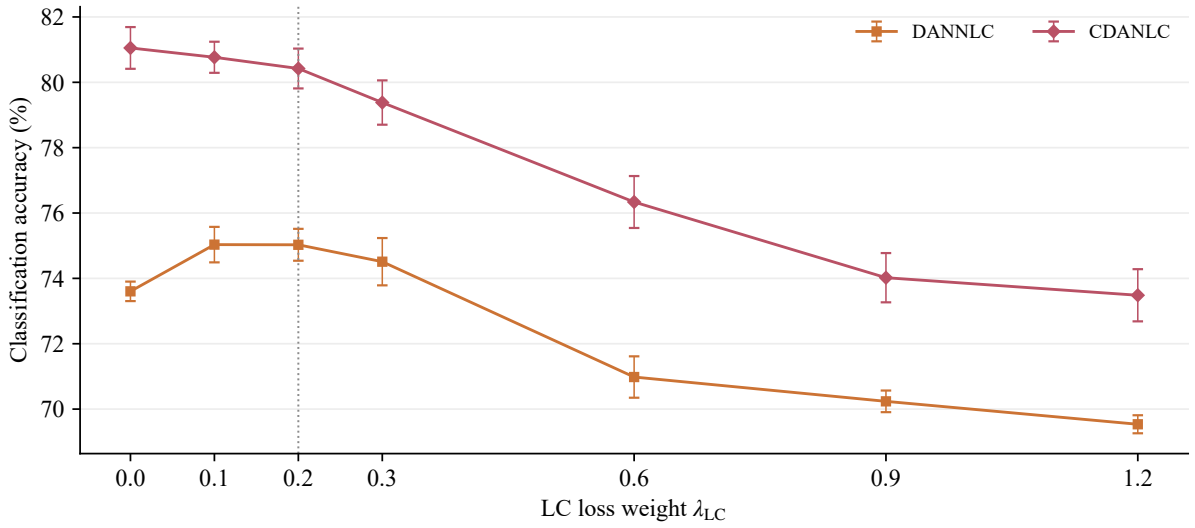


Figure 6.10 LC-weight sensitivity of seven-member MSPA under Setting B, with $\lambda_d = 0.8$. Error bars show one sample standard deviation across the 20 groups. The dotted line marks $\lambda_{LC} = 0.2$.

Figure 6.10 shows that the effect of LC remained model-dependent after aggregation. DANN-based ensembles reached similar accuracies at $\lambda_{LC} = 0.1$ and 0.2 (75.03 % at both weights after rounding), exceeding the zero-LC result of 73.60 %. In the CDAN family, however, every tested positive LC weight remained below the zero-LC ensemble. Larger LC weights reduced accuracy in both families.

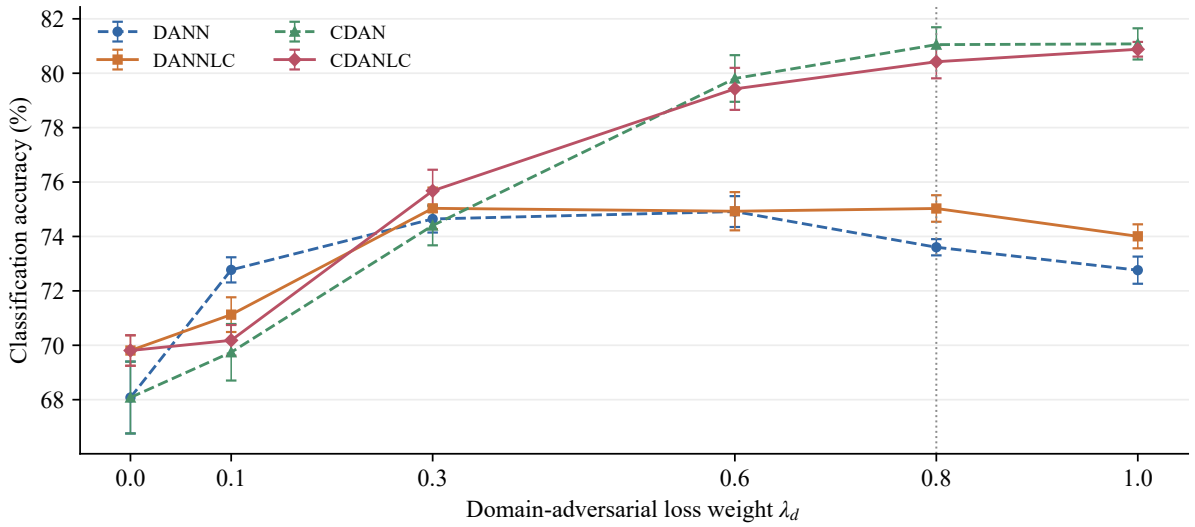


Figure 6.11 Domain-loss sensitivity of seven-member MSPA under Setting B. Solid curves show LC variants with $\lambda_{LC} = 0.2$; dashed curves show DANN and CDAN without LC. Error bars show one sample standard deviation across the 20 groups. The dotted line marks $\lambda_d = 0.8$.

Figure 6.11 shows different responses to the domain-loss weight. DANNLC remained near 75 % for $\lambda_d = 0.3$–0.8 before declining at 1.0, whereas CDANLC improved across the tested range, reaching 80.88 % at 1.0 compared with 80.42 % at 0.8. The baseline curves also varied with the domain-loss weight. Setting $\lambda_d = 0$ removes the domain-adversarial loss, but target samples still enter shared BN

and, for LC variants, the LC loss. LC did not consistently outperform its corresponding baseline across the tested weights. These observations show that MSPA does not remove sensitivity to the objectives used to train its members. The main-experiment settings were retained.

6.4.5 Analysis of Long-Term MSPA Performance

MSPA improved the mean accuracy of all four methods under long-term drift. Increasing M generally brought further gains, with diminishing returns for larger ensembles (Figure 6.9). Within the seven-member groups, approximately 65 % of samples were classified correctly by every member and roughly 20 % received mixed correct and incorrect predictions (Figure 6.6). This mixed subset may help explain the gains, as correct predictions from some members could compensate for errors from others. Seven-member ensembles retained much of the observed improvement, while adding more members increased inference cost for progressively smaller gains.

The benefits of averaging and LC differed within the matched comparisons. DANNLC retained an advantage over DANN after MSPA, whereas CDANLC fell below CDAN (Table 6.6). The loss-weight sweeps also showed that stronger LC could reduce ensemble accuracy (Figure 6.10). These results suggest that combining predictions can reduce some errors across seeds while retaining the influence of the members' training objectives. An improvement from MSPA therefore does not establish an additional benefit from LC in the same model family.

The acetaldehyde results show a further limit. In the separate analysis using all 20 seeds, very few samples were classified correctly by every model, and many were misclassified by all models (Figure 6.8). Recognition remained poor after MSPA despite a substantial mixed subset. The benefit depends on the probabilities assigned to the correct and incorrect classes: some correct member predictions may be outweighed by confident errors from other members. The overall improvement therefore coexisted with a shared difficulty in recognizing acetaldehyde. Improving that class requires more reliable member predictions, rather than relying on additional seeds alone.

6.5 Chapter Summary

For KEQ 6.1, LC produced small positive mean gains for DANN and CDAN under long-term drift, but its effects varied across tasks and seeds. Increasing the LC weight beyond the main setting reduced mean accuracy in both model families. The short-term benefit of stronger class-center alignment therefore did not carry over consistently to these fixed-source adaptation tasks.

For KEQ 6.2, target prediction errors persisted during training and differed considerably across gases. Some incorrect predictions also became highly confident. Entropy conditioning improved DANN and DANNLC on average but reduced CDAN and CDANLC performance. Together, these results show that prediction confidence alone is insufficient to identify reliable class information for alignment.

For KEQ 6.3, MSPA improved mean accuracy for all four methods. Seven-member ensembles retained much of the observed benefit, and larger ensembles provided smaller additional gains at greater inference cost. LC remained beneficial after aggregation for DANN, but its small single-model advantage for CDAN did not persist. Acetaldehyde remained difficult to classify, showing that improved overall accuracy can coexist with a shared class-specific weakness.

This chapter connects the limited LC gains with the observed unreliability of target predictions and shows where averaging can help. MSPA improved final classification while leaving the members' training objectives unchanged; it did not resolve the persistent acetaldehyde errors. The next chapter examines whether these patterns also appear when each target batch is paired with the immediately preceding source batch.

7 Medium-Term Drift Compensation

This chapter investigates medium-term sensor drift under Setting C, in which each batch serves as the labeled source for the immediately following target batch. Unlike the fixed-source tasks in Chapter 6, each task uses a different labeled source batch. A separate model was trained independently for each task; the tasks do not form a sequence of updates to a single model. Setting C therefore provides a further test of whether the effects of domain-adversarial learning and latent class-center alignment (LC) remain consistent across different source–target relationships.

The chapter first compares the four models and their sensitivity to the loss weights, then evaluates MSPA separately for each model type using the method in Section 3.6. Error overlap and parameter sensitivity are used to interpret the MSPA results.

This chapter addresses RQ1 and RQ2 through single-model comparisons and loss-weight sensitivity, and RQ3 through the MSPA evaluation. The following KEQs organize these experiments:

KEQ 7.1: Does LC improve classification accuracy under medium-term drift, and how does performance change with the loss weights? (Section 7.1)

KEQ 7.2: To what extent does MSPA improve classification under medium-term drift? (Section 7.2)

7.1 Single-Model Evaluation

Except for the source–target pairing, all model architectures, preprocessing procedures, training hyperparameters, and implementation settings were identical to those in Section 6.1.1. Setting C comprises nine adaptation tasks, $B_k \to B_{k+1}$ for $k = 1, \ldots, 9$, with a separate model trained from scratch for each task. The standardization parameters were fitted to each task's source batch after the inverse hyperbolic sine (asinh) transformation. Source labels were used for training; target labels were reserved for evaluation. All neural models retained six output classes, including when a source batch did not contain every class.

The main experiments retained $\lambda_d = 0.8$ and, for the LC variants, $\lambda_{\mathrm{LC}} = 0.2$. Each model was trained for 450 updates with seeds 0–19. For each seed, accuracy was first averaged equally across the nine tasks. The overall mean and sample standard deviation were then calculated across the 20 seeds.

7.1.1 Classification Results

Table 7.1 presents the results for the four models. CDANLC achieved the highest overall mean among these models, followed by CDAN, DANNLC, and DANN. The differences between the two models within each family were small compared with the substantial variation in task difficulty.

Table 7.1 Classification accuracy (%) of the four models under Setting C.

Method	B_2	B_3	B_4	B_5	B_6	B_7	B_8	B_9	B_{10}	Average
DANN	88.05	97.71	75.93	95.79	55.33	73.92	87.09	72.87	38.77	76.16 ± 1.29
DANNLC	88.14	97.60	76.71	95.53	57.39	73.68	88.04	75.32	39.10	76.84 ± 1.15
CDAN	89.56	98.42	79.84	96.32	62.12	71.00	94.42	68.23	38.87	77.64 ± 1.01
CDANLC	89.83	98.24	75.06	96.80	61.36	71.62	95.03	72.21	40.06	77.80 ± 1.12

Note: Column B_k denotes task $B_{k-1} \to B_k$. Task columns are means across seeds 0–19. Average reports the mean $\pm$ sample standard deviation of each seed's equal-weighted mean accuracy across the nine tasks.

Using matched seeds, LC increased mean accuracy by 0.67 ± 1.47 percentage points for DANN and 0.16 ± 0.93 points for CDAN. Values are reported as the mean $\pm$ sample standard deviation of the paired gains. Positive paired gains occurred for 15 of 20 seeds in DANN and 10 of 20 in CDAN.

The adjacent-batch setting also retained difficult transfer tasks. All four models exceeded 95 % on $B_2 \to B_3$ and $B_4 \to B_5$, whereas their means on $B_9 \to B_{10}$ were only 38.77 %–40.06 %. Class support is another limitation: B_5 lacks toluene (class 6 in the original UCI labels), which reappears in B_6 with 467 of its 2,300 samples. All four models had zero recall for this class in that task. These observations show that temporal adjacency does not by itself ensure that the source data provide sufficient support for every target class.

7.1.2 Parameter Sensitivity Analysis

The loss weights were varied one at a time, with all other settings unchanged. The LC sweep fixed $\lambda_d = 0.8$ and used weights 0, 0.1, 0.2, 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, 2.1, and 2.4. At zero LC weight, each LC variant reduces to its corresponding baseline. The domain-loss sweep included all four models at weights 0, 0.1, 0.3, 0.6, 0.8, and 1.0; LC weights remained at 0.2 for LC variants and zero for the baselines. Every point used seeds 0–19 and the same equal-task averaging as the main experiment.

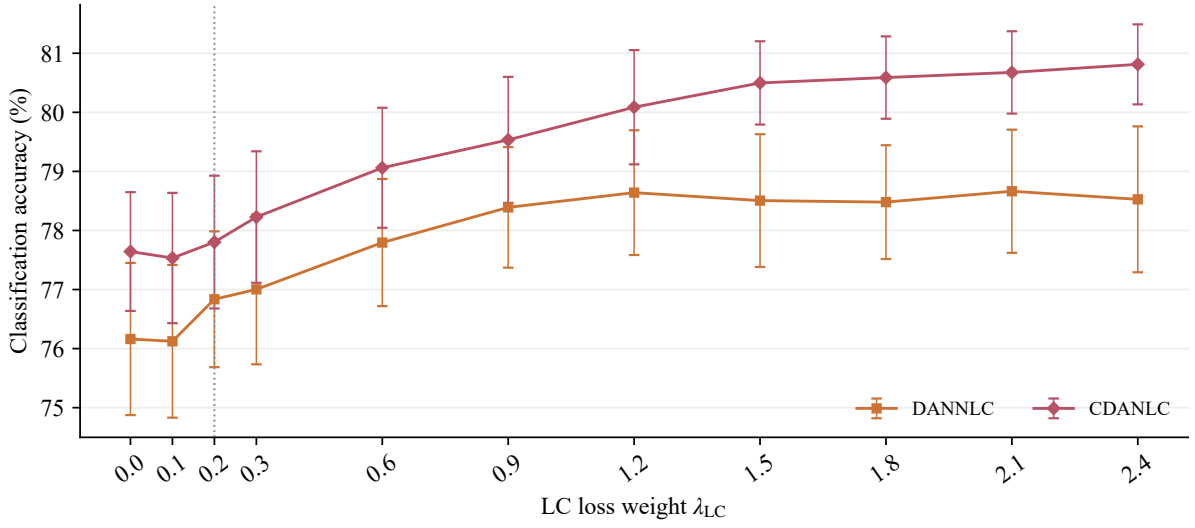


Figure 7.1 Single-model LC-weight sensitivity under Setting C, with $\lambda_d = 0.8$. Error bars show one sample standard deviation over seeds 0–19. The dotted line marks $\lambda_{LC} = 0.2$.

Figure 7.1 shows that the response to stronger LC differed between the model families. DANNLC increased from 76.84 % at the default weight of 0.2 to 78.64 % at 1.2, after which its mean fluctuated around a similar level, reaching 78.53 % at 2.4. CDANLC continued to improve over this range, from 77.80 % at 0.2 to 80.09 % at 1.2 and 80.81 % at 2.4.

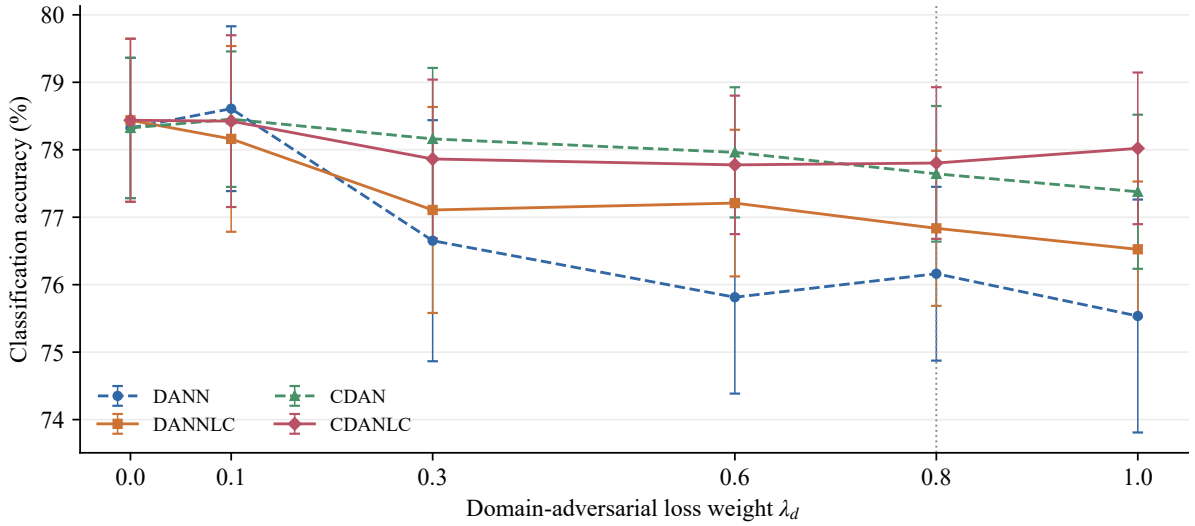


Figure 7.2 Single-model domain-loss sensitivity under Setting C. LC variants use $\lambda_{\text{LC}} = 0.2$; dashed curves show models without LC. Error bars show one sample standard deviation over seeds 0–19. The dotted line marks $\lambda_d = 0.8$.

The domain-weight results in Figure 7.2 do not show a uniform benefit from stronger adversarial alignment. DANN reached 78.61 % at $\lambda_d = 0.1$, compared with 76.16 % at 0.8. CDAN varied less, but its mean also decreased from 78.45 % at 0.1 to 77.64 % at 0.8. DANNLC and CDANLC both achieved 78.44 % at zero domain weight, exceeding their respective default results. At $\lambda_d = 0$, target samples still contribute through joint source–target batch normalization and, when enabled, LC.

7.2 Multi-Seed Probability Averaging

7.2.1 Evaluation Setups

The second part of the analysis applies MSPA using the probability-averaging rule in Equation 3.15. The main configuration and group construction follow Section 6.4.1: seven models of the same method are selected from the fixed pool of seeds 0–19, their softmax probabilities are averaged equally, and the class with the largest averaged probability is predicted. The same 20 overlapping groups were used for each task and method. All members used their final checkpoints, and no training objective was changed.

7.2.2 Overlap of Prediction Errors

Target samples were divided within each seven-member group into three categories: all models predict correctly; some predict correctly and others incorrectly; or all models predict incorrectly. The proportions were calculated within each task, averaged equally over the nine tasks, and then averaged over the 20 groups. Ground-truth target labels were used only for this post-training analysis.

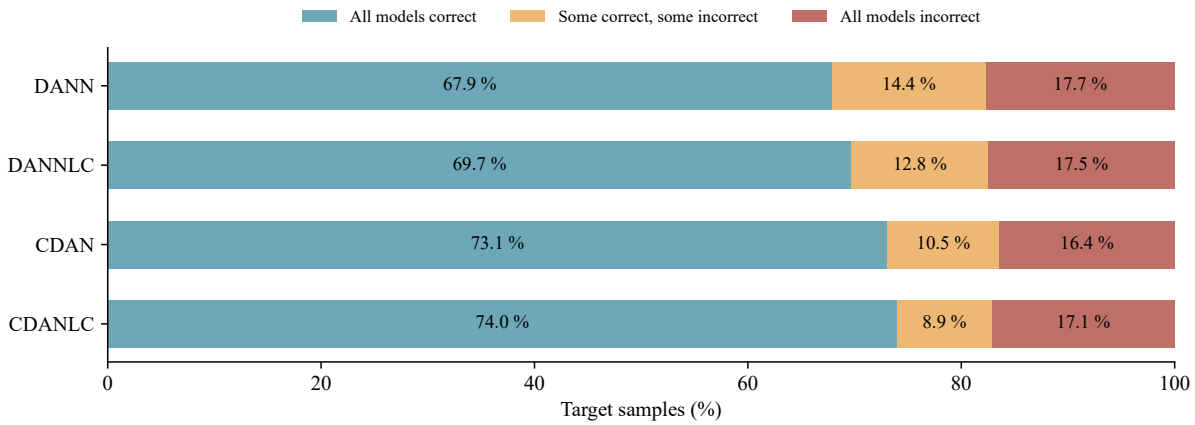


Figure 7.3 Overlap of individual-member prediction errors under Setting C, before MSPA. Each group contains seven models of the same method.

Figure 7.3 shows mixed correct and incorrect predictions for 8.9 %–14.4 % of target samples, indicating that some correct predictions are available from other seeds when an individual model fails. This proportion was smaller than the 17.6–23.3 % observed with the same seven-member grouping under Setting B (Figure 6.6). Meanwhile, 16.4–17.7 % of samples were misclassified by every member under Setting C. Models in the all-wrong category may predict different incorrect classes.

7.2.3 Classification Results

Table 7.2 presents the seven-member MSPA results. MSPA improved the DANN variants on average but slightly reduced the accuracy of the CDAN variants.

Table 7.2 Seven-member MSPA classification accuracy (%) under Setting C.

Method	B_2	B_3	B_4	B_5	B_6	B_7	B_8	B_9	B_{10}	Average	Gain (pp)
DANN	88.95	97.82	76.65	97.18	55.39	74.96	92.89	73.93	39.30	77.45 ± 0.44	+1.29
DANNLC	88.85	97.73	77.08	97.23	57.73	74.39	91.51	76.38	39.14	77.78 ± 0.56	+0.95
CDAN	89.63	98.56	79.57	96.73	59.57	70.92	95.05	65.77	38.49	77.14 ± 0.30	-0.50
CDANLC	89.63	98.35	75.00	96.95	60.18	70.89	95.51	71.74	40.04	77.59 ± 0.57	-0.21

Note: Column B_k denotes task $B_{k-1} \to B_k$. Task columns are means across 20 overlapping seven-member groups. Average reports the mean $\pm$ sample standard deviation of the group accuracies averaged equally across the nine tasks. Gain is relative to the corresponding single-model mean and is expressed in percentage points (pp).

The ensemble group standard deviations were 0.30–0.57 percentage points, smaller than the single-model seed standard deviations. These statistics describe different sources of variation: individual training seeds and overlapping groups within a fixed seed pool.

LC remained favorable in the matched MSPA comparisons: DANNLC exceeded DANN by 0.33 percentage points, and CDANLC exceeded CDAN by 0.45 points. These are comparisons between separately trained model families after aggregation.

7.2.4 Sensitivity Analysis

Number of ensemble members. The ensemble-size analysis followed the nested group construction in Section 6.4.4, using $M \in \{1, 3, 5, 7, 10, 12, 15, 20\}$. For $M < 20$, each point averages the same 20 overlapping group constructions; $M = 1$ corresponds to the individual seeds. At $M = 20$, all members belong to one unique ensemble. Figure 7.4 shows mean accuracy in separate panels for the two model families; the dotted line marks $M = 7$.

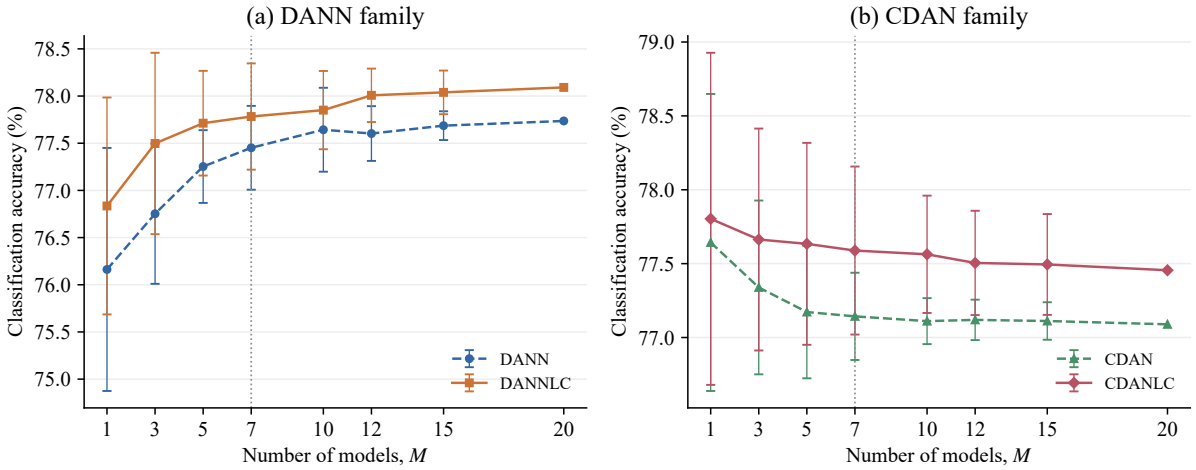


Figure 7.4 Effect of ensemble size on accuracy under Setting C. Error bars show one sample standard deviation across the 20 groups for $M < 20$; at $M = 20$, only one ensemble exists. At $M = 1$, groups are individual seeds. The panels use different vertical scales; the dotted line marks $M = 7$.

DANN and DANNLC generally benefited from additional members, reaching 77.74 % and 78.09 % at $M = 20$, respectively. CDAN and CDANLC followed a different trajectory: their means at $M = 20$ were 77.09 % and 77.46 %, below their single-model means. Increasing ensemble size therefore did not remove the model-dependent effect observed at $M = 7$. The seven-member setting was retained for comparability with Chapter 6.

Loss weights with seven-member MSPA. The second sensitivity analysis fixed $M = 7$ and used the loss-weight grids in Section 7.1.2. The LC sweep evaluated DANNLC and CDANLC, whereas the domain-loss sweep included all four methods. At each parameter setting, probabilities were averaged across members trained with that setting. All 20 seeds, nine tasks, and group memberships were retained. The LC curves extend to 2.4. The domain sweep fixed $\lambda_{\mathrm{LC}} = 0.2$ for the LC variants and zero for DANN and CDAN. Accuracy was calculated from the averaged probability vectors within each group, then averaged equally across tasks and across the 20 groups.

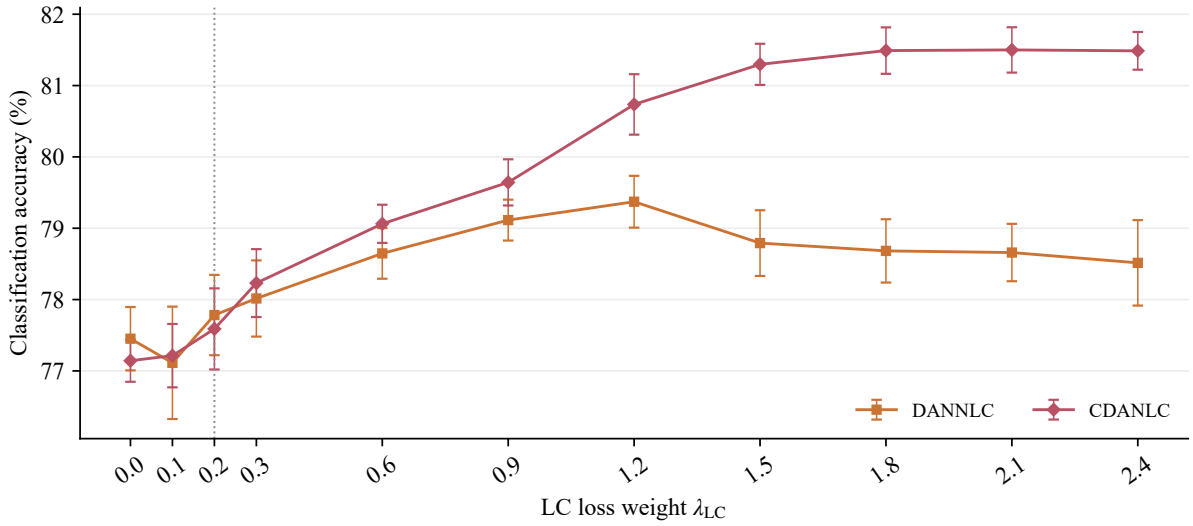


Figure 7.5 LC-weight sensitivity of seven-member MSPA under Setting C, with $\lambda_d = 0.8$. Error bars show one sample standard deviation across the 20 groups. The dotted line marks $\lambda_{LC} = 0.2$.

Figure 7.5 shows the LC-weight sensitivity of both seven-member ensembles. DANNLC + MSPA reached its highest tested mean of 79.37 % at $\lambda_{LC} = 1.2$, compared with 77.78 % at the default weight of 0.2, and declined to 78.51 % at 2.4. CDANLC + MSPA improved from 77.59 % at 0.2 to 80.74 % at 1.2 and approached a plateau near 81.5 % over 1.8–2.4. The main experiment retained $\lambda_{LC} = 0.2$ for comparability.

Over the common tested range up to $\lambda_{LC} = 1.2$, stronger LC produced different responses in Settings B and C: both Setting B ensembles declined at larger weights, whereas both Setting C ensembles improved from the default weight.

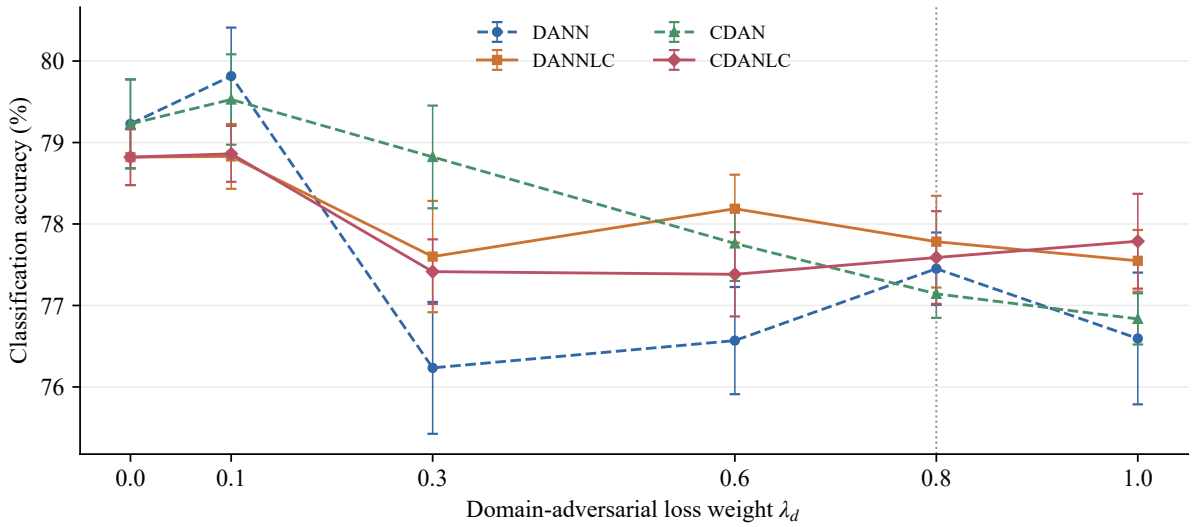


Figure 7.6 Domain-loss sensitivity of seven-member MSPA under Setting C. Solid curves show LC variants with $\lambda_{LC} = 0.2$; dashed curves show DANN and CDAN without LC. Error bars show one sample standard deviation across the 20 groups. The dotted line marks $\lambda_d = 0.8$.

Figure 7.6 shows that probability averaging did not remove sensitivity to domain alignment. Across the tested domain weights, DANNLC + MSPA ranged from 77.55 % to 78.83 %, while CDANLC + MSPA

ranged from 77.38 % to 78.86 %. Both means were higher at $\lambda_d = 0.1$ than at the retained weight of 0.8. Neither LC variant consistently exceeded its corresponding baseline across the tested domain weights. The main training configuration was retained.

7.2.5 Analysis of Medium-Term MSPA Performance

As in the short-term analysis, most samples were classified correctly by every member, leaving a relatively small mixed subset. Within the medium-term seven-member groups, DANN and DANNLC had larger mixed proportions than CDAN and CDANLC (Figure 7.3). This difference may help explain why DANN and DANNLC benefited from MSPA and generally improved as M increased: more samples had correct member predictions available to compensate for other members' errors.

CDAN and CDANLC had smaller mixed proportions, and increasing M did not raise their ensemble means above their single-model means (Figure 7.4). At $M = 7$, accuracy decreased by 0.50 and 0.21 percentage points, respectively. These small reductions suggest that adding members did not provide enough useful complementary information to improve classification under the tested configuration. The result also depends on the probability vectors: a correct prediction from one member may carry too little probability to change the ensemble decision when other members favor an incorrect class.

The loss-weight experiments identify a separate influence on ensemble performance. With seven members, stronger LC improved CDANLC over the tested range, while DANNLC reached its highest tested ensemble mean before declining (Figure 7.5). The CDANLC ensemble could therefore improve through changes to member training even though adding members at the main weights did not improve on the corresponding single-model mean. These findings distinguish the quality of the learned predictions from the number of predictions being averaged. Under medium-term drift, the choice of training objective remained important for the final MSPA result.

7.3 Chapter Summary

For KEQ 7.1, LC produced small positive mean gains at the main loss weights, with effects that varied across tasks and seeds. The sensitivity experiments showed a broader response: stronger LC improved accuracy over part of the tested range, with diminishing returns for DANNLC and improvements over a wider range for CDANLC. The domain-loss results showed no uniform advantage from stronger adversarial alignment. These findings indicate that the main-weight comparison captures only part of LC's behavior under adjacent-batch adaptation.

For KEQ 7.2, MSPA improved DANN and DANNLC on average but slightly reduced CDAN and CDANLC accuracy. Increasing the ensemble size did not reverse this difference. The larger mixed subsets for DANN and DANNLC may help explain their gains, while the loss-weight sweeps showed that ensemble performance also depended on member training. As in the long-term experiments, increasing the number of members and changing the training objective affected different aspects of the final prediction.

Adjacent batches still included difficult transfers and a target gas absent from the corresponding source batch. Temporal proximity alone therefore did not ensure sufficient class support or reliable classification. Taken together, the results show that LC and MSPA depend on the source–target relationship, the model, and the training weights. The next chapter combines the three experimental settings to answer the global RQs and distinguish the benefits of class-center alignment from those of probability averaging.

8 Discussion and Conclusions

This chapter synthesizes the effects of latent class-center alignment (LC) and multi-seed probability averaging (MSPA) across the three drift settings. It discusses how the observed gains changed with the adaptation task and loss weights, before presenting the limitations, directions for future work, and final conclusions.

8.1 Discussion Across the Three Drift Settings

Table 8.1 summarizes the effects of LC and seven-member MSPA across the three drift settings.

Table 8.1 Summary of LC and seven-member MSPA effects across the three drift settings.

Drift setting	Effect of LC	Effect of MSPA
Short-term	Positive gains of 1.23–3.79 pp across the three protocols	Very small gains under Protocol 1, with CDAN almost unchanged
Long-term	$+0.89$ pp for DANN and $+0.58$ pp for CDAN at the retained weights	Positive gains of 0.49–3.34 pp for all four methods
Medium-term	$+0.67$ pp for DANN and $+0.16$ pp for CDAN at the retained weights	Gains of approximately 1 pp for DANN and DANNLC; slight decreases for CDAN and CDANLC

Note: LC effects compare each LC variant with its corresponding baseline. MSPA effects compare seven-member probability averaging with the corresponding single-model mean. Values are changes in classification accuracy expressed in percentage points (pp).

For RQ1, LC improved both DANN and CDAN across all three short-term target-data protocols, with supporting evidence from the class-balanced metrics and feature analysis. Its gains were smaller at the main long- and medium-term weights and varied across tasks. Thus, the most consistent benefit occurred in the short-term experiments.

For RQ2, access to all target batches in advance was not necessary for the short-term LC gains, and no protocol was best for every model. Stronger LC improved accuracy over part of the short- and medium-term ranges but reduced it beyond the main long-term weight. Stronger domain alignment also gave no uniform improvement. The long-term confidence analysis suggests a further limitation: incorrect target predictions can bias the class-center estimates used by LC, even when those predictions have high confidence.

For RQ3, MSPA improved all four methods on average under long-term drift. Under short-term drift, it gave small gains for DANN, DANNLC, and CDANLC, while CDAN remained almost unchanged. Under medium-term drift, it improved DANN and DANNLC but slightly reduced CDAN and CDANLC accuracy. The larger long-term gains were particularly evident for CDAN and CDANLC (Table 8.1).

Differences in usable complementarity may help explain these results. Most short- and medium-term samples were classified correctly by every member in their respective analyses, leaving relatively small mixed subsets (Figures 5.8 and 7.3). The short-term analysis used all 20 seeds, so its proportions are not directly comparable with the seven-member results. With the same seven-member grouping, long-term drift had a lower all-correct proportion and a higher mixed proportion than medium-term drift (Figure 6.6), providing more opportunities for correct member predictions to compensate for errors.

The acetaldehyde case shows why mixed predictions alone are insufficient. In the separate 20-seed analysis, the mixed proportion ranged from about 39 % to 53 %, exceeding half for CDAN and CDANLC, but many samples were misclassified by every seed. Recognition remained poor after MSPA (Figures 6.8 and 6.7). Some correct predictions do not ensure that the correct class receives enough probability to dominate the average. MSPA therefore depends on both prediction quality and usable complementarity.

8.2 Limitations and Future Work

CDAN and LC depend on the reliability of target predictions used as soft pseudo-labels. CDAN uses these predictions to condition domain discrimination, while LC uses them to estimate target class centers. Incorrect predictions can therefore misguide alignment and reduce classification accuracy.

MSPA was evaluated as a way to combine predictions across random seeds and reduce some of these errors. It averages prediction probabilities after training, so it does not directly correct the pseudo-labels used to train the members. Its benefit remains limited by the quality and complementarity of their predictions. When members make similar errors or consistently fail to recognize a gas class, averaging alone may offer little improvement.

Future work should reduce the influence of incorrect pseudo-labels during training. Teacher–student knowledge distillation could use teacher-generated soft pseudo-labels to supervise further student training. Prediction confidence and agreement could guide label selection and weighting, although high-confidence teacher errors may still propagate to the student. Other directions could include consistency constraints across perturbed inputs and class-specific reliability estimates.

8.3 Final Conclusions

This thesis evaluated LC and MSPA as two additions to domain-adversarial gas sensor drift compensation. For RQ1, LC improved DANN and CDAN most consistently across the short-term target-data protocols. The class-balanced metrics and feature analyses supported improved class discrimination alongside the accuracy gains. At the main long- and medium-term weights, its average gains were smaller and varied across tasks. Explicit class-center alignment can therefore complement both overall and conditional domain alignment, but its contribution depends on the adaptation conditions.

For RQ2, the short-term LC gains did not require access to all target batches in advance. Target-data organization and loss weights affected performance, and stronger alignment did not uniformly improve accuracy. The long-term analysis also revealed persistent incorrect target predictions, including high-confidence errors and pronounced differences between gases. These findings identify prediction reliability as a concern for methods that use predicted class information to guide alignment. Confidence-based weighting produced different effects across the model families, showing that greater prediction certainty alone does not ensure more accurate adaptation.

For RQ3, MSPA improved all four methods on average under long-term drift. Under short-term drift, it provided small gains for DANN, DANNLC, and CDANLC while leaving CDAN almost unchanged; under medium-term drift, it improved DANN and DANNLC but slightly reduced CDAN and CDANLC accuracy. The overlap and ensemble-size analyses suggest that averaging benefits depend on both prediction quality and usable complementarity. A high all-correct proportion leaves limited room for improvement, while the acetaldehyde results show that mixed predictions alone may be insufficient when members share a class-specific weakness.

LC changes the representations learned during adaptation, whereas MSPA combines the predictions of the trained members. The experiments show that improvements at these two stages need to be assessed separately: averaging can improve a model without preserving an additional advantage from LC. Together, the findings support examining class-specific errors and prediction complementarity alongside overall accuracy when evaluating drift compensation. Future work should improve the reliability of the class information used during training, so that aggregation has more accurate member predictions to combine.

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